The biregular case of a spectral-inclusion conjecture: the weighted plane class and its obstructions

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Abstract

The conjecture that every root of the d-matching polynomial $\mu_{d,G}$ lies in $\mathrm{spec}(T)$, T the universal cover, is false in general [27]. Its biregular case survives, contains Problem 1 of Song, Fan and Miao, and is the subject of this paper. At $b = 2$ the object is the *weighted plane class*: families $\{(c_k, V_k)\}$ of weighted 2-planes with $\sum_k c_k P_{V_k} \preceq aI$, whose matching polynomial is the mixed characteristic polynomial of Marcus, Spielman and Srivastava.

We prove a positive margin for the complete bipartite family, so no $K_{d,q}$ refutes the repaired statement however wide its gap is made, and we show the margin decays like $n^{-2/3}$, the soft-edge exponent, so the statement is true but tight and no size-free bound can prove it. We give an exact vertex recursion for the plane class, decompose its cross-term into pieces whose nonnegativity would give the band, and prove two unconditional bounds from real-rootedness alone.

The obstructions are the substance. A rank-blind barrier reads the Marchenko–Pastur law and cannot reach the tree band at either edge; compressions cannot reach the inner end at all; the coefficient ladder's reach is unbounded in the moment order, but an input at a uniform rate is the conclusion it would prove, so the dimension restriction is intrinsic; and the tree moment bound, the natural a priori input, is false off the coordinate case. A ratio route escapes these and settles the inner edge on a nontrivial class. Everything asserted is machine-checked in Lean 4/Mathlib unless labelled otherwise.

Status conventions. Claims here are of three kinds, and the text says which each time rather than leaving it to the reader. A result described as *formalized*, or cited by a Lean declaration name, is machine-checked in Lean 4/Mathlib: the sources are in `RamaLean/` of the accompanying repository, carry no `sorry`, and rest only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A claim described as *measured*, *verified to* a stated tolerance, or *checked numerically* is computational evidence and is not a proof; the script producing it is named, and every named script is run and its output compared against a recorded baseline (`code/cite_check.py`). A claim called a *conjecture* is unproved, and is called that even where the evidence is strong.

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1 Preliminaries, from the companion paper

This paper is a sequel and does not repeat what it rests on. The conjecture, the closed form at first Betti number one, the subdivision theorem, the complete bipartite case and the minimum-degree repair are proved in [27]; they are restated here without proof, in the form used below. Numbering is local to this paper.

Conjecture 1. *For every finite graph G and every $d \geq 1$, all roots of $\mu_{d,G}$ lie in* $\operatorname{spec}(T)$*, the spectrum of the universal cover of G.*

Conjecture 2 (D3)**.** *Every finite graph of minimum degree at least three satisfies* $\operatorname{Zeros}(\mu_G) \subseteq \operatorname{spec}(T_G)$.

Theorem 3. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) \;=\; U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) \;=\; \underbrace{\big(2T_n(x/2) - 2\big)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}\big(T_n(x/2)\big)}_{\Psi_r(x)}.$$

Corollary 4. $\displaystyle\sum_{d\geq 0} \mu_{d,C_n}(x)\, z^d = \frac{1}{1 - 2T_n(x/2)\, z + z^2}$, *and* $\displaystyle\sum_{r\geq 0} \Phi_{n,r}(x)\, z^r = \frac{(1-z)^2}{1 - 2T_n(x/2)\, z + z^2}$.

Corollary 5 (real-rootedness)**.** *All roots of $\Phi_{n,r}$ lie in $[-2, 2]$: with $x = 2\cos\theta$, χ_{C_n} vanishes at $\theta = 2\pi m/n$ and $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta = m\pi/(rn)$ with $r \nmid m$ (when $r \mid m$ the denominator vanishes too and $\Psi_r = \pm r \neq 0$); this leaves exactly $n(r-1)$ roots, matching* $\deg \Psi_r$.

Corollary 6 (Fibonacci–Lucas evaluation)**.** $\Phi_{n,r}(3) = (L_{2n} - 2)\, F_{2nr}/F_{2n}$*, since $T_n(3/2) = L_{2n}/2$ and $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$.*

Corollary 7 (parity)**.** $\Psi_r(-x) = (-1)^{n(r-1)}\Psi_r(x)$*: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite C_n it is an even polynomial for every r; at $r = 2$ it is the classical matching polynomial $2T_n(x/2)$ of C_n, consistent with Godsil–Gutman [3].)*

Theorem 8. *Let H be D-regular, $D \geq 2$, and let $S(H)$ be its subdivision, so that the universal cover of $S(H)$ is the $(D, 2)$-biregular tree with spectral gap $\big(0, \sqrt{D-1} - 1\big)$. Then every nonzero root of $\mu_{S(H)}$ has absolute value at least $\sqrt{D-1} - 1$; that is, Conjecture 1 holds for $S(H)$ at $d = 1$. The constant is the spectral gap of the universal cover; we do not claim it is attained, and by the strictness of Heilmann–Lieb on finite graphs it is not.*

Theorem 9. *Let $2 \leq p \leq q$. Every root of $\mu_{K_{p,q}}$ lies in*

$$\{0\} \cup \pm\big[\sqrt{q-1} - \sqrt{p-1},\ \sqrt{q-1} + \sqrt{p-1}\big],$$

the spectrum of the (q, p)-biregular tree. So Conjecture 1 holds for every $K_{p,q}$ at $d = 1$.

Five displays are used by number below and are reproduced here. The averaging identity

$$\Phi_{n,r}(x) \;=\; \mathbb{E}_{\sigma\in S_r} \prod_{\ell\,\in\,\mathrm{cycles}(\sigma)} \bigl(2T_{n\ell}(x/2)-2\bigr). \tag{1}$$

the recurrence carrying the closed form

$$V_0 = 1, \qquad V_1 = \mu_G, \qquad V_d = \mu_G\, V_{d-1} - \mu_{G-V(C)}^2\, V_{d-2}. \tag{2}$$

Newton's identity in the form used for the coefficients

$$r\,\Phi_r = \sum_{k=1}^{r} f(k)\,\Phi_{r-k}, \qquad \Phi_0 = 1, \tag{3}$$

the arithmetic–geometric mean step

$$\sqrt{u(u+\alpha)} + 1 \;\le\; \sqrt{(u+1)(u+1+\alpha)} \tag{4}$$

and the two-sided statement

$$\text{every root of } g \text{ lies in } \bigl[-2\sqrt{m},\, 2\sqrt{m}\bigr], \tag{5}$$

That is everything borrowed. From here the paper is self-contained, and it proceeds in three movements: what the biregular case has going for it, what the plane class looks like when the question is posed operator-theoretically, and what the obstructions are.

2 Why the biregular case survives, and how far

The refutation leaves the biregular case standing, and this section says why and measures how much room it has: a positive margin for the complete bipartite family, the decay of that margin, the two obstructions that bar compressions and rank-blind barriers from the inner end, and the ratio route that escapes them.

2.1 Why the biregular case survives

For an (a, b)-biregular graph the universal cover is the (a, b)-biregular tree, whose spectrum is $\{0\}\cup\{x : \tau \le |x| \le \rho\}$ with $\tau = |\sqrt{a-1}-\sqrt{b-1}|$ and $\rho = \sqrt{a-1}+\sqrt{b-1}$. Since [4] already places every root in $[-\rho, \rho]$, Conjecture 1 for such a graph is *equivalent* to the assertion that no nonzero root satisfies $|x| < \tau$, which is Problem 1 of Song, Fan and Miao. That reduction is machine-checked as `BiregularBlocking.biregular_reduction`, with the tree spectrum and the Hall–Puder–Sawin interval as the two hypotheses; neither is in Mathlib. So the biregular case is not one fragment of the conjecture among others, it is the whole of what survives.

Both mechanisms of [27] are structurally blocked there, and for different reasons. The subcubic construction hangs branches on a tree skeleton, and a graph in which every vertex has degree at least two is not a tree, by handshaking against the edge count of a tree (`no_tree_of_two_le_degree`); a biregular graph with $a, b \ge 2$ has minimum degree at least two and so admits no such skeleton. Hall's construction needs a wide spread of degrees, and a biregular graph has exactly two. Beyond that, the two constructions place their roots in *bulk* gaps, at 1.20, 2.24 and 2.85 in the examples above, whereas the only gap available in the biregular case is the one at the origin: a counterexample there would need a root just above zero, which is a question about near-deficiency of matchings rather than about band structure.

There is also a quantitative obstruction, in the direction one would want, though it is not complete. At a cut vertex with p isomorphic branches the matching polynomial factors through $F = x\mu_H - p\mu_{H-v}$, and $F(0) = -p\,\mu_{H-v}(0)$, so a Lipschitz bound gives

$$|\theta| \;\geq\; \frac{p\,|\mu_{H-v}(0)|}{M}, \qquad M = \sup |F'|,$$

for every root θ (`branch_root_sep`). The separation grows linearly in the number of branches: strengthening the mechanism drives its roots away from zero, which is the opposite of what a biregular counterexample requires. The bound is conditional on $\mu_{H-v}(0) \neq 0$, that is on $H - v$ having a perfect matching, and Hall's own branch fails that, since there $H - v$ is a star of odd order. It therefore blocks the subcubic mechanism and not his, and on the subcubic branch it gives $|\theta| \geq 1/6$ against a true smallest root of 0.662: valid, and far from sharp.

The vacuity is removable. If $F(0) = 0$, divide out the zero: writing $F = x^m G$ with $G(0) \neq 0$, the nonzero roots of F are the roots of G, and G is what should be bounded. For Hall's branch $F = x^5(x^4 - 16x^2 + 55)$, so $G(0) = 55$ and the obstruction disappears; the bound then gives 0.991 against a true root of $\sqrt{5}$. Carried out on the matching polynomial itself, where $\mu_G(x) = \pm x^{n-2\nu} g(x^2)$ with ν the matching number and $g(0) = \pm m_\nu$, the classical Cauchy estimate becomes a statement purely about matching numbers.

Proposition 10. *Let G be a finite graph with matching number ν and k-matching numbers m_k. Then every nonzero root θ of μ_G satisfies*

$$\theta^2 \;\geq\; \frac{m_\nu}{m_\nu + \max_{k<\nu} m_k}.$$

The estimate is Cauchy's and is classical; only its reading in terms of matching numbers is being recorded here. It is machine-checked as `RootSeparation.cauchy_lower` and `matching_root_sep`, with no hypothesis on the graph.

What makes it worth stating is that the inequality points the useful way. By `biregular_reduction`, Conjecture 1 for an (a, b)-biregular graph is exactly the absence of a nonzero root below τ, so any graph whose bound reaches τ satisfies the conjecture, unconditionally and with no appeal to [25]; that implication is `RootSeparation.sfm_of_separation`. Run on random biregular graphs (`code/rootsep.py`) it settles three of nine, the smaller ones, the bound falling as $0.63, 0.45, 0.28$ against a fixed $\tau = 0.318$ for $(3, 4)$-biregular as the graph grows. So it decides small cases outright and is too weak for large ones.

The weakness is not intrinsic: Cauchy's estimate uses only the moduli of the coefficients and discards the one structural fact available. By Heilmann–Lieb μ_G is real-rooted, so g has all roots real and *positive*. For a finite family of positive reals the ℓ^m norm of the reciprocals decreases to the ℓ^∞ norm, so with $P_m = \sum_i y_i^{-m}$,

$$\min_i y_i \;\geq\; P_m^{-1/m},$$

sharpening as m grows and exact in the limit. The case $m = 1$ is the harmonic bound $m_\nu/m_{\nu-1}$, already better than Cauchy; and the P_m are computed exactly from the matching numbers by Newton's identities on the reversed polynomial, so every quantity is rational.

Proposition 11. *Let G be biregular with $\tau = |\sqrt{a-1} - \sqrt{b-1}|$, and let $q \geq \tau^2$ be rational. If $P_m\, q^m \leq 1$ for some $m \geq 1$, then no nonzero root of μ_G has $|\theta| < \tau$, so Conjecture 1 holds for G.*

Proof. $y_i^{-m} \leq P_m \leq q^{-m}$ for each i, hence $y_i \geq q \geq \tau^2$. □

Machine-checked as `PowerSumCertificate.roots_ge_of_powersum`, with division and m-th roots avoided so that the check is exact rational arithmetic. Measured in `code/powersum.py`, $m = 8$ already agrees with the true smallest root to three or four decimal places, and the certificate settles all nine biregular graphs tested, including every size at which the Cauchy and harmonic forms fail.

Run at scale it certifies 417 of 417 random biregular graphs across nine degree pairs and sizes up to 40 vertices, in exact rational arithmetic throughout (`code/certify_biregular.py`). Each of those is an unconditional proof of Conjecture 1 for that graph.

It is a decision procedure and not a proof of the conjecture, and the distinction is not a matter of effort. For a finite family of positive reals the ℓ^m norm of the reciprocals *decreases to* the ℓ^∞ norm, so

$$\sup_m P_m^{-1/m} = y_1$$

exactly: optimising the certificate over m computes the smallest root rather than bounding it. Numerically the agreement is already to five decimal places by $m = 16$ (`code/powersum_limit.py`). Consequently the certificate succeeds for a graph precisely when $y_1 > \tau^2$, which is Song–Fan–Miao for that graph, and a bound on P_m uniform in the size of the graph strong enough to give the conjecture would already be the conjecture. The difficulty has not moved, and the route is recorded as closed rather than promising. What the certificate does supply is a complete and exact decision procedure, which is worth having and is all it is.

Empirically the statement is robust. Over 180 random (a, b)-biregular bipartite graphs on up to 40 vertices, for (a, b) ranging over eight pairs, no root falls below τ; the smallest nonzero root stays a factor 1.25 to 2.2 above it (`code/biregular.py`). Random sampling is the wrong instrument for a structured phenomenon, so an adversarial local search was run as well, descending on the smallest nonzero root by degree-preserving edge swaps, which stay inside the biregular class exactly. It barely moves the quantity, from 0.6076 to 0.6005 over 1200 swaps in the hardest case, and the final ratios to τ run from 1.42 to 1.97 (`code/biregular_anneal.py`). The smallest nonzero root appears to be rigid inside the biregular class.

That, together with the first Betti number one case where the conjecture is a theorem, is where the remaining interest lies.

2.2 A positive margin for the complete bipartite family

The biregular case is most exposed at large degree contrast, since the gap $(0, \tau)$ widens without bound with q. For $K_{d,q}$ the question is decidable in closed form. A k-matching picks

k vertices on each side and matches them, so

$$\mu_{K_{d,q}} = x^{q-d} P(x^2), \qquad P(y) = \sum_{k \le d} (-1)^k \binom{d}{k} (q)_k \, y^{d-k}, \tag{6}$$

with $(q)_k$ the falling factorial. Here P is a Laguerre polynomial, and as $q \to \infty$ its zeros satisfy $(y-q)/\sqrt{q} \to$ the zeros of the probabilists' Hermite polynomial He_d. Writing h_d for the largest of those, $x_{\min} = \sqrt{q} - \frac{h_d}{2} + o(1)$ against $\tau = \sqrt{q} - \sqrt{d-1} + o(1)$, so the margin tends to $\sqrt{d-1} - h_d/2$; at $q = 10^6$ this agrees to four decimals for every d from 3 to 20 (`code/d3attack.py`).

Theorem 12. *For every $d \ge 3$ the limiting margin is positive: $\sqrt{d-1} - h_d/2 \ \ge\ \frac{1}{2}\big(\sqrt{d-1} - \sqrt{d-2}\big) > 0$.*

Proof. By the recurrence $\mathrm{He}_{n+1} = y\,\mathrm{He}_n - n\,\mathrm{He}_{n-1}$, the zeros of He_d are the eigenvalues of the Jacobi matrix with zero diagonal and off-diagonal entries $\sqrt{1}, \dots, \sqrt{d-1}$. Its absolute row sums are $\sqrt{i-1} + \sqrt{i}$, largest in the last row, so Gershgorin gives $h_d \le \sqrt{d-2} + \sqrt{d-1}$. □

The Gershgorin step is `CompleteBipartiteMargin.abs_eigenvalue_le_max_rowsum`, proved from the eigenvalue equation at an index where the eigenvector is largest; the consequence is `margin_pos_of_rowsum`, and `no_root_below_gap_edge` closes the chain. No complete bipartite graph violates Conjecture 1, however wide the gap is made.

2.3 The margin vanishes, so the statement is tight

$K_{d,q}$ is not extremal. Random (d,q)-biregular graphs have smaller margins, and the margin falls as they grow: for $(3,6)$ the minimum over twelve samples falls monotonically across fourteen sizes, from 0.5301 at $n = 12$ to 0.2009 at $n = 51$, fitting a power law at $R^2 \ge 0.999$ in each of twelve families spanning $d = 3$ to 6 (`code/softedge2.py`). Matchings are counted exactly, by a bitmask permanent over the smaller side, which is what makes the range reachable.

The rate is the soft-edge one. The cavity resolvents on the biregular tree satisfy $X = 1/(z - A_1 Y)$, $Y = 1/(z - B_1 X)$ with $A_1 = d-1$, $B_1 = q-1$, so $P = XY$ solves $A_1 B_1 P^2 + (A_1 + B_1 - z^2)P + 1 = 0$; writing $a = \sqrt{A_1}$, $b = \sqrt{B_1}$, its discriminant factors,

$$(z^2 - a^2 - b^2)^2 - 4a^2b^2 = \big(z^2 - (a+b)^2\big)\big(z^2 - (b-a)^2\big), \tag{7}$$

locating the band edges at $b \pm a$ (`SoftEdge.discriminant_factors`). The inner factor has a simple zero there (`inner_edge_simple`), so the density vanishes like $\sqrt{x-\tau}$ and the density exponent is $\frac{1}{2}$ for every (d,q). The expected root count within δ of the edge is then $CN\delta^{3/2}$, and the extreme root sits where that count is one, giving $\delta = (CN)^{-2/3}$ (`edge_scale`, `sqrt_edge_exponent`). Writing $\rho(x) = \kappa\sqrt{x-\tau} + O(x-\tau)$, the asymptotic margin is explicit,

$$\delta = \left(\frac{3}{2\kappa n}\right)^{2/3} \tag{8}$$

(`sqrt_edge_quantile`), and agrees with the analytic quantile to four significant figures at $n = 10^6$ in all twelve families. The density itself passes a sharp check: $n \int_\tau^\rho \rho = r$ to eight decimals, r being the exact number of positive roots.

The consequences are structural, and they disqualify a class of proofs including Theorem 12. A quantity positive at every size may admit no uniform positive lower bound, so any argument producing a constant independent of n is insufficient in general (`no_uniform_lower_bound`); the Gershgorin constant is such a constant. A lower bound may not decay more slowly than the true one, so a proof must reach the exponent (`exponent_floor`). And the picture is a Friedman-type one: the edge is the greatest lower bound of the smallest positive roots and is never attained (`inf_not_attained`), with no constant above it bounding them below (`edge_unimprovable`). There is no ε to spare, and the statement cannot be softened.

Two limits on this. The fitted exponents vary with the aspect ratio q/d at the sizes reachable, from -0.67 at $q = 2d$ to -0.61 at $q = 4d$, but so does the exponent of the analytic quantile, which is $-2/3$ by construction and reaches it only near $n = 10^6$; the variation is a finite-size effect (`code/quantile.py`). And the observed smallest roots sit where the tree measure places about 0.5 roots, a count that drifts as $n^{+0.10}$; that departure is neither the girth, which is constant at 4 throughout, nor the estimator, since the unbiased sample mean drifts equally, and it is unexplained (`code/nobs.py`).

2.4 Compression cannot reach the inner end

There is a structural reason the outer end of (5) is a theorem and the inner end is not. The outer bound is a statement about quadratic forms: every Rayleigh quotient of a diagonal form lies between its least and greatest entry (`SoftEdge.rayleigh_between`), and being a statement about forms it passes to every subspace. That is exactly the path-tree argument, the path tree being an induced subtree of T and its adjacency form a compression.

A gap is not a statement about quadratic forms and is not inherited. With entries -1 and 1 the spectrum is $\{-1, 1\}$ and $(-1, 1)$ is a gap, yet the vector $(1, 1)$ has Rayleigh quotient 0, strictly inside it (`rayleigh_in_gap`). So a compression can take values the original excludes, and no argument phrased in Rayleigh quotients or compressions can rule them out however refined. This is the P_8-inside-the-$(3, 2)$-biregular-tree phenomenon of [27] at its smallest, and it is why Heilmann–Lieb, Godsil's path tree and the Marcus–Spielman–Srivastava interlacing machinery all deliver one end of the interval and are silent about the other.

2.5 The ratio route

What is not barred is the ratio recursion, which bounds ratios of matching polynomials rather than Rayleigh quotients. By Godsil μ_G divides μ_P for the path tree P, and on a tree the deletion recurrence is exact and local, $F_v = \lambda - \sum_u 1/F_u$ over the children u of v with $F = \lambda$ at leaves, so $\mu_G(\lambda) \neq 0$ follows once no F vanishes.

A positive invariant interval cannot do it: invariance would need $a \leq \lambda - k/a$, which has a positive solution only for $\lambda \geq 2\sqrt{k}$, the Heilmann–Lieb bound. The certificate is instead sign-alternating. On path trees of (d, q)-biregular graphs at gap points, over twenty-two thousand path-tree vertices, the signs separate by vertex type with no exceptions: every degree-d vertex has $F > 0$ and every degree-q vertex has $F < 0$ (`code/pathratio.py`).

Proposition 13. *Let $\lambda > 0$.*

(i) If $F_1, \dots, F_k \in [-C, -c]$ with $c > 0$ then $\lambda \leq \lambda - \sum_j 1/F_j \leq \lambda + k/c$.

(ii) If $F_1, \dots, F_k \in [a, B]$ with $0 < a \le B$ and $k > \lambda B$, or if $0 < F_j < k/\lambda$ for all j, then $\lambda - \sum_j 1/F_j < 0$.

These are `RatioRoute.left_step`, `right_step` and `right_step_sharp`; (i) holds for every k, so a path-tree vertex missing some of its children is still covered, and the sharp form of (ii) is what the path trees satisfy, with slack 0.42 to 4.47 and no violations over every vertex measured (`code/childbound.py`).

The combinatorial input is an alternation count. Let π end at $w \in R$, put $k = |N(w) \setminus \pi|$, $p_L = |\pi \cap L|$, $p_R = |\pi \cap R|$, and let u be a child of w with child count k_u.

Lemma 14. *$q - r \le k$ and $k_u \le k - (q - r)$.*

Proof. The $q - k$ blocked neighbours of w lie in L, so $q - k \le p_L$; the path alternates and ends in R, so $p_L \le p_R \le r$, giving $q - r \le k$. For the child, $N(u) \subseteq R$ and an unblocked neighbour must avoid $\pi \cap R$, so $k_u \le r - p_R \le r - q + k$. □

The three counting steps are formalized in `PathCount`: `blocked_le_left` and `unblocked_le` are the set-counting steps, and `alt_count` is the alternation count, proved on lists recording which side each path vertex lies on, with `alt_count_last` transferring it from the first vertex to the last. At the minimum $k = q - r$ the lemma forces $k_u = 0$, so the children are leaves of ratio exactly λ and the parent is $\lambda - k/\lambda < 0$ once $k > \lambda^2$.

Assembling both sides gives a system in one constant $m > 0$, the lower bound on $|F|$ at right-type vertices. With $\kappa(t) = \lambda + t/m$ bounding left-type ratios at child count t, and Lemma 14 capping a right-type vertex's children at $k - \Delta$ each where $\Delta = q - r$, the parent is at most $\lambda - k/\kappa(k - \Delta)$ (`RatioRoute.right_step_bound`); the invariant propagates when that is at most $-m$, which is

$$\lambda + m \;\le\; \frac{k}{\kappa(k - \Delta)} \qquad \text{for all } \Delta \le k \le q - 1 \tag{9}$$

(`closure_iff`, `induction_step`). Equation (9) is a fixed-point condition on m with no closed form; iterating $m \mapsto \min_k\bigl(k/(\lambda + (k - \Delta)/m) - \lambda\bigr)$ from above gives the largest admissible m.

Theorem 15. *Let G be (d, q)-biregular with r vertices of degree q, and $\lambda \in (0, \tau)$. If (9) admits some $m > 0$, then $\mu_G(\lambda) \neq 0$.*

The recursion consuming the two steps is `PathTreeInduction.invariant`, by strong induction on a height function, with `no_vanishing` concluding that no ratio vanishes; the path tree enters only through an abstraction carrying a side map, a children map and a height that strictly decreases along edges. A single constant is more than the argument needs: the alternation count applies at every depth, a path of ℓ vertices ending in L having $\lfloor \ell/2 \rfloor$ vertices in R, which caps a left-type vertex's children by $\min(d-1, r - \lfloor \ell/2 \rfloor)$ and bounds a right-type vertex's below by $q - \lfloor \ell/2 \rfloor$. Indexing κ and m by depth turns (9) into a finite backward recursion, and Theorem 15 holds verbatim in that form (`invariant_depth`).

Solving numerically, the condition holds throughout the gap for (d, q, r) equal to $(3, 6, 4)$, $(3, 9, 4)$, $(3, 12, 4)$, $(4, 8, 4)$, $(4, 12, 4)$ and $(5, 10, 4)$, and to 69% of the gap for $(3, 6, 5)$ (`code/depthind.py`); §2.6 removes that last restriction by bounding each child at its own count rather than at the worst one. This is the first proof of the inner end of (5) for a nontrivial class.

This has to be reconciled with §12, which rules out cavity recursions. The obstruction there is that every b-regular bipartite graph is an induced subgraph of some (a, b)-biregular graph, and over b-regular bipartite graphs the least root tends to zero, so on the closure of the biregular graphs under vertex deletion the inequality to be proved is false; an induction carrying the target statement through that closure cannot work. Theorem 15 is an induction over vertex-deleted subgraphs, but it does not carry the target statement through them. It carries the interval bounds, and those are guarded by $\Delta = q - r \geq 1$: when $\Delta \leq 0$ a right-type path-tree vertex may have no children at all, the sign alternation fails at once, and (9) admits no positive m. A b-regular bipartite graph with parts of equal size has $\Delta = b - r < 0$, so the family witnessing the obstruction is exactly what the hypothesis excludes. The condition $\Delta \geq 1$ is therefore not a technical convenience; it is what §12 forces, and it is the reason the route reaches a nontrivial class rather than nothing or everything.

2.6 The bound is per-vertex, and the saturation is only of constants

The limit at $(3, 6, 5)$ is real but narrower than it looks. What is saturated there is the *uniform* certificate: both bounds are attained with ratio 1.0000, so no choice of constants does better (`saturation_blocks_improvement`). The refinement below is not a choice of constants, and it escapes for that reason.

The requirement (9) caps every child by the same κ, evaluated at the worst child count $d-1$. But Proposition 13(i) already gives a left-type vertex with j children the sharper bound $\kappa(j) = \lambda + j/m$, and a right-type vertex with very few children sits where the path tree is nearly exhausted, so its children are child-poor too and carry a much smaller κ. Instantiating the bound at each child's own count rather than globally, the requirement at a right-type vertex becomes

$$k \; > \; \lambda\, \kappa(j_{\max}), \qquad j_{\max} = \max_i j_i,$$

which is strictly weaker whenever $j_{\max} < d - 1$ (`RatioRoute.refined_weaker`, `right_step_refined`). Measured on the path trees, it fails nowhere, including at the two $(3, 6, 5)$ points where the uniform requirement fails at eight vertices (`code/twolevel.py`). The saturation result is untouched: it says the constants cannot be improved, and this does not improve them.

Substituting Lemma 14 for $j_{\max}$ and asking the requirement for every admissible k removes the path tree altogether. With $\Delta = q - r$ it reduces, by one cancellation, to two inequalities:

Theorem 16. *If $\lambda < m$ and $\lambda^2 < \Delta$, then $\lambda\, \kappa(j) < k$ for every $k \geq \Delta$ and every $j \leq k - \Delta$.*

Proof. $\Delta(m - \lambda) > \lambda(\lambda m - \Delta)$ is equivalent to $\Delta > \lambda^2$ after cancelling m; since $m > \lambda$ and $k \geq \Delta$ the left side only grows with k (`RatioRoute.structural_closure`). □

Neither hypothesis mentions a path tree, a family or a graph. The second is the binding case already noted, where the children are leaves of ratio exactly λ; the first is what the per-vertex bound buys. Over 354 parameter points (d, q, r) that are *realizable* — a (d, q)-biregular graph with r vertices of degree q needs qr/d to be a positive integer — the pair holds at 326, or 92.1%, failing only as λ approaches the gap edge (`code/universal_close.py`). The elementary sharpening $k_u \le \min(d-1, k-\Delta)$, using that a child lies in L and has degree d, gains nothing at any of them (`code/sharp_close.py`): the obstruction is the two conditions themselves and not the counting.

2.7 The path tree inherits the gap

That obstruction is the invariant's and not the method's, and the reason is worth recording because it says the route is not exhausted. The recursion computes $F_v = \mu_{T_v}/\mu_{T_v - v}$, so "no F vanishes" is exactly $\mu_P(\lambda) \ne 0$ — strictly stronger than $\mu_G(\lambda) \ne 0$, since μ_P carries its own roots as well as those of μ_G. Were μ_P to vanish somewhere in the gap, no certificate of this kind could succeed there, whatever invariant it used.

It does not. On a tree $\mu_P = \prod_v F_v$ and the F_v are the pivots of the LDL factorization of $\lambda I - A_P$, so by Sylvester's law the number of negative F_v counts the eigenvalues of P above λ; comparing the count at the two ends of the gap counts the eigenvalues between, in time linear in $|P|$ and with no matrix. Over four (d, q, r) families that are *not* complete bipartite, with path trees of 3457, 17241, 22402 and 291429 vertices, the count is equal at both ends in every case: the path tree has no eigenvalue in the gap (`code/pathtree_inertia.py`). This is 'HEURISTIC', four families, and it is stronger than Godsil's divisibility, which places the roots of μ_G among those of μ_P and says nothing about where the others go.

So the statement the route proves is true throughout the gap, including where the sign-alternating certificate cannot hold. An argument therefore exists in principle at every λ in the gap. What does not exist is a better certificate of *this* family, and that is worth separating from the first point.

Two things settle it. Measuring $\min_v |F_v|$ across the gap suggests the sign-free alternative $|F_v| \ge \lambda$: the minimum equals λ exactly, attained at a leaf, at every fraction tested in two of three families, falling to 0.83λ and 0.51λ only at $(3, 6, 5)$ near the edge (`code/minF_scan.py`). But it is the harder condition, not a weaker one. With every child positive, $F_v = \lambda - \sum_u 1/F_u$ falls below λ automatically, and $|F_v| \ge \lambda$ additionally forces the sum past 2λ, which at a right-type vertex is $k \gtrsim 2\lambda B$ against the $k > \lambda B$ the sign condition needs — a factor of two in the child count.

And there is no cancellation to repair. The path tree is bipartite by depth, so the children of a vertex all lie on one side and are of one type, and same-type siblings never disagree in sign: over three families at six gap fractions each, including 0.99, the number of vertices whose children have mixed signs is *zero* in every case, out of 865, 4185 and 3220 parents (`code/signmix.py`).

2.8 The same measurement reformulates the conjecture

The inheritance of §2.7 says more than that the ratio route's target is true. It turns the conjecture into a statement about a finite tree.

Godsil gives $\mu_G \mid \mu_P$, and P is a tree, where the matching polynomial *is* the characteristic polynomial. So every root of μ_G is an eigenvalue of P, and

Proposition 17. $\mathrm{spec}(P) \subseteq \mathrm{spec}(T)$ *implies* $\mathrm{Zeros}(\mu_G) \subseteq \mathrm{spec}(T)$. *Conversely a root outside* $\mathrm{spec}(T)$ *is an eigenvalue of* P *outside it, so the containment fails exactly where the conjecture does.*

Proof. Immediate from $\mu_G \mid \mu_P$ in both directions (`PathTreeRoute.conj_of_pathtree`, `pathtree_fails_of_violation`). □

The hypothesis is strictly stronger than the conclusion, since μ_P carries its own roots as well, and that is the point rather than a defect: it is a statement about the spectrum of one explicit finite tree with no matchings in it. It is also not vacuous in either direction. It fails for Hall's graph, necessarily — $\sqrt{5}$ is a root of μ_G, hence an eigenvalue of P, and the ratio system at $\sqrt{5}$ has decay 0.9636233789, so it is outside spec(T) — and a hypothesis provable in general would be worthless, the conjecture being false. And it holds wherever it has been measured, over the four families of §2.7.

The strictness of the hypothesis is decisive, and against it. At minimum degree three the containment is *false*. Over 86 graphs with $\delta \geq 3$ carrying a gap of spec(T), all but one inherit it; the exception is a 12-vertex graph with degrees $\{3, 4, 10\}$ whose path tree, on 3944 vertices, carries twelve eigenvalues inside the gap $[2.060, 2.160]$ while no root of μ_G lies there — counted by Sylvester's law, with the near-zero-pivot flag honoured, so the count is trustworthy (`code/pathtree_delta3.py`).

So spec$(P) \subseteq$ spec(T) fails exactly where Conjecture 1 holds, and Proposition 17 cannot carry a proof of the minimum-degree conjecture: no argument can pass through a hypothesis that is false on the graphs in question. What survives is the biregular reading. There the containment is measured to hold (§2.7), and there the proposition does reduce the inner end to a statement about one explicit finite tree. We record the negative result because it closes a route that looks plausible from the biregular evidence alone, and because it was the only reformulation five searches produced.

So the sign-alternating certificate is the best of its family, and its reach is exactly $\lambda^2 < \Delta$ — the binding case, where the children at the minimum count are leaves of ratio λ and the parent is $\lambda - \Delta/\lambda$. **That boundary is sharp rather than an artefact of the certificate's shape**, and extending the route past it needs an argument that is not an interval invariant on the ratios at all, since the target it would prove is true there and every certificate of this kind provably is not.

Finally, the sign separation is a consequence of bipartiteness: on wheels no natural class determines the sign of a path-tree ratio, endpoint degree, depth parity and child count each being mixed (`code/d3ratio.py`), so the certificate does not transfer to Conjecture 2 in general.

The biregular case is what survives, and §2.5 settles it on a nontrivial class by the ratio recursion. This part develops a second route to the same statement, through counting, the coefficient ladder and the operator form, together with the obstructions each of them meets. It uses the framework of Part I throughout; nothing in Part I depends on it, except that §2.5 and [27] each cite one result proved here.

3 A counting refinement

GAPCOUNT is refuted along with the conjecture it refines, and this section is kept because most of it does not depend on GAPCOUNT being true. What survives, and is what the section is now for: the implication GAPCOUNT ⇒ localization and the boundary conditions, machine-checked; BAND, which is a theorem about the abelian shadow and independent of either conjecture; the settled cases at feedback vertex number one and, conditionally, two; and the two barriers, which close routes that would otherwise look open. The sweeps are retained as a record of where the failure is *not* visible. A reader interested only in the biregular case may go straight to §2 and the operator form.

Conjecture 1 asserts a non-vanishing, which offers no handle. This section records a strictly stronger statement that equates two integers, the cases it settles, and the two barriers that close off the obvious routes to the rest. Labels are stated throughout: what is machine-checked, what is proved modulo a cited classical theorem, and what is open.

Let T be the universal cover of a finite connected loopless G on n vertices, and $b = b_1(G)$. Writing A_G for the universal-cover adjacency operator as a matrix over $C^*_r(F_b)$, a real x lies outside $\operatorname{spec}(T)$ exactly when $xI - A_G$ is invertible there, and then the trace of its negative spectral projection,

$$\kappa(x) = \tau_n\big(\mathbf{1}_{(-\infty,0)}(xI - A_G)\big),$$

is an integer in $[0, n]$, because $K_0(C^*_r(F_b)) = \mathbb{Z} \cdot [1]$ by Pimsner–Voiculescu.

Conjecture 18 (GAPCOUNT)**.** *For every $x \notin \operatorname{spec}(T)$, the number of roots of μ_G strictly above x, with multiplicity, equals $\kappa(x)$.*

Conjecture 18 implies Conjecture 1 at $d = 1$: κ is constant across a gap while the root count jumps at any root, so a gap contains none. By the counterexample of [23] the implication now runs the other way and Conjecture 18 is false as well, on the same graph: the root $\sqrt{5}$ crosses inside a gap on which κ is constant. We state it as it was, because the implication and the boundary conditions are what the sweeps below actually test, and because the counting refinement remains the right object for whichever class of graphs turns out to satisfy the localization. That implication, the fact that an integer-valued locally constant function is determined on a connected gap, and the boundary conditions $\kappa = 0$ above $\rho(T)$ and $\kappa = n$ below $-\rho(T)$, are machine-checked in `RamaLean/GapLabel.lean`. All the content is in what happens as a band is crossed.

3.1 The abelian shadow

Put a circle variable on each cotree edge and let $A_G(z)$ be the magnetic adjacency matrix, $B_k = \lambda_k(\mathbb{T}^b)$ the k-th Floquet band, and $\theta_1 \geq \cdots \geq \theta_n$ the roots of μ_G.

Proposition 19 (BAND)**.** *$\theta_k \in B_k$ for every k. Consequently, for $x \notin \operatorname{spec}(G^{\mathrm{ab}})$ the number of roots above x equals the number of bands above x.*

Proof. Godsil–Gutman writes μ_G as the average characteristic polynomial over ± 1 edge signings, and Marcus–Spielman–Srivastava [19] make those signings an interlacing family. The common interlacing lemma brackets the average at every index, not only the largest, so $\lambda_k(A_s) \leq \theta_k \leq \lambda_k(A_{s'})$ for two signings. Both are points of $\mathbb{T}^b$, and B_k is the continuous image of a connected space, hence an interval. □

Everything downstream of those two classical inputs is machine-checked in `RamaLean/BandTheorem.lean`: that bands are order connected (`ordConnected_range_of_continuous`), the squeeze (`mem_of_squeezed`), the count identity (`countAbove_eq_countBands`) and the recovery of the unicyclic case (`conj10_of_bands_subset`). The inputs themselves enter `band_of` as hypotheses.

The core of the common interlacing lemma is machine-checked in `RamaLean/Interlacing.lean`: a sign argument fixes the parity of a root count and nothing more, an interlacing confines that count to a window of width two, and parity with such a window determines the value. What is not formalized, and is cited to [19], is that the signings have a common interlacing at all.

Proposition 19 settles Conjecture 1 whenever every band lies in $\operatorname{spec}(T)$, which is exactly $b_1(G) = 1$, since $F_1 = \mathbb{Z}$ makes the maximal abelian cover the universal cover. That recovers [27] by a different route and sharpens it to $\theta_k \in B_k$. The hypothesis fails at $b_1 \geq 2$: for K_4 the top band carries the Perron value 3 while $\rho(T) = 2\sqrt{2}$.

The counting half of Proposition 19 does not need [19]. If x avoids every $\lambda_k(z)$ then $\{z : \lambda_k(z) > x\}$ is clopen, so on a connected parameter space the eigenvalue count above x is the same integer for every z; every member of the family then has the same sign at x, so the average does too. Parity follows at once, and parity upgrades to equality by counting: crossing one component of the spectrum, each component carries an odd number of roots, and n odd positive integers summing to n are all 1. Both steps are machine-checked in `RamaLean/BandCount.lean`.

3.2 Feedback vertex number one

Suppose some vertex v has $F = G - v$ a forest, so every cycle passes through v; the first Betti number is unrestricted, and flowers, theta graphs and every $K_{2,q}$ qualify.

Proposition 20. *Conjecture 18, hence Conjecture 1 at $d = 1$, holds for every such G, modulo Pimsner–Voiculescu and Godsil's path tree.*

Proof. Eliminating the preimage of F gives a Schur complement $S_v(x)$ which is a *finite* sum of group elements, so it lies in the group algebra inside $C^*_r(F_b)$ and not merely in $L(F_b)$; that distinction is what makes projectionlessness applicable, since $L(F_b)$ is a II_1 factor full of projections. Haynsworth additivity gives $\kappa(x) = N_F(x) + \delta(x)$ with $\delta(x) \in \{0, 1\}$, and $\tau(S_v(x)) = \mu_G(x)/\mu_{G-v}(x)$, so definiteness fixes the parity of $N_G - N_F$. Godsil's path tree makes μ_{G-v} interlace μ_G, confining $N_G - N_F$ to $\{0, 1\}$. A window of width two and a parity determine the value. □

The assembly is `FeedbackGapCount.gapcount_fvs_one_of`, with the three classical inputs as explicit hypotheses, and `conj10_fvs_one_of` composes it with the skeleton of Section 3.

Three remarks on that proof. The combinatorial core, that in an acyclic graph two distinct neighbours of v cannot be joined without passing through v, is machine-checked (`FeedbackVertex.no_bypass`); it is why a walk returning to the same lift must enter and leave at the same attachment point. The trace formula requires G *simple*: the argument uses that a lifted attachment point has exactly one neighbouring lift of v. Haynsworth additivity needs no appeal to Sylvester's law in a von Neumann algebra: the block matrix is congruent to the direct sum of the corner and the Schur complement by explicit unipotent factors, and scaling the off-diagonal part keeps every point of the path invertible, so the trace of the negative spectral projection is constant along it. That is machine-checked in `RamaLean/Congruence.lean`.

3.3 Feedback vertex number two, and where it stops

At $k = 2$ the argument breaks in a locatable place: $M_2(C_r^*(F_b))$ is *not* projectionless, $\mathrm{diag}(1,0)$ being a counterexample, so $S(x)$ can be indefinite. What survives is the K-theory, which keeps $\delta(x) \in \{0,1,2\}$. Two deletions give a window of width three, so parity leaves a two-way choice between $\delta = 0$ and $\delta = 2$, and one further bit is needed. That bit exists.

Proposition 21. *Let $F = G - v_1 - v_2$ and let a, b_1, b_2, c count the roots above x of $\mu_G, \mu_{G-v_1}, \mu_{G-v_2}, \mu_F$. If $a - c$ is even then $b_1 = b_2$, and*

$$a = c \iff \mu_F(x) \text{ and } \mu_{G-v_1}(x) + \mu_{G-v_2}(x) \text{ agree in sign.}$$

Proof. Vertex-deletion interlacing gives $a - b_i \in \{0,1\}$ and $b_i - c \in \{0,1\}$. If $a = c$ then $c \le b_i \le a = c$; if $a = c+2$ then $c+1 = a-1 \le b_i \le c+1$. Either way $b_1 = b_2$, so the two polynomials share a sign at x and so does their sum. □

Proposition 21 is machine-checked in `RamaLean/TieBreak.lean`, and it needs no hypothesis about $\mathrm{spec}(T)$. It is not enough. The remaining gap is the parity itself: at $k = 1$ it came free from $\tau(S_v) = \mu_G/\mu_F$ and definiteness, while at $k = 2$ the corresponding object is the constant term of the abelianised determinant of a 2×2 matrix over a noncommutative algebra, whose sign has no evident reason to be $(-1)^\delta$. So feedback vertex number two is *reduced, not proved.* The arithmetic that would close it once the parity is supplied, a window of width three together with a parity and one tie-break bit, is `FeedbackTwo.eq_of_window3`, and `gapcount_fvs_two_sign` discharges the tie-break using Proposition 21, leaving the parity as the sole remaining hypothesis.

What it reduces to is now quantitative. Since μ_G/μ_F is the average over $\mathbb{T}^b$ of $\det S(x,z)$, and $\mathrm{sign}\det S(x,z) = (-1)^{\delta_{\mathrm{ab}}(x,z)}$, writing I_j for the integral of $|\det S|$ over $\{\delta_{\mathrm{ab}} = j\}$ gives $\mu_G/\mu_F = I_0 - I_1 + I_2$, and the required sign is exactly the assertion that the parity class matching δ dominates the others *in integral.* Measurement (`code/inertia_split.py`) shows the split is lopsided but not degenerate: at two triangles joined by an edge, in a genuine gap with $\delta = 1$, the class $\delta_{\mathrm{ab}} = 1$ occupies 0.987 of the torus and $\delta_{\mathrm{ab}} = 2$ occupies 0.013. The integrand therefore does change sign, so no argument asserting that it does not can work. The decomposition and the criterion are recorded in `RamaLean/InertiaSplit.lean`, with a witness that measure alone does not decide it.

3.4 Two barriers

Both close off routes that look available.

Proposition 22. *For $b \ge 2$ there is no unital $*$-homomorphism $C_r^*(F_b) \to C(\mathbb{T}^b)$ carrying the generators to the coordinates.*

Proof. Such a map would carry units to units, so $S(x)$ invertible would force $S(x,z)$ invertible for every z, hence $\det(xI - A_G(z)) = \mu_F(x)\det S(x,z)$ nonzero for every z, hence $\mathrm{spec}(G^{\mathrm{ab}}) \subseteq \mathrm{spec}(T) \cup \{\mu_F = 0\}$. For K_4 with a two-element feedback set that fails at $x = 3$: the Perron value lies in $\mathrm{spec}(G^{\mathrm{ab}})$, while $\rho(T) = 2\sqrt{2} < 3$ and $\mu_{K_2}(3) = 8$. □

The refutation shape is machine-checked in `RamaLean/NoQuotient.lean`. The underlying reason is that a quotient descends to reduced C^*-algebras when the *kernel* is amenable, and $[F_b, F_b]$ is free of infinite rank.

The second barrier is narrower than it first appeared. Every argument that runs on the torus family needs x outside $\mathrm{spec}(G^{\mathrm{ab}})$, and on the smallest examples that fails: for K_4, two triangles joined by an edge, the theta graph and K_4 with pendants, the internal gaps of $\mathrm{spec}(T)$ are swallowed by bands of the abelian cover. It would be wrong to read that as a general obstruction. A certified search over two growing families (`code/torus_gg.py`) finds the opposite behaviour as soon as the graphs are larger: ten of fourteen gap midpoints lie outside $\mathrm{spec}(G^{\mathrm{ab}})$, with a margin that grows with the number of vertices: for two c-cycles joined by an edge it runs 1.18, 7.52, 21.50, 51.35 as c goes from four to seven. So the torus family is vacuous on the smallest graphs and informative as they grow, the reverse of what the first computation suggested, and the localization of [27] settles Conjecture 1 outright at every such point, for every first Betti number and with no feedback vertex hypothesis.

Nor is that settlement subsumed by Heilmann–Lieb. Those graphs have maximum degree three, so Heilmann–Lieb excludes roots only outside $[-2\sqrt{2}, 2\sqrt{2}]$, while the certified points sit at $\pm 1.93, \pm 1.99, \pm 2.04, \pm 2.06$, all interior. The certification cannot be a grid minimum, which proves nothing when the zero set of $\det(xI - A_G(z))$ is a curve: that determinant has degree one in each z_e, so its 3^{b_1} Fourier coefficients are recovered exactly by a three-point transform, they bound the gradient, and a grid search together with that Lipschitz constant certifies a positive lower bound, while a sign change on the grid certifies the opposite verdict.

3.5 Evidence

`code/universal_cover.py` computes $\mathrm{spec}(T)$ and κ for an arbitrary finite graph, by the non-backtracking cavity equations on the directed edges. Its regression check is K_4 with a pendant at each vertex, where $\mathrm{spec}(T) = \pm[\sqrt{3} - \sqrt{2}, \sqrt{3} + \sqrt{2}]$ in closed form; the solver returns $[-3.1456, -0.3163]$ and $[0.3184, 3.1477]$. Runs whose density fails to integrate to one are discarded, since point spectrum makes the cavity fixed point singular and a diverged κ reads exactly like a violated Conjecture 18. That the cavity map preserves the lower half plane, so the iteration is total and the density non-negative, is machine-checked in `RamaLean/Cavity.lean`; convergence is not claimed.

A Rust sweep (`code/covercheck`) applies this to every connected graph of feedback vertex number at most two on up to seven vertices, computing the root count by Budan–Fourier, which is exact for a real-rooted polynomial, and κ by the cavity equations. Over 55 131 graphs and 99 487 gap points there is no failure of Conjecture 1 and none of Conjecture 18, no run is discarded by the validity gate, and κ is never further than 0.205 from an integer. That range could not have found the counterexample and should not be read as having looked: Hall's graph has 41 vertices and feedback vertex number five, so it lies outside the sweep in both parameters. What the sweep establishes is that the failure is not visible at small size, which is why it went unnoticed, and not that it is rare.

The measurement also shows what an estimate would have to exploit. The wrong-parity class is not merely small in measure, it is suppressed quadratically in integral: at two triangles joined by an edge, in the gap at $x = -1.767$, it occupies 0.013 of the torus and contributes 0.0007 of the integral of $|\det S|$, a margin of 0.9987. The reason is structural: the wrong-parity region is a shell around $\{z : \det S(x, z) = 0\}$, where one eigenvalue has just crossed and $|\det S|$ is correspondingly small. Small measure and small integrand together, which is why the competition is not close.

Scanning a whole gap rather than one point confirms the exponent. Across both internal gaps of two triangles joined by an edge, $I_{\rm wrong}/m^2_{\rm wrong}$ stays between 2.4 and 5.2 and the margin never falls below 0.977 (`code/margin_scan.py`); there is no blow-up at the gap edges, where the wrong-parity class is largest and the ratio is smallest. The chain that would turn this into an estimate is machine-checked in `RamaLean/ShellBound.lean`. Its first step is an identity rather than a bound: for a 2×2 Hermitian matrix the operator norm is the larger eigenvalue in absolute value, so $|\det S| = \|S\| \cdot \min(|\lambda_1|, |\lambda_2|)$ (`abs_mul_eq_max_mul_min`), and on the shell only the crossing eigenvalue matters. Weyl makes that eigenvalue Lipschitz in the phase, giving $|\det S(z)| \leq ML \operatorname{dist}(z, \partial)$ and hence $I_{\rm wrong} \leq m_{\rm wrong} ML\,\text{reach}$, where $\text{reach} = \sup_{z\in A} \operatorname{dist}(z, \partial A)$ is the inradius (`det_le_on_shell`). The relevant quantity is the inradius and not the diameter: a shell wrapping the torus has diameter of order one however thin it is.

The inradius is then bounded for free. A region of inradius r contains a ball of radius r, so $c_b r^b \leq m$ and $\text{reach} \leq (m/c_b)^{1/b}$, giving

$$I_{\rm wrong} \;\leq\; ML\, c_b^{-1/b}\, m_{\rm wrong}^{1+1/b},$$

which at $b = 2$ is $m^{3/2}$ (`reach_le_sqrt`, `integral_le_three_halves`). The upper half of the estimate is therefore unconditional. The region is not a thin shell but a blob, with m of order reach^2: measurement gives $\text{reach}\,/m$ running 2.60, 3.10, 4.63, 5.94, 9.80 as m falls from 0.052 to 0.0053. The exponent 3/2 is accordingly weaker than the m^2 the numerics suggest, and stronger than anything needed. The lower half, a bound $I_{\rm right} \geq c > 0$, is open, and its natural route is precisely what Proposition 22 rules out.

The obvious substitute is circular, and it is worth saying why. Splitting the average by inertia class gives $\mu_G/\mu_F = I_0 - I_1 + I_2$ *identically*, so $I_{\rm right} - I_{\rm wrong} = |\mu_G/\mu_F|$ and bounding $I_{\rm right}$ below by $|\mu_G/\mu_F|$ assumes exactly the sign at issue. No independent lower bound on $|\mu_G(x)|$ is available inside a gap, since Heilmann–Lieb gives root-free intervals only outside $[-\rho, \rho]$, where the conjecture is already known.

A substitute that is not circular is the geometric mean. Jensen gives

$$I_{\rm right} + I_{\rm wrong} \;=\; \int_{\mathbb{T}^b} |\det S| \;\geq\; \exp\Big(\int_{\mathbb{T}^b} \log|\det S|\Big) \;=:\; \Delta,$$

a Mahler measure of the abelian cover, which owes nothing to the sign of μ_G. Hence $I_{\rm right} \geq \Delta - I_{\rm wrong}$, and domination holds as soon as $\Delta > 2I_{\rm wrong}$. Combined with the unconditional upper half this reduces feedback vertex number two to a single inequality, $\Delta > 2MLc^{-1/2}m^{3/2}$ (`RamaLean/JensenRoute.lean`). The geometric mean, which appeared above as the source of the obstruction, is here the tool.

Measured across both internal gaps of two triangles joined by an edge, Δ lies between 0.759 and 0.823 while $I_{\rm wrong}$ runs from 10^{-4} to 10^{-2}, so $\Delta/2I_{\rm wrong}$ runs between 41 and 3927 (`code/jensen_route.py`); the same script verifies the circularity identity to 10^{-15}. A sweep over 1475 gap points of graphs of feedback vertex number at most two with $2 \leq b \leq 4$ records no failure of $\Delta > 2I_{\rm wrong}$ and a worst ratio of 298, and the ratio does not degrade with b: its minimum is 298 at $b = 2$, 2070 at $b = 3$ and 1740 at $b = 4$ (`code/jensen_sweep.py`). What is a Mahler measure over $|\mu_F(x)|$ is recorded in `RamaLean/MahlerRoute.lean`.

3.6 Why the feedback vertex set should be removed

The inequality $\Delta > 2MLc^{-1/2}m^{3/2}$ is not the right target, and the reason is an artefact of the Schur complement rather than a feature of the conjecture. Both $M = \sup\|S\|$ and the Lipschitz constant L scale like $1/\mu_F(x)$, while Δ scales like $1/\mu_F(x)$; so as x approaches a root of μ_F inside a gap the left side grows one whole factor of μ_F faster than the right, and the inequality fails at points where nothing is wrong. Roots of μ_F are not excluded from gaps of $\operatorname{spec}(T)$.

The Schur complement is a computational device and can be dropped. With $P_x(z) = \det(xI - A_G(z)) = \prod_k (x - \lambda_k(z))$ and $N(z) = \#\{k : \lambda_k(z) > x\}$, the sign of P_x is $(-1)^{N(z)}$, and the localization of [27] gives

$$\mu_G(x) = \int_{\mathbb{T}^b} P_x = I_{\text{even}} - I_{\text{odd}}, \qquad I_p = \int_{\{N \equiv p\ (2)\}} |P_x|,$$

with no feedback vertex hypothesis, no restriction on b, and no μ_F. It also makes the Lipschitz constant explicit instead of measured, and better than the obvious bound. Crudely, $\partial A/\partial\theta_j$ has a single entry of modulus one together with its conjugate, so its operator norm is 1 and $\|A(z) - A(w)\| \le \sum_j |\Delta\theta_j|$, which costs a factor b. First-order perturbation theory does better: $\partial\lambda_k/\partial\theta_j$ is the quadratic form of that one edge against a unit eigenvector v, so it is at most $|v_{u_j}|^2 + |v_{v_j}|^2$, and summing over cotree edges charges each vertex once per incident cotree edge. Hence

$$\sum_j \left|\frac{\partial\lambda_k}{\partial\theta_j}\right| \;\le\; D,$$

with D the largest number of cotree edges at a vertex. $D \le 2$ whenever the cotree edges can be chosen vertex-disjoint and $D \le \Delta(G)$ always, so D does not grow with the graph while b does. The per-edge estimate (`BandLipschitz.quad_form_bound`), the double count (`sum_deg_bound`) and the assembly (`grad_l1_bound`) are machine-checked; first-order perturbation theory is not in Mathlib and enters `grad_l1_bound` as a hypothesis in the shape the argument consumes.

Let κ be the number of bands whose range contains x. If $\kappa = 0$ the localization settles x. Over 149 764 residue points from 40 146 graphs on up to eight vertices, $|K| = 1$ at 140 661 and $|K| = 2$ at 9103, and no point has $|K| \ge 3$ (`code/jsweep`). How often $|K| = 0$ happens depends sharply on b: 80.2% of 486 midpoints with $b = 2$ lie outside $\operatorname{spec}(G^{\text{ab}})$, against 3.4% of 493 midpoints with $b = 3$, where a further 69.8% are certified inside and 26.8% undecided, so at most 30% of the $b = 3$ points can be settled this way (`code/residue_sweep.py`, `code/residue_b3.py`). More torus directions fill the gaps in. Over 5971 residue points, with $\kappa = 1$ at 5397 and $\kappa = 2$ at 574 and no point with $\kappa \ge 3$, the minority-parity region has measure at most 0.0674 and $\min(I_{\text{even}}, I_{\text{odd}})/\max(I_{\text{even}}, I_{\text{odd}})$ is at most 0.0169 (`code/minority.py`), so the two classes are not close to cancelling.

Proposition 23. *Suppose exactly one band function λ_{k_0} attains x on $\mathbb{T}^b$. Then $Q = \prod_{k \ne k_0}(x - \lambda_k)$ has constant sign s, and*

$$\mu_G(x) = s \int_{\mathbb{T}^b} \big(x - \lambda_{k_0}(z)\big)\, |Q(z)|\, dz.$$

Consequently $\mu_G(x) = 0$ if and only if x is the $|Q|$-weighted mean of λ_{k_0}.

Proof. The factors other than k_0 never vanish and $\mathbb{T}^b$ is connected, so Q has one sign. The integral is affine in x, so it vanishes exactly when $x\int|Q| = \int \lambda_{k_0}|Q|$. $\square$

Machine-checked as `ParitySplit.zero_iff_weighted_mean` and `ne_zero_iff_ne_weighted_mean`, with the identity verified to 10^{-15} in `code/kap.py`. A weighted mean lies between the bounds of the function averaged (`mean_mem_Icc`), so an x outside the range of the crossing band cannot be that mean: `ne_zero_of_outside_band` recovers the localization in this case with no estimate at all. Conjecture 1 at a point with $\kappa = 1$ becomes the statement that the $|Q|$-weighted mean of the crossing band lies in $\mathrm{spec}(T)$. Measured means are $-2.149, -1.483, 1.483, 2.143, 2.150$ on gaps whose interiors sit near ± 1.77, so they are far from x and inside $\mathrm{spec}(T)$.

Two caveats keep this in proportion. First, over 4900 gap midpoints $\kappa = 1$ at 2206 residue points and $\kappa \geq 2$ at 183, so Proposition 23 covers 92.3% of the residue and not all of it. Second, the statement it reduces to,

$$\text{the } |Q|\text{-weighted mean of the crossing band lies in } \mathrm{spec}(T),$$

is a conjecture and is labelled as one; what can be said is that a deliberate attempt to break it failed. Over all 2206 points with $\kappa = 1$ the weighted mean lies in $\mathrm{spec}(T)$ every time and is never closer than 0.167 to x (`code/wmean_test.py`), and a single coincidence there would have refuted Conjecture 1 outright.

3.7 Separating the band geometry from the weight

Proposition 23 mixes two unrelated things: where the crossing band sits relative to x, and how much the remaining factors vary. Separating them removes the restriction to one crossing band and gives a criterion whose two halves can be attacked independently.

Let $K = \{k : x \in B_k\}$ be the crossing set and put

$$\pi(z) = \prod_{k\in K}\bigl(x - \lambda_k(z)\bigr), \qquad w(z) = \Bigl|\prod_{k\notin K}\bigl(x - \lambda_k(z)\bigr)\Bigr|.$$

The factors outside K never vanish, so their product has a constant sign s and $\mu_G(x) = s\int \pi w$.

Proposition 24. *Write $\pi = \pi_+ - \pi_-$ and $J_\pm = \int_{\mathbb{T}^b}\pi_\pm$, and suppose $0 < q_{\mathrm{lo}} \leq w \leq q_{\mathrm{hi}}$. If*

$$\frac{J_+}{J_-} \;>\; \Lambda := \frac{q_{\mathrm{hi}}}{q_{\mathrm{lo}}},$$

then $\mu_G(x) \neq 0$. No hypothesis on $|K|$ is used.

Proof. $\int \pi w = \int \pi_+ w - \int \pi_- w \geq q_{\mathrm{lo}}J_+ - q_{\mathrm{hi}}J_- > 0$. $\square$

Machine-checked as `CrossingSplit.ne_zero_of_split`, with the ratio form as `crit_of_ratio`. The point is that $J_\pm$ involve only the crossing bands and Λ only the rest, so a bound on either half is useful on its own; the weighted mean of Proposition 23 admits no such division. It is also not the wasteful estimate: bounding the whole integrand by $\sup w$

loses the four orders of magnitude that $\sup w/\inf w$ can reach, whereas here $\inf w$ is applied to the majority term and $\sup w$ only to the minority term.

The criterion is not always satisfied. Over 149 764 residue points drawn from 40 146 graphs on up to eight vertices it holds 149 511 times and fails 253 times, with worst margin 0.0616 (`code/jsweep`). A sample of 208 points had shown it holding every time, which was too small to see the failures.

Two facts sharpen what is left. Every failure has exactly one crossing band: at $|K| = 2$ the criterion holds at all 9103 points, and no point with $|K| \geq 3$ occurs in the corpus at all. And Conjecture 1 is untouched at the failing points, the true ratio $\min(I_+, I_-)/\max(I_+, I_-)$ never exceeding 0.0233 over the whole corpus, so the conclusion holds comfortably where this sufficient condition does not see it. The worst case is $n = 8$, $b = 3$, $|K| = 1$, $x = 2.0607$ on the graph with edges $01, 02, 03, 04, 14, 25, 36, 37, 56, 57$.

What the criterion buys is therefore a reduction covering 99.83% of the residue, together with a target that has structure: J_+/J_- is large because x sits near an edge of the crossing band, the minority measure never exceeding 0.0674, and Λ is a product of band-distance ratios over the non-crossing bands. Bounding either one is a self-contained problem, and the 253 exceptions are a small explicit list to be understood rather than a diffuse obstruction.

The weight can be removed from the criterion entirely. A product is bounded above by the product of the upper bounds of its factors and below by the product of the lower bounds, and for a band that does not cross x the range $B_k = [\ell_k, h_k]$ lies on one side, so the corresponding factor is exactly $1 + \operatorname{width}(B_k)/\operatorname{dist}(x, B_k)$. Hence

$$\Lambda \;\leq\; \prod_{k \notin K} \Big(1 + \frac{\operatorname{width}(B_k)}{\operatorname{dist}(x, B_k)}\Big),$$

machine-checked as `WeightBound.crit_of_band_geometry` with the per-band identity as `factor_below` and `factor_above`. The criterion then reads entirely in the band ranges of the abelian cover, in the same currency as J_+/J_-.

That bound is not sufficient on its own, and the failure is measured. The only thing discarded is the correlation between bands, since $\sup \prod \leq \prod \sup$ is an equality only when the extrema align, and that costs a median factor 4.73 and up to 28.7. Over 208 residue points the criterion with the true Λ holds every time, worst margin 2.149, while the criterion with the product in place of Λ holds 207 times and fails once, at $J_+/J_- = 1213$ against a product of 2298. The failing point is the one where x sits closest to the edge of the crossing band, minority measure 0.0171, which is not a coincidence: proximity to a band edge makes J_+/J_- large and also tends to bring a second band close to x, which makes the product large. Decoupling the two halves therefore costs most exactly where the margin is thinnest.

Difficulty is conserved and it is worth saying where it went. A weighted mean lies in the convex hull of the range of λ_{k_0}, which is the band containing x, and that band is not inside $\operatorname{spec}(T)$ precisely because x is in it and outside $\operatorname{spec}(T)$. So the containment is not automatic. What has changed is the shape of the remaining work: one scalar in one interval in place of two integrals of comparable size. Two crude routes to it are already closed. Replacing $|Q|$ by its extremes fails because $\sup |Q|/\inf |Q|$ reaches $5.4 \cdot 10^4$ on the same examples, which is far more than the small minority measure buys. And deforming the polytorus radii, which leaves the constant Fourier coefficient unchanged and so preserves $\mu_G(x)$, cannot help either: at radius one the image of P_x lies on a line and spans an arc of exactly π, while any other radius makes A_G non-Hermitian and opens the image into two dimensions, so the arc only grows and the half-plane criterion is never reached (`code/halfplane.py`).

4 The coefficient ladder

This also explains the boundary between what is proved and what is not, and the explanation is exact.

Theorem 25. *Let G be (a,b)-biregular bipartite with parts of sizes $p \le q$, degree a on the p-side and b on the q-side, and let $g(z) = f_G(z+c)$ with $c = (a-1)+(b-1)$. Then*

$$[z^{p-1}]\, g \;=\; p\,(b-2).$$

Consequently the roots of g have mean $-(b-2)$, and g is the matching polynomial of a graph if and only if $b = 2$.

Proof. Counting edges by their smaller side gives $pa = qb$, so $p \le q$ forces $a \ge b$ and $b = \min(a,b)$. The polynomial f_G is monic of degree p, and its next coefficient is $-m_1$, where m_1 is the number of 1-matchings, that is $|E| = pa$. Hence

$$[z^{p-1}]\, f_G(z+c) = \binom{p}{1}c - pa = p(a+b-2) - pa = p(b-2).$$

The mean of the roots is $-[z^{p-1}]g/p = -(b-2)$. A matching polynomial of degree p involves only every other power of z, so its coefficient at z^{p-1} vanishes; thus $b = 2$ is necessary. It is sufficient: for $b = 2$ the graph is $S(H)$ for an a-regular H and $g = \mu_H$ by [8]. □

So for $b = 2$ the polynomial g *is* a matching polynomial, Heilmann–Lieb applies verbatim, and one recovers Theorem 8 and the subdivision identity as the case $c = (a-1)+1$ of the change of variable. For $b \ge 3$ it provably is not, and Theorem 25 says more than that it fails: it locates the failure. The roots of g are not centred at the band centre but sit a distance $b-2$ below it, towards the lower band edge, which is exactly the end of (5) that is open. The asymmetry of the problem and the departure from the matching-polynomial class are the same phenomenon, measured by the single quantity $b-2$. Examples:

$$g_{K_{3,4}} = z^3+3z^2-9z-19, \quad g_{K_{3,5}} = z^3+3z^2-12z-24, \quad g_{K_{4,5}} = z^4+8z^3-6z^2-128z-139,$$

with subleading coefficients $3 = 3(3-2)$, $3 = 3(3-2)$ and $8 = 4(4-2)$ as the theorem requires.

The same method determines more than the one coefficient, and shows that the graph itself does not enter until surprisingly late.

Theorem 26. *With notation as in Theorem 25, the coefficients of z^p, z^{p-1}, z^{p-2} and z^{p-3} in g depend only on (p,a,b), not on G. Explicitly the first three are 1, $p(b-2)$ and $\frac{p}{2}\big[(p-1)(b-2)^2 - a(b-1)\big]$.*

Proof. The coefficients of g up to z^{p-j} are determined by $m_1,\dots,m_j$, so it suffices that m_1, m_2, m_3 are functions of (p,a,b) alone. Write $N = |E| = pa = qb$. Then $m_1 = N$. Two distinct edges of a bipartite graph meet in at most one vertex, so $m_2 = \binom{N}{2} - p\binom{a}{2} - q\binom{b}{2}$, which after substituting $q = pa/b$ gives $m_2 = \frac{pa}{2}(pa - a - b + 1)$; the stated value of $[z^{p-2}]g$ follows from $\binom{p}{2}c^2 - (p-1)Nc + m_2$ with $c = a + b - 2$.

For m_3, let L be the graph on E in which two edges are adjacent when they meet, so m_3 is the number of independent 3-sets of L. Every vertex of L has degree $(a-1)+(b-1) = c$. Three pairwise meeting edges of a bipartite graph must share a vertex, since otherwise they form a triangle in G; hence the number of triangles of L is $t = p\binom{a}{3} + q\binom{b}{3}$, and the number of 2-edge paths is $N\binom{c}{2} - 3t$. The standard count $\binom{N}{3} - \frac{Nc}{2}(N-2) + \big(N\binom{c}{2} - 3t\big) + 2t$ therefore depends only on (p, a, b). □

The graph enters at the very next coefficient, and it does so through a single invariant, with coefficient one.

Theorem 27. *Let $C_4(G)$ denote the number of 4-cycles of G. Then*

$$[z^{p-4}]\,g \;=\; U(p,a,b) + C_4(G),$$

where, writing $\beta = b - 2$ and $A = a(b-1)$,

$$U = \binom{p}{4}\beta^4 - \tfrac{3}{2}A\beta^2\binom{p}{3} + \frac{A\,(3A + 4\beta^2)}{12}\binom{p}{2} - \frac{A\,(9A + 2b^2 - 14b + 14)}{24}\,p.$$

Proof. As in Theorem 26, m_4 is the number of independent 4-sets of L, and inclusion–exclusion expresses it through the *non-induced* counts in L of the ten graphs on at most four vertices with no isolated vertex. Nine of the ten are universal, by the clique structure of L: each vertex of L is a cell of the $p \times q$ biadjacency matrix, and two cells are adjacent exactly when they share a row or share a column, never both, so every edge of L has a well-defined type.

Only the count of C_4 in L is not universal. Reading the cyclic word of types around a four-cycle of L, the words $RRRC$ and $RCCC$ are impossible, since three consecutive edges of one type force all four cells into a single line and then the fourth edge has that type too; and $RRCC$ is impossible, since it would require a cell sharing a column with two distinct cells of one row. This leaves $RRRR$ and $CCCC$, whose four cells lie in one line and which therefore number $3\big[p\binom{a}{4} + q\binom{b}{4}\big]$, and $RCRC$, whose four cells are exactly the four entries of a 2×2 all-ones submatrix, that is a 4-cycle of G, each contributing exactly one. Substituting and applying the shift $[z^{p-4}](z+c)^{p-k} = \binom{p-k}{p-4}c^{4-k}$ gives the stated U. □

Together with Theorem 26 this determines the top five coefficients of g completely:

$$[z^{p-1}]g = p\beta, \qquad [z^{p-2}]g = \binom{p}{2}\beta^2 - \tfrac{1}{2}Ap,$$

$$[z^{p-3}]g = \binom{p}{3}\beta^3 - A\beta\binom{p}{2} + \tfrac{1}{6}A\beta p, \quad [z^{p-4}]g = U + C_4(G).$$

For the Heawood graph ($p = 7$, $a = b = 3$, $C_4 = 0$) the last reads $35 - 315 + 231 - 77 = -126$. So the conjecture is insensitive to the graph until girth enters, and it enters at the first opportunity: four pairwise disjoint edges are first constrained by a 4-cycle.

Theorem 26 has a consequence for what a proof of Conjecture 1 can look like. The universal coefficients determine the first two power sums of the roots of g, namely $\sum \rho_i = -p(b-2)$ and $\sum \rho_i^2 = p\big[(b-2)^2 + a(b-1)\big]$, and those two numbers already imply a bound on the least root.

Proposition 28. *Every root of g satisfies $\rho \geq -(b-2) - \sqrt{(p-1)a(b-1)}$, and this is the sharpest bound implied by the coefficients of z^p, z^{p-1}, z^{p-2}.*

Proof. The roots are real. Fix a root ρ and apply Cauchy–Schwarz to the other $p-1$: $\left(\sum \rho_i - \rho\right)^2 \leq (p-1)\left(\sum \rho_i^2 - \rho^2\right)$. With $e = \sum \rho_i$ and $S = \sum \rho_i^2$ this is $p\rho^2 - 2e\rho + e^2 - (p-1)S \leq 0$, so $\rho \geq \left[e - \sqrt{(p-1)(pS - e^2)}\right]/p$. Substituting the two power sums gives $pS - e^2 = p^2 a(b-1)$ and the stated bound. Sharpness is the equality case of Cauchy–Schwarz. □

The bound of Proposition 28 grows like $\sqrt{p}$, whereas the conjectured bound $-2\sqrt{m}$ does not depend on p at all. So for all but the smallest graphs the moment bound is far weaker, and since by Theorem 26 the coefficients it uses do not see G, no argument resting on them can distinguish one (a, b)-biregular graph from another. Any proof of Conjecture 1 must therefore use m_4 or beyond, that is, it must be sensitive to the cycle structure of G. This is consistent with the place the conjecture occupies: bounds of the shape $2\sqrt{(a-1)(b-1)}$ are non-backtracking bounds, and the non-backtracking operator of a bipartite graph is exactly the object that records cycles. That object can be exhibited, but only in averaged form, and the reason for the averaging is not incidental.

5 An expected non-backtracking polynomial

Lemma 29 (Chernyak–Chertkov; Watanabe–Fukumizu)**.** *Let G be any graph, A_s its adjacency matrix with edges signed by $s \in \{\pm 1\}^E$, and D the degree matrix. Then, with s uniform,*

$$\mathbb{E}_s \det\bigl(I + u^2(D - I) - uA_s\bigr) = \sum_M (-u^2)^{|M|} \prod_{v \notin V(M)} \bigl(1 + (d_v - 1)u^2\bigr),$$

the sum being over all matchings M of G. If G is (a, b)-biregular bipartite with $P = a - 1$ and $Q = b - 1$, the right side is $\mu_G(\lambda)$ up to normalization at $\lambda = \sqrt{(1 + Pu^2)(1 + Qu^2)}/u$, and then, with $t = u^2$,

$$\lambda^2 - c = \frac{1}{t} + mt. \tag{10}$$

The identity is not new and we claim none of it. Chernyak and Chertkov [11, Eqs. (2), (6)] prove that for an arbitrary matrix H the average of $\det H(\sigma)$ over $\mathbb{Z}_2$ edge gaugings equals a monomer–dimer sum with weights $w_a = H_{aa}$ and $w_{ab} = -H_{ab}H_{ba}$; taking $H = I + u^2(D - I) - uA$ gives the display, since then $w_a = 1 + (d_a - 1)u^2$ and $w_{ab} = -u^2$. Equivalently, the averaging step alone is the Godsil–Gutman expansion [3], valid for any diagonal matrix in place of $I + u^2(D - I)$, and the right-hand side with those degree-dependent weights is the monomer–dimer partition function $\Xi_G(-u^2, \lambda)$, $\lambda_v = 1 + (d_v - 1)u^2$, of Watanabe and Fukumizu [14, Thm. 5.5]; on a $(q + 1)$-regular graph they reduce it to $u^{|V|}\mu_G(1/u + qu)$ [14, Eq. (5.3)], attributing the regular case to Nagle, an attribution we take from [14], not having consulted the original, and record the consequent band statement, that the roots then lie on the circle of radius $1/q$. Only (10), the biregular substitution, is not in those sources.

What we do wish to point out is how the left-hand side should be read. By the Ihara–Bass formula for signed graphs, $\mathbb{E}_s \det(I - uW_s)$ is $(1 - u^2)^{|E|-|V|}$ times the left side above, so the polynomial in Lemma 29 is an *expected non-backtracking characteristic polynomial.* This reading is absent from the sources just cited, and not by oversight in an obvious direction: [11] obtain the average but explicitly set the Ihara spectral parameter to zero, and [14] display the Bass matrix $I - uA_G + u^2(D_G - I)$ next to their Theorem 5.9, a sum over deletions of vertex-disjoint cycles, and remark in print that its relation to the graph zeta function is unknown. The sign average is exactly what annihilates those cycle terms. The closest antecedent we have found is Watanabe and Chertkov [12, Thm. 4], who do average a non-backtracking determinant over ± 1 edge signs and convert it by Ihara–Bass, but for the permanent, that is for perfect matchings on $K_{N,N}$, with no degree-dependent vertex weights.

The band condition has a clean reading on that polynomial, and it sharpens a published bound. By (10), a root β of ω_G coming from a nonzero root λ of μ_G satisfies $m\beta^2 - \rho\beta + 1 = 0$ with $\rho = \lambda^2 - c$, so the two roots of that quadratic have product $1/m$; they are complex conjugates, hence both of modulus $m^{-1/2}$, exactly when $|\rho| \leq 2\sqrt{m}$, and otherwise real with one inside and one outside that circle. Watanabe and Fukumizu prove [14, Cor. 5.8] that every root of ω_G lies in the annulus $1/(d_M - 1) \leq |\beta| \leq 1/(d_m - 1)$, which for an (a,b)-biregular graph reads $1/(a-1) \leq |\beta| \leq 1/(b-1)$. Conjecture 1 at $d = 1$ asserts that all roots except those coming from $\lambda = 0$ lie on the single circle $|\beta| = m^{-1/2}$, the geometric mean of the two radii they bound by. The roots at $\lambda = 0$ are $\beta = -1/(a-1)$ and $\beta = -1/(b-1)$, so they sit at the two ends of that annulus, which is why it is attained and cannot be narrowed without excluding them.

Since $1/t + mt = 2\sqrt{m}\cos\varphi$ exactly when $|t| = 1/\sqrt{m}$:

Corollary 30. *For (a,b)-biregular bipartite G, Conjecture 1 at $d = 1$ holds for G if and only if every root u of the expected non-backtracking characteristic polynomial of G, other than the trivial ones, satisfies $|u| = m^{-1/4}$.*

So the conjecture is a Ramanujan condition after all, but for an *expected* polynomial rather than for a spectrum. That is forced. If μ_G were, up to the substitution and prefactor above, the determinant of a single locally described operator on G, its coefficients could be recovered by interpolation from polynomially many determinant evaluations; but those coefficients are the matching numbers m_k, whose computation is $\#P$-hard even for planar graphs of bounded degree [13]. No unaveraged Bass-type identity for the matching polynomial can therefore exist unless FP = #P. This is also why the roots of μ_G are not the spectrum of any operator on G, and hence why the available technology for such statements is the method of interlacing families [19], which works with expected polynomials, rather than a spectral argument.

6 How close the bound comes to being attained

Proposition 28 also explains why the interval is not filled. The mean square of the roots is $\big[(b-2)^2 + a(b-1)\big]$, so the fraction of the band $[-2\sqrt{m}, 2\sqrt{m}]$ occupied in mean square is

$$\frac{(b-2)^2 + a(b-1)}{4(a-1)(b-1)},$$

which is 7/16 for $(a,b) = (3,3)$, 1/2 for $(2,2)$, 3/8 for $(4,3)$, and tends to 1/4 as $a \to \infty$. The bulk of the spectrum sits well inside the band; only the extremes are at issue, which is why moments cannot decide the question.

The extremes behave as Theorem 25 predicts. Writing the least and greatest roots as percentages of the two band edges:

G	girth	least root	greatest root
$K_{3,3}$	4	89.6%	57.2%
$K_{4,4}$	4	94.6%	56.6%
circulant $6+6$	4	96.4%	74.2%
Heawood	6	97.5%	79.9%
Pappus	6	98.6%	83.4%
C_{12}	6	96.6%	96.6%
$3C_4$	4	70.7%	70.7%

The first block has $b = 3$, and the least root is consistently far closer to its edge than the greatest is to its, the roots are displaced downwards by $b-2$, exactly as Theorem 25 requires. The second block has $b = 2$, the displacement vanishes, and the two ends are used equally. That is what the table shows, and it is all it shows.

It does not bear on sharpness, for a reason worth stating because it is easy to miss. Every graph in the table has $a = b$, and there $c = 2(a-1) = 2\sqrt{m}$, so $\rho = y - c \geq -c = -2\sqrt{m}$ holds automatically from $y = x^2 \geq 0$: the lower bound is an identity, not a theorem. This is the triviality that Song, Fan and Miao note for $d = r$ [7]. The percentages above therefore measure only how close the least root of μ_G comes to 0, and 100% would require $\mu_G(0) = 0$, that is a graph with no perfect matching, which by Kőnig's theorem no regular bipartite graph is. The conjecture has content exactly when $a \neq b$, where $c > 2\sqrt{m}$ strictly by the arithmetic–geometric mean inequality.

Where it has content the bound holds with room to spare, and is not approached. For $(4,2)$-biregular graphs the conjecture asserts $y \geq c - 2\sqrt{m} = 4 - 2\sqrt{3} = 0.5359$; a girth-4 family plateaus at $y_{\min} = 1.1716$ and a girth-8 family at 0.7392. In the trivial regime the same plateauing is visible, and there the limit is an algebraic number one can name. For a cubic bipartite circulant on $n+n$ vertices with connection set S, the matching counts satisfy $f_n(y) = \operatorname{tr} A(y)^n$ for a transfer matrix $A(y)$ with entries linear in y, so by the theorem of Beraha, Kahane and Weiss the roots accumulate where the two dominant eigenvalues of $A(y)$ have equal modulus. That happens by a genuine collision of two real eigenvalues, below y^* the pair is real and distinct, above it a complex-conjugate pair, so the locus is cut out by $\operatorname{disc}_\lambda \chi_{A(y)} = 0$ and y^* is the left edge of the accumulation band. For $S = \{0,1,2\}$, girth 4,

$$\operatorname{disc}_\lambda \chi = 4y^2\big(y^4 - 12y^3 + 46y^2 - 84y + 5\big),$$

and y^* is the least positive root of that quartic, which is irreducible over $\mathbb{Q}$; then $y^* = 0.0615663466041589\ldots$ and the limit is $100 - 25y^* = 98.4608413349\ldots\%$ of the lower edge. For $S = \{0,1,3\}$, girth 6, the discriminant is $y^3F(y)$ with F irreducible of degree 17, giving $y^* = 0.0059255959422562\ldots$ and $99.8518601014\ldots\%$. Both stop strictly short of 100%. The finite-n values approach these from above with a $1/n^2$ band-edge law, and $n^2\big(y_{\min}(n) - y^*\big)$ is constant to five digits at $n = 200, 300, 500$, which identifies the extrapolated plateau with the algebraic number.

One half of the limit is certified exactly, without appeal to Beraha–Kahane–Weiss. On the whole of $[0, y^*)$ the largest eigenvalue modulus of $A(y)$ is attained by exactly one eigenvalue, which is real, simple and negative: the discriminant has no zero there, so real branches stay real; no two eigenvalues are negatives of one another, by a resultant of the even and odd parts of χ which has no zero on $(0, y^*]$; and for the octic a separating radius 5/4, which no eigenvalue ever attains, together with one exact Schur count, pins the number of eigenvalues outside it at two. Hence $|f_n| \geq |\lambda_1|^n(1-(N-1)(1+\delta)^{-n})$ on any compact subinterval, giving $\liminf_n y_{\min}(n) \geq y^*$. The reverse inequality does use Beraha–Kahane–Weiss: just above y^* the two branches form a conjugate pair of equal and maximal modulus, and one needs that theorem to conclude that the zeros accumulate there. This is what one should expect: a family with a fixed local structure has a fixed Benjamini–Schramm limit, which is not a tree, so its matching measure retains a gap at 0. Non-attainment of this kind was already visible to Heilmann and Lieb, who tabulate the largest zero against their own bound for six lattices, 11.24 against 12, 17.86 against 20, 41.32 against 44, and remark that equality "cannot be expected to be true in general" [10, Table 1 and Remark 4.3]. Girth does help monotonically, at 30 vertices, girth 4, 6 and 8 give 98.166%, 99.311% and 99.410%, but no family of bounded girth can approach the edge, and every cubic bipartite circulant has girth at most 6, since the closed hexagon $0, s_1, s_1-s_2, s_1-s_2+s_3, s_3-s_2, s_3$ is always present. We therefore make no claim that $2\sqrt{m}$ is attained or approached. The roots of these circulants do satisfy (5), with maxima $4.06, 4.48, 6.26$ against bounds $4.90, 5.66, 6.93$.

7 The multivariate polynomial and the cavity recursion

The polynomials g admit a clean description. Since a k-matching leaves $p-k$ vertices of A unmatched,

$$f_G(y) = \sum_k (-1)^k m_k\, y^{p-k} = \sum_M (-1)^{|M|} \prod_{v\in A\setminus V(M)} y = \mathcal{M}\big(G;\, x_A = y,\ x_B = 1\big), \qquad (11)$$

the multivariate matching polynomial of Heilmann and Lieb under the asymmetric specialization that puts the variable on one side of the bipartition and 1 on the other. So the g are not exotic: they are shifts of matching polynomials in the multivariate sense. This is a reformulation and not an advance: for bipartite G every edge of a matching covers one vertex of each side, so $\mathcal{M}(G; x_A = y, x_B = z)$ depends only on the product yz, and the asymmetric specialization therefore carries exactly the information of the univariate matching polynomial. In particular no quantitative Heilmann–Lieb theorem for non-uniform specializations can help here, and none is available: the multivariate bounds of [10] apply a single graph-wide threshold to every variable.

Deleting a vertex now gives two different recursions, because only A-vertices carry the variable:

$$v \in A: \quad f_G = y\, f_{G-v} - \sum_{u\sim v} f_{G-v-u}, \qquad u \in B: \quad f_G = f_{G-u} - \sum_{v\sim u} f_{G-u-v},$$

whence the cavity recursion $R_A = y - \sum 1/R_B$, $R_B = 1 - \sum 1/R_A$. On the (a,b)-biregular tree, with $P = a-1$, $Q = b-1$, its fixed point satisfies $R^2 + (P-Q-y)R + yQ = 0$, of discriminant

$$D(y) = (P-Q-y)^2 - 4yQ, \qquad D(y) = 0 \iff y = (\sqrt{P} \pm \sqrt{Q})^2.$$

The two roots are exactly the spectrum edges in the variable $y = x^2$, and $D < 0$ strictly between them: the fixed point is complex precisely on the spectrum and real precisely off it, as it must be. Both recursions and the discriminant have been checked exactly.

Conjecture 1 for biregular G is therefore the statement that the asymmetric recursion, started from the leaves of the path tree, never produces $R = 0$ while $D > 0$. The natural invariant region is

$$R_A \le -\alpha < 0, \qquad R_B \in [\,1,\ 1 + Q/\alpha\,],$$

for a parameter $\alpha > 0$. A B-vertex with children in the region has $R_B = 1 + \sum 1/|R_A| \in (1, 1 + Q/\alpha]$, and a B-leaf carries $R_B = 1$, exactly the lower boundary. An A-vertex with its full complement of P children has $R_A \le y - P\alpha/(\alpha + Q)$, so the region is closed precisely when $y \le h(\alpha)$, where $h(\alpha) = \alpha(P - Q - \alpha)/(\alpha + Q)$. The following is elementary and is machine-checked in Lean 4 (`region_closure_bound`, `gap_edge_identity`):

$$(s - t)^2(\alpha + t^2) - \alpha(s^2 - t^2 - \alpha) = \big(\alpha - t(s - t)\big)^2, \tag{12}$$

so $h(\alpha) \le (s-t)^2$ for every α, with equality at $\alpha = t(s-t) = \sqrt{PQ} - Q$. The region therefore closes for every y strictly below the gap edge and for no y above it: the invariant region is calibrated exactly to the spectrum, and (12) is the reason.

8 The reflection, and why the band edges are never roots

One structural fact explains why $b = 2$ is the case that closes, and it is a reflection. The subdivision identity $\mu_{S(H)}(x) = x^{|E|-|V|}\,\mathcal{LM}(H, x^2)$ is due to Yan and Yeh [8], who also treat the regular and semiregular cases; see [6], Corollary 2.3, for a restatement through the multivariate matching polynomial. For H D-regular it reads $f_G(y) = \mu_H(y - D)$, and $\mu_H(-z) = (-1)^{|V(H)|}\mu_H(z)$, so

$$f_G(2c - y) = (-1)^p f_G(y), \qquad c = (a-1) + (b-1), \tag{13}$$

and c is the midpoint of $[(s-t)^2, (s+t)^2]$ because $(s \pm t)^2 = c \pm 2\sqrt{m}$. The roots of f_G are therefore symmetric about the centre of the band, so for $b = 2$ the lower bound in (5) is *equivalent* to the upper one, and Theorem 8 is that equivalence. The reflection (13) fails as soon as both degrees exceed 2: it fails for $K_{3,4}$, $K_{3,5}$ and already for the regular $K_{3,3}$, so regularity does not rescue it. Since $b = 2$ holds exactly when G is the subdivision of an a-regular multigraph, (13) describes the whole of the phenomenon, and for $b \ge 3$ the lower bound is not a repackaging of the upper bound but carries information the upper bound does not. We note that (13) has no adjacency analogue: the companion identity $\varphi(S(H), x) = x^{|E|-|V|}\varphi(H, x^2 - D)$ holds as well, but $\varphi(H, \cdot)$ has a parity only for bipartite H, whereas μ_H always does. For the Petersen graph the matching-side values are symmetric about $D = 3$ and the adjacency-side values are not.

A consequence of the first is that the band edges are never roots, by an argument of a different kind.

Proposition 31. *Let G be (a, b)-biregular bipartite with $a, b \ge 2$, and suppose $m = (a-1)(b-1)$ is not a perfect square. Then neither $(s-t)^2$ nor $(s+t)^2$ is a root of f_G.*

Proof. f_G has integer coefficients and $(s \mp t)^2 = (a + b - 2) \mp 2\sqrt{m}$, so for m not a square the two band edges are conjugate over $\mathbb{Q}$; if one is a root of f_G, so is the other. Heilmann and Lieb already give a *strict* bound, $|a| < 2\sqrt{B_1}$ with $B_1 = \Delta - 1$ for unit edge weights [10, Thm. 4.3], but at the max-degree radius, which for $a \neq b$ exceeds $s+t$; strictness at the biregular radius therefore has to be argued separately. Every root of μ_G is bounded by the spectral radius of a path tree $T_u(G)$ [22], and $T_u(G)$ is finite and embeds in the universal cover of G, the (a,b)-biregular tree $T_{a,b}$, with degrees bounded by those of $T_{a,b}$. Now $\rho(T_{a,b}) = s+t$ is the supremum of ρ over finite subtrees of $T_{a,b}$, and that supremum is not attained: any finite subtree is a proper subgraph of a larger connected finite subtree, and ρ is strictly monotone under proper subgraph containment for finite connected graphs. Hence $\rho(T_u(G)) < s+t$, so $(s+t)^2$ is not a root of f_G, and therefore neither is $(s-t)^2$. $\square$

The hypothesis on m is needed for the argument, not merely for convenience: when m is a square the two band edges are rational and independent, and conjugacy says nothing. The smallest such case is $(a,b) = (5,2)$, where $m=4$ and the edges are 1 and 9. There the conclusion nevertheless holds, but for a different reason, the Heilmann–Lieb bound is strict for finite graphs, so Theorem 8 already excludes the lower edge. Proposition 31 is the statement for the cases those methods do not reach.

Evaluating f_G in $\mathbb{Z}[\sqrt{m}]$ confirms this and exhibits the conjugacy: for $K_{3,4}$, $K_{3,5}$, $K_{3,6}$, $K_{4,5}$ the two edge values are $53 \mp 30\sqrt{6}$, $72 \mp 40\sqrt{8}$, $91 \mp 50\sqrt{10}$, $1877 \mp 512\sqrt{12}$. For $b=2$ the irrational parts vanish, as (13) requires. Proposition 31 is worth isolating because it uses no induction on vertex deletion, and so is not subject to the obstruction of Section 12; it is the only tool we have found with that property, although it controls only the edges and not the interior of the gap.

9 The operator form

9.1 The mixed characteristic polynomial

The multivariate object that does carry the biregular information is the mixed characteristic polynomial of Marcus, Spielman and Srivastava: with $P_k = \operatorname{diag}(\mathbf{1}_{N(k)})$ of rank b and $\sum_k P_k = aI_p$, one has $\nu_G = \mu[P_1, \ldots, P_q]$, since $\sum_k z_k P_k$ is diagonal and each ∂_{z_k} must strike a distinct linear factor, so the surviving terms are the injections, that is the matchings. This is the stability route done properly, and it locates the shortfall quantitatively: a barrier argument that sees only the rank, the trace and $\sum_k P_k = aI$ is reading the Marchenko–Pastur law, and yields the interval $\left[(\sqrt{a}-\sqrt{b})^2, (\sqrt{a}+\sqrt{b})^2\right]$, which is strictly wider than the band, the discrepancy being $2\left[\sqrt{ab} - \sqrt{(a-1)(b-1)} - 1\right] \geq 0$ with equality only at $a=b$. So the standard potentials do not even recover the known upper bound. That is not a deficiency of the analysis but of the hypotheses, and it can be seen exactly. Take $A_k = (b/p)I_p$: these are positive semidefinite with $\operatorname{tr} A_k = b$ and $\sum_k A_k = (qb/p)I = aI$, so they satisfy every hypothesis the barrier method uses, and $\mu[A_1, \ldots, A_q]$ is a Laguerre polynomial whose roots converge to the Marchenko–Pastur edges. At $(a,b) = (3,2)$ the upper edge is already exceeded at $p=4$, where the largest root is 6.229 against $(s+t)^2 = 5.828$, and the lower edge at $p=48$, where the smallest is 0.154 against $(s-t)^2 = 0.172$. The Marchenko–Pastur interval is therefore attained on those hypotheses, and *no* barrier argument that does not see $\operatorname{rank} A_k \leq b$ can reach the tree band at either edge. Numerically the rank condition is what carries the statement: dropping idempotency but keeping $\operatorname{rank} A_k \leq b$ and $\operatorname{tr} A_k = b$ produced no violation in search, which suggests that class, rather than the projections, as the natural target. What is missing is a barrier built from the Kesten measure of the (a,b)-biregular tree rather than from Marchenko–Pastur, which is to say one that uses that the P_k are 0/1 diagonal, exactly the combinatorial information free probability discards.

That raises the question of how much of the hypothesis is really needed, and the answer delimits the problem usefully. Asking for the same conclusion from *arbitrary* positive semidefinite A_k of rank b with $\sum_k A_k = aI_p$ is false. Take $p=2$, $q=3$, $(a,b)=(3,2)$, and $A_1 = A_2 = \frac{u}{2}I_2$, $A_3 = (3-u)I_2$: each has rank $2=b$ and the sum is $3I = aI$, while $\mu[A_1,A_2,A_3](y) = y^2 - 6y + \left(6u - \frac{3}{2}u^2\right)$. At $u = \frac{1}{10}$ the constant term is $\frac{117}{200}$ and the roots are 0.09914 and 5.90086, outside the band $[3-2\sqrt{2}, 3+2\sqrt{2}] = [0.17157, 5.82843]$ at both ends; as $u \to 0$ the roots tend to 0 and 6. The counterexample is diagonal, hence commuting, so noncommutativity is not what fails. What fails is that the traces are unconstrained: $\operatorname{tr} A_3 = 5.8 \gg b$. Imposing $\operatorname{tr} A_k = b$ restores the second coefficient, since it then depends on the family only through $\sum_k \|A_k\|_F^2$, which by Cauchy–Schwarz is at least b with equality exactly for idempotents. Projections are thus the extreme point already at the first nontrivial coefficient.

For rank-b orthogonal projections the statement survives, once one adds the hypothesis $a \ge b$, equivalently $p \le q$, which the graph setting supplies automatically. Without it the statement is false for a trivial reason: if $p > q$ then y^{p-q} divides $\mu[P_1,\ldots,P_q]$, so 0 is a root while the lower band edge is positive. Taking $p=6$, $q=4$, $(a,b)=(2,3)$ with $P_1 = P_2$ and $P_3 = P_4$ the projections onto two complementary 3-spaces gives a double root at 0 against a lower edge of $(\sqrt{2}-1)^2$. A search over more than 10^5 feasible families, real and complex projections, and the intermediate class rank $\le b$, $\operatorname{tr} = b$, produced no violation, the smallest observed margin being 3.6% of the band width. That figure needs a caveat, though, and it cuts against the aggregate impression: every tight case in those searches has $b=2$, hence lies in the regime settled by Proposition 53. In the genuinely open range $b \ge 3$ the observed margins are wide, at $(p,q,a,b) = (6,8,4,3)$ the least root is 0.486 against a band edge of 0.101, a factor of nearly five. The numerical evidence for the open part of the statement is thinner than the total counts suggest. Two features make this the right generalization to record. First, commuting rank-b projections with $\sum_k P_k = aI$ are simultaneously diagonalizable and therefore 0/1 diagonal, so they are *exactly* the biregular incidence families: the projection statement is precisely the noncommutative extension of Conjecture 1. Second, the universality of Theorem 26 persists in that setting, for projections the coefficients $E_0,\ldots,E_3$ depend only on (p,q,a,b), because idempotency collapses every trace of a word of length at most three with a repeated letter, while at length four $\operatorname{tr}(A_iA_jA_iA_k)$ does not reduce. The first free invariant sits at level four on both sides of the analogy, matching the 4-cycle count of Theorem 27.

9.2 A determinantal representation

The mixed characteristic polynomial admits a probabilistic representation which gives an unconditional bound with no barrier or stability input. Write $P_k = W_kW_k^*$ with $W_k^*W_k = I_b$, set $U = [W_1|\cdots|W_q]/\sqrt{a}$, and let $\Pi = U^*U$, a rank-p orthogonal projection on $\mathbb{R}^n$, $n = pa = qb$, whose q diagonal $b \times b$ blocks are all $\frac{1}{a}I_b$. Let S be a sample from the determinantal process with kernel Π and $s_k(S)$ the number of slots of block k it uses. Then

$$y^{\,q-p}\,\mu[P_1,\ldots,P_q](y) = \mathbb{E}_S \prod_{k=1}^{q} \bigl(y - a\,s_k(S)\bigr), \tag{14}$$

by expanding $\det(yI + \sum_k z_k P_k)$ over subsets and applying $\prod_k (1 - \partial_{z_k})$, which annihilates every term using a block twice. Here $|S| = p$ always, so $\sum_k s_k = p$, and each s_k is exactly $\mathrm{Binomial}(b, 1/a)$; for a diagonal family S is simply an independent uniform choice of one incident edge at each vertex of the p-side.

Since $0 \le s_k \le b$, every factor in (14) is positive once $y > ab$, so the expectation is, and there is no root there.

Proposition 32. *Every root of $\mu[P_1, \dots, P_q]$ is at most ab. Moreover, with $m = (a-1)(b-1)$,*

$$ab - \left(\sqrt{a-1} + \sqrt{b-1}\right)^2 = \left(\sqrt{m} - 1\right)^2.$$

So the bound overshoots the conjectured edge by exactly $(\sqrt{m} - 1)^2$, and is lossless precisely when $m = 1$. At $a = b = 2$ the overshoot vanishes and $ab = 4$ is the upper edge; the roots are nonnegative because the matching numbers are, so **the band is proved outright at $a = b = 2$.**

Where ab stands relative to what is known needs care, and we record it exactly. Xu, Xu and Zhu [2, Thm. 1.9] prove, for positive semidefinite $X_1, \dots, X_q$ of rank at most k with $\sum X_i \preceq I$ and $\varepsilon = \max \operatorname{tr} X_i$, that $\operatorname{maxroot} \mu \le \left(\sqrt{1 - \varepsilon/(k-1)} + \sqrt{\varepsilon}\right)^2$ whenever $\varepsilon \le (k-1)^2/k$. Normalizing a rank-$b$ projection family with $\sum_k P_k = aI$ by a puts it in exactly that form with $k = b$ and $\varepsilon = b/a$, and since the roots scale with the family this gives

$$\lambda_{\max} \le a\left(\sqrt{1 - \tfrac{b}{a(b-1)}} + \sqrt{\tfrac{b}{a}}\right)^2, \qquad a \ge \frac{b^2}{(b-1)^2}, \tag{15}$$

which at $b = 2$ is $a + 2\sqrt{2a-4}$ for $a \ge 4$. That is strictly better than ab for every $a > 4$, indeed $a + 2\sqrt{2a-4} \le 2a$ with equality only at $a = 4$, formalized as `xxz_le_ab` and `xxz_eq_ab_iff`, and better than Marchenko–Pastur throughout. So ab is the best available bound only where the hypothesis of (15) fails, which at $b = 2$ means $a < 4$: at $a = b = 2$, where it is sharp and settles the band, and at $(a, b) = (3, 2)$, where it gives 6 against a Marchenko–Pastur edge of 9.899 and a band edge of 5.828. Elsewhere (15) is the state of the art:

(a,b)	(15)	ab	Marchenko–Pastur	$a + 2\sqrt{a}$	band
$(3,2)$	n/a	6.000	9.899	6.464	5.828
$(4,2)$	8.000	8.000	11.657	8.000	7.464
$(9,2)$	16.483	18.000	19.485	15.000	14.657
$(100,2)$	128.000	200.000	130.284	120.000	119.900
$(4,3)$	10.977	12.000	13.928	–	9.899
$(25,4)$	47.126	100.000	49.000	–	43.971

The comparison also says what the target of this section is worth: at $b = 2$ the bound $a + 2\sqrt{a}$ of Proposition 39 would improve on (15) for every $a > 4$, since $\sqrt{a} \le \sqrt{2a-4}$ there (`target_le_xxz`). It is therefore not a weaker target than the literature already provides, but a stronger one everywhere the barrier hypothesis holds. Proposition 32 is formalized in `RamaLean/ProductBound.lean` as `no_root_above` and `edge_gap`.

The representation also supports an exact recursion on the underlying kernels. Writing N_K for the transversal expansion attached to a kernel K, one has the border identity

$$a\, N_{M + ww^\mathsf{T}} = (a - y)\, N_M + N_{\left(\begin{smallmatrix} 0 & w^\mathsf{T} \\ w & M \end{smallmatrix}\right)},$$

and Schur complements of admissible kernels remain admissible, with $\delta_e \le 1 - c_e$ forced by $K^2 \preceq K$. This makes an induction on kernel size available in principle. It does not close the upper edge: the scalar cavity invariants that carry the classical argument on graphs are false on the kernel class, witnessed by explicit rational kernels on which the graph unfolding identity fails as an inequality by a y-independent amount. What survives is a reduction of the upper edge on the whole class to a single inequality on bordered kernels, proved for at most two blocks at $b = 2$ by monotonicity of transversal determinants under downdates, which follows from positivity of the adjugate of a positive semidefinite matrix.

9.3 The case of rank two

At $b = 2$ the representation (14) collapses further. Only $s_k \in \{0, 1, 2\}$ occurs, and $\sum_k s_k = p$ forces the counts of 0s and 1s to be determined by $n_2 = \#\{k : s_k = 2\}$, so with $\Psi(u) = \mathbb{E}\, u^{n_2}$ one has

$$\mu(y) = (y - a)^p\, \Psi\big(\theta(y)\big), \qquad \theta(y) = 1 - \frac{a^2}{(y - a)^2}.$$

Real-rootedness of μ forces Ψ to be real-rooted with nonpositive roots, hence n_2 a sum of independent Bernoulli variables, and the extreme roots of μ are $a\big(1 \pm \sqrt{\pi_{\max}}\big)$ for $\pi_{\max}$ the largest of those parameters. Since the band at $b = 2$ is $[a - 2\sqrt{a - 1},\, a + 2\sqrt{a - 1}]$, both edges say $a\sqrt{\pi_{\max}} \le 2\sqrt{a - 1}$:

Proposition 33. *For $b = 2$, Conjecture 1 for the projection family is equivalent to*

$$\pi_{\max} \;\le\; \frac{4(a - 1)}{a^2} \;=\; 1 - \Big(\frac{a - 2}{a}\Big)^2.$$

The equivalence of the two edges with each other and with this inequality is formalized in `RamaLean/ProductBound.lean` as `band_iff_pi`; that they coincide is Proposition 53 seen through θ, which is two-to-one about $y = a$.

9.4 The matching form

Recentring at a turns μ into a matching polynomial, and the coefficients have a basis-free form. They also have a Grassmann form: writing $A_k = b_{k1}b_{k1}^{\mathsf{T}} + b_{k2}b_{k2}^{\mathsf{T}}$ and $\omega_k = b_{k1} \wedge b_{k2} \in \Lambda^2\mathbb{R}^p$, the Cauchy–Binet formula gives $e_{2r}\big(\sum_{k\in T} A_k\big) = \big\|\bigwedge_{k\in T}\omega_k\big\|^2$, so

$$\mu(x + a) = \sum_{T\subseteq[q]} (-1)^{|T|} \Big\|\bigwedge_{k\in T}\omega_k\Big\|^2 x^{\,p-2|T|},$$

and M_r is a sum of squared Plücker norms. In the commuting case each norm is 0 or 1 and one recovers the matching numbers.

Theorem 34. *Let $P_1, \ldots, P_q$ be rank-two orthogonal projections on $\mathbb{R}^p$ with $\sum_k P_k = aI$. Then*

$$\mu(x + a) = \sum_{r\ge 0} (-1)^r M_r\, x^{\,p-2r}, \qquad M_r = \sum_{|T|=r} e_{2r}\Big(\sum_{k\in T} P_k\Big),$$

where e_m denotes the m-th elementary symmetric function of the spectrum. The M_r are nonnegative, $M_0 = 1$, $M_1 = q$, and $M_r \le \binom{q}{r}$.

The expression for M_r involves no dilation and no choice of basis for the ranges of the P_k. When the P_k commute, $\sum_{k\in T} P_k$ is a 0/1 diagonal matrix of rank $2r$ exactly when T is a matching of the multigraph whose edges are the blocks, and has rank below $2r$ otherwise; so e_{2r} is the indicator of that event and M_r is the number of r-matchings. Thus at $b = 2$ the recentred mixed characteristic polynomial *is* the matching polynomial in the commuting case, and M_r is its natural noncommutative extension. Verified exactly on $S(K_4)$, $S(K_{3,3})$, the cube, and on noncommuting families including the icosahedral one, where the commutators have norm 0.57.

Two consequences change how the remaining gap should be described. First, in the commuting case Proposition 33 *is* Heilmann–Lieb, so it is a theorem there, and it is sharp: the extremal families are the large-girth a-regular graphs, for which the ratio of $\pi_{\max}$ to its bound rises monotonically through 0.866 at $p = 10$ to 0.949 at $p = 38$. No argument with slack can close it. Second, and more to the point, in that formulation Proposition 33 says exactly that the roots of a mixed characteristic polynomial lie inside the support of the corresponding free convolution, which is the known open sharpening of the Marcus–Spielman–Srivastava bound. So $b = 2$ is not an easier case awaiting a short argument; it is that open problem in another notation.

9.5 The coordinate case at rank b, and the sharpness of Xu's constant

Theorem 34 identifies the recentred mixed characteristic polynomial with the matching polynomial in the commuting case at $b = 2$. That identification survives to every rank, and the object it produces is again an ordinary matching polynomial, on a different graph. The consequence is that Xu's constant is not merely an upper bound at $b \geq 3$: it is exactly the truth, and the class attains it.

Let H be an a-regular b-uniform hypergraph on n vertices with q hyperedges, and let P_e be the coordinate projection onto $\operatorname{span}\{e_v : v \in e\}$. These have rank b and satisfy $\sum_e P_e = aI$ exactly, so they are an admissible tight family. Write $I(H)$ for the incidence bipartite graph, with one side the n vertices and the other the q hyperedges; it is (a, b)-biregular.

Proposition 35. *With μ the mixed characteristic polynomial of $\{P_e\}$,*

$$\mu(x) = \sum_{k\geq 0} (-1)^k\, m_k\big(I(H)\big)\, x^{n-k}, \qquad \textit{equivalently} \qquad \mu(t^2) = t^{n-q}\, \mu_{I(H)}(t),$$

where m_k is the number of k-matchings. In particular the positive roots of μ are exactly the squares of the positive roots of $\mu_{I(H)}$.

Proof. The matrix $xI + \sum_e z_e P_e$ is diagonal, with v-th entry $x + \sum_{e\ni v} z_e$, so $\det = \prod_v \big(x + \sum_{e\ni v} z_e\big)$. Each factor is affine in every z_e, so for $S \subseteq [q]$ the derivative $\partial_S \det$ evaluated at $z = 0$ is obtained by choosing, for each $e \in S$, one factor to differentiate, that is one vertex $v \in e$; two hyperedges choosing the same vertex differentiate one affine factor twice and contribute 0. Hence $\partial_S \det$ at $z = 0$ counts the injective maps $S \to V$ with $e \mapsto v \in e$, each contributing $x^{n-|S|}$. Applying $\prod_e(1 - \partial_{z_e})$ and grouping by $|S| = k$, the coefficient is the number of systems of distinct representatives of k hyperedges, which is the number of k-matchings of $I(H)$ saturating k hyperedge-nodes. The second display follows from $\mu_{I(H)}(t) = \sum_k (-1)^k m_k t^{n+q-2k}$ on substituting $x = t^2$. $\square$

Corollary 36. *For every coordinate family as above, at every $b \geq 2$,*

$$\operatorname{maxroot} \mu \;\leq\; \left(\sqrt{a-1}+\sqrt{b-1}\right)^2.$$

Proof. $I(H)$ is (a, b)-biregular, so its universal cover is the (a, b)-biregular tree, of spectral radius $\rho = \sqrt{a-1}+\sqrt{b-1}$. Godsil's bound [22] puts every root of $\mu_{I(H)}$ in $[-\rho, \rho]$, and Proposition 35 squares them. □

So the inequality Xu conjectures is a theorem on the coordinate locus, at every rank, and not only at $b = 2$ where it is Heilmann–Lieb. The question that matters for the conjecture is whether the constant is the right one there, and it is.

Proposition 37. *Fix $a, b \geq 2$. Over coordinate families with parameters (a, b),*

$$\sup \frac{\operatorname{maxroot} \mu}{\left(\sqrt{a-1}+\sqrt{b-1}\right)^2} \;=\; 1.$$

Proof. At most 1 by Corollary 36. For the other direction, let B_r be the ball of radius r in the (a, b)-biregular tree. If $\operatorname{girth} I(H) > 2r$ then the r-ball of $I(H)$ about any a-side vertex is acyclic and, by biregularity, isomorphic to B_r, so $B_r \subseteq I(H)$ as a subgraph. The greatest root of a matching polynomial is monotone under subgraphs [10, 22], and B_r is a tree, where the matching polynomial is the characteristic polynomial; hence $\operatorname{maxroot} \mu_{I(H)} \geq \lambda_{\max}(B_r)$, and squaring, $\operatorname{maxroot} \mu \geq \lambda_{\max}(B_r)^2$. The balls exhaust the tree and $\lambda_{\max}(B_r) \uparrow \rho$. Finally (a, b)-biregular bipartite graphs of arbitrarily large girth exist: random N-lifts of any one of them preserve biregularity and have girth tending to infinity in probability. Taking $r \to \infty$ gives the claim. □

The ladder is measured rather than assumed, and it also supplies an explicit floor for any graph whose girth is known. Writing the ratio in the μ variable, $\lambda_{\max}(B_r)^2/\rho^2$ is

(a, b)	$r = 2$	4	6	8	10	12
$(3, 2)$	0.6863	0.8579	0.9167	0.9443	0.9596	0.9692
$(4, 3)$	0.6061	0.8082	0.8854	0.9234	0.9450	0.9585
$(5, 4)$	0.5744	0.7898	0.8747	0.9168	0.9396	
$(4, 4)$	0.5833	0.7951	0.8777	0.9186	0.9418	0.9505
$(6, 3)$	0.6004	0.8031	0.8826	0.9219	0.9442	0.9489

increasing in r at every pair, $b \geq 3$ included; the missing entry is where the ball passed the 400,000 vertex cap, and a truncated ball is a proper subtree of the true one, hence still a subgraph of any graph of girth $> 2r$, so the floor it gives remains valid. On named incidence graphs, with girth recomputed by breadth-first search rather than quoted, the realized ratios are 0.8995 for Heawood, 0.9171 for Pappus and 0.9453 for Tutte–Coxeter at $(3, 3)$; 0.9098 for AG$(2, 3)$ at $(4, 3)$; and 0.9097 for PG$(2, 3)$ at $(4, 4)$. Every one exceeds its ball floor, which is the containment argument of Proposition 37 checked against the object it is about (`code/xu_sharp.py`).

Two things follow, and the second is the reason for the subsection.

First, the plateaus found by the adversarial search of `code/adversarial.py` and `code/rl_push.py` — 0.739 at $(4, 3)$ and 0.726 at $(5, 4)$, against 0.939 at $(3, 2)$, none of them a violation — are an artefact of that search, not a property of the class. It sampled generic subspaces, and generic subspaces behave like short girth. The coordinate construction reaches 0.9098 at $(4,3)$ on twenty-one vertices and is bounded below by 0.9585 at girth 26, well past where the search stalled. There is no evidence of a smaller true constant at $b \geq 3$, and the reading that suggested one is closed.

Second, the constant is now fixed. Any proof of Xu's conjecture in general must produce exactly $(\sqrt{a-1}+\sqrt{b-1})^2$; any attempted improvement of it is false, by Proposition 37, before it is attempted. Equally, the whole content of the conjecture lies off the coordinate locus: the bridge of Proposition 35 fails for non-coordinate rank-b projections, which is checked as a control rather than asserted, and it is that failure which leaves the general statement open.

The identification in Proposition 35 is elementary and we make no claim of novelty for it; we have not found it stated, but the search caveats of §12 apply verbatim. It should not be confused with the hypergraph Heilmann–Lieb theorem of Wan, Wang and Fan [1], which concerns a different polynomial — the matching polynomial of the adjacency hypermatrix, whose extreme root scales as $\Delta^{1/k}$ — and neither implies the other. Both routes of Proposition 35 are computed in `code/xu_sharp.py`, the mixed characteristic polynomial from the definition $\prod_e (1-\partial_{z_e}) \det(xI+\sum_e z_e P_e)$ and the matching counts by subset dynamic programming, sharing no code and agreeing coefficient by coefficient in exact arithmetic on four hypergraphs at $b = 2, 3, 4$. The skeleton of the argument is machine-checked: the root identification of Proposition 35 as `XuSharp.bridge_roots`, the passage from a ceiling and a converging ladder to the supremum as `XuSharp.sharp_of_ladder`, and the consequence that no smaller constant can hold as `XuSharp.no_smaller_constant`. Godsil's bound and subgraph monotonicity enter there as hypotheses, being classical.

9.6 The plane class, and two unconditional bounds

This subsection is the longest in the note, so its outcome is stated first. It delivers an exact vertex recursion for the plane class (Theorem 38), checked numerically to 10^{-13}; a decomposition of its cross-term into pieces C_r (Proposition 39, Theorems 40 and 41) whose nonnegativity would give the band $[a-2\sqrt{a},\ a+2\sqrt{a}]$ for all weighted plane families, supported numerically but conjectural; and two bounds following unconditionally from real-rootedness alone, giving the band whenever $m \leq 8$ and whenever $m \leq 16 + \left(\sum_k c_k^2\right)/a^2$, the latter reproducing the known two-moment bound for projections and extending it to all weighted plane families. The general case is not closed. A reader who wants only that summary may skip to §11.

What the problem is, at $b = 2$, can be said exactly. Both $\mu(x+a)$ and the operator $\mathrm{Adj}(A) := \sum_k e_2(A_k) P_{\mathrm{range}\, A_k}$ depend on the family only through the pairs $\big(e_2(A_k), \mathrm{range}\, A_k\big)$. So the class is the class of weighted 2-plane families $\{(c_k, V_k) : c_k \geq 0,\ \sum_k c_k P_{V_k} \preceq aI\}$, and $\mu(x+a)$ is the matching polynomial of such a family; for coordinate planes it is the ordinary weighted matching polynomial. Two points about that identification need care, and both are invisible in the projection case.

First, the invariance holds within the tight class and not outside it: with $\sum_k A_k$ a multiple of the identity, families sharing the pairs $\big(e_2(A_k), \mathrm{range}\, A_k\big)$ have the same recentred μ even when the A_k are strongly anisotropic, verified to 10^{-13} at eigenvalue ratios up to 129 (`code/rl_conj.py`); without tightness the recentring has no meaning.

Second, the recentring constant is that tight constant, and it is *not* the constant in $\mathrm{Adj}(A) \preceq aI$. Plane data with $\mathrm{Adj}(A) = aI$ admits a realization with $\sum_k A_k = a'I$ — solvable to 10^{-16} — but with $a' > a$ strictly: $a' = 3.860$ against $a = 3$ at $m = 4$, and 4.385 against 3 and 5.120 against 4 at $m = 6$ (`code/rl_conj.py`). So for weighted families $F_A(x) = \mu(x+a')$ with $a' \neq a$, and the identity as stated holds exactly when the two constants agree, which is the projection case $A_k = P_k$, where $e_2 \equiv 1$ gives $\sum_k A_k = \mathrm{Adj}(A) = aI$. Everything below is derived from the wedge coefficients M_r and from $\mathrm{Adj}(A)$ directly and is unaffected; what does not transfer automatically to the weighted class is a result stated for mixed characteristic polynomials, since the two constants there are different.

Consequently the band is stated below on the roots of F_A, as $|x| \leq 2\sqrt{a}$ with a the constant of $\mathrm{Adj}(A) \preceq aI$, and that is the form the unconditional bounds establish, through $\lambda^2 \leq M_1 \leq am/2$. Written in the μ variable it becomes $[a' - 2\sqrt{a},\ a' + 2\sqrt{a}]$: the radius is set by the Adj constant and the centre by the tight one. Both the shift $y = x + a'$ and the bound $|x| \leq 2\sqrt{a}$ are checked directly (`code/rl_conj.py`); the two coincide precisely for projection families, where $a' = a$, and every μ-variable statement of the band in this paper should be read with that proviso.

The band being sought is the sharp weighted Heilmann–Lieb constant for plane families, and that constant is not $2\sqrt{a}$. On projections the conjectural sharp value is $2\sqrt{a-1}$, the spectral radius of the a-regular tree, and $2\sqrt{a-1} < 2\sqrt{a}$ strictly (`XuBound.xu_lt_note_target`); it is the case $b = 2$ of the bound $\left(\sqrt{a-1} + \sqrt{b-1}\right)^2$ on maxroot μ conjectured by Xu[1], which is exactly the upper edge of the (a,b)-biregular tree band. So $2\sqrt{a}$ is the ceiling of the cavity argument used here, not the truth, and an argument of this kind cannot be sharp. That the sharp value is $(\sqrt{a-1} + \sqrt{b-1})^2$ and not something smaller is no longer conjectural on the coordinate locus: it is Proposition 37, at every b.

In that language the vertex step has an exact form. Setting $\Theta^{(r)} := \sum_{|T|=r} M_{\omega_T} \succeq 0$ and $W(x) := \sum_{r\geq 1} (-1)^r \Theta^{(r)} x^{m-2r}$, one has $\langle e, W(x)e\rangle = F_A(x) - x\, F_{A^{(e)}}(x)$ for every unit vector e, where $F_A(x) = \mu(x+a)$ and $A^{(e)}$ is the compression to $e^\perp$; consequently $\operatorname{tr} W = mF_A - xF_A'$ and $F_A' = \sum_i F_{A^{(e_i)}}$ for any orthonormal basis. The cavity hypothesis is then equivalent to a single matrix inequality: when $\mathrm{Adj}(A) = aI$, the band $[a - 2\sqrt{a},\ a + 2\sqrt{a}]$ holds if and only if

$$W(2\sqrt{a}) + F_A(2\sqrt{a})\, I \ \succeq\ 0,$$

a coordinate-free statement with no recursion in it.

The matrix W has a closed form. If $\omega_T = c_1 \wedge \cdots \wedge c_{2r}$ is decomposable with factor matrix $C = [c_1 | \cdots | c_{2r}]$ and Gram matrix $G = C^{\mathsf{T}}C$, then $\langle \omega_{\hat{\jmath}}, \omega_{\hat{l}}\rangle$ is the (l,j) cofactor of G, so

$$M_{\omega_T} = C\ \mathrm{adj}(G)\, C^{\mathsf{T}}, \tag{16}$$

whence $\operatorname{tr} M_{\omega_T} = 2r\|\omega_T\|^2$ and $\operatorname{tr} \Theta^{(r)} = 2rM_r$, which is the identity $\operatorname{tr} W = mF_A - xF_A'$ again. Two of the levels are forced to be scalar: $\Theta^{(1)} = \mathrm{Adj}(A) = aI$ by hypothesis, and when m is even $\Theta^{(m/2)} = M_{m/2}\, I$, since a top-degree form is a multiple of the volume form and $\|\iota_v \omega\|^2 = \|\omega\|^2\|v\|^2$ for such. Hence for $m \leq 4$ every level is scalar, $W(x)$ is a scalar matrix, and the matrix inequality degenerates to a scalar one; from $m = 5$ on it does not, and that is where the content begins.

Behind the matrix inequality there is a vertex recursion valid in every direction, which the graph case sees only along the coordinate axes.

[1]Personal communication, August 2026, deriving it from a generalisation of [24, Conjecture 1].

Theorem 38. *Let A be a weighted 2-plane family on $\mathbb{R}^m$ with bivectors ω_k, and let e be a unit vector. Put $f_k = \iota_e\omega_k$, so that*

$$\omega_k = e \wedge f_k + \omega_k', \qquad f_k,\ \omega_k' \in e^\perp,$$

and ω_k' is the bivector of the compressed family $A^{(e)}$. Then for every $r \geq 1$

$$M_r(A) = M_r\big(A^{(e)}\big) + \sum_{|T|=r} \Big\| \sum_{k\in T} f_k \wedge \omega'_{T\setminus k} \Big\|^2,$$

and consequently, with $N_r(e)$ denoting that sum of squares,

$$F_A(x) = x\, F_{A^{(e)}}(x) - N_e(x), \qquad N_e(x) = \sum_{r\geq 1} (-1)^{r-1} N_r(e)\, x^{\,m-2r}.$$

Proof. Since e is a unit vector, $\iota_e(\omega_k - e\wedge f_k) = f_k - f_k = 0$, so $\omega_k - e \wedge f_k$ lies in $\Lambda^2(e^\perp)$; writing $b_i = \beta_i e + b_i'$ identifies it as $b_1' \wedge b_2'$, the bivector of the compressed block. As $e \wedge f_k$ is a 2-form it commutes with the other factors, and $e \wedge e = 0$ kills every term using two of them, so $\omega_T = \omega_T' + e \wedge \sum_{k\in T} f_k \wedge \omega'_{T\setminus k}$. The two summands lie in $\Lambda^{2r}(e^\perp)$ and $e \wedge \Lambda^{2r-1}(e^\perp)$, which are orthogonal, so the norms add; summing over $|T| = r$ gives the first display, and the second is that display multiplied by $(-1)^r x^{\,m-2r}$ and summed. ☐

We have looked for Theorem 38 in the literature and not found it. The whole arXiv corpus mentioning "mixed characteristic polynomial" is seven papers, all working through the barrier function or interlacing families; the generalizations of Heilmann–Lieb we can find are to hypergraphs [1], to spanning subgraphs with degree constraints, and to multivariate independence polynomials, all of them graph-theoretic; and a search of arXiv for "matching polynomial" together with any of Grassmann, Plücker, bivector or exterior algebra returns nothing. That is weak evidence: the Gram form below is elementary enough to be folklore, the pre-arXiv matching-polynomial literature is not covered by such a search, and the statement could well appear under different language, hyperbolic or Lorentzian polynomials, or the theory of stable polynomials, where Plücker coordinates do occur but in the different role of characterizing stability by total nonnegativity. We claim only that we have not found it.

The exterior algebra can be dispensed with entirely, and doing so identifies the theorem with a known lemma. Let C be the $m \times 2r$ matrix whose columns are the factors of ω_T, so that $\|\omega_T\|^2 = \det(C^\mathsf{T} C)$. Splitting off e is the decomposition $C = eg^\mathsf{T} + C'$ with $g = C^\mathsf{T} e$ and $C' = (I - ee^\mathsf{T})C$ the compressed factor matrix; since $e^\mathsf{T} C' = 0$ and $\|e\| = 1$,

$$C^\mathsf{T} C = C'^\mathsf{T} C' + gg^\mathsf{T}.$$

So the content of Theorem 38 is the *matrix determinant lemma*

$$\det\big(A + gg^\mathsf{T}\big) = \det A + g^\mathsf{T} \operatorname{adj}(A)\, g, \qquad A = C'^\mathsf{T} C', \tag{17}$$

and the nonnegativity of the cavity term is nothing but positivity of the adjugate of a positive semidefinite matrix. In particular $M_r(A) \geq M_r(A^{(e)})$ needs no computation at all. Splitting g into its r blocks of two identifies the pieces: the block-diagonal part of the quadratic form

(17) is $\sum_k \theta_k M''_{k,r-1}$ and the off-block-diagonal part is C_r, which is how the cross terms should be read.

For a coordinate family at a coordinate direction $e = e_v$ one has $f_k = \pm e_u$ for the edges $k = uv$ and $f_k = 0$ otherwise, the cross terms vanish, and $N_r(e_v) = \sum_{u\sim v} m_{r-1}(G-v-u)$: the theorem is then the Heilmann–Lieb vertex recursion. At a generic direction, or for a family that is not diagonal, the cross terms do not vanish and the recursion is not the classical one. Two things follow at once. First, taking r arbitrary, $M_r(A) \geq M_r(A^{(e)})$ for every unit e: the matching numbers of a plane family decrease under compression in every direction, which for graphs is $m_r(G) \geq m_r(G-v)$. Second, the whole band reduces to the sign of the cross terms.

Proposition 39. *Write $X_e := N_e - \sum_k \theta_k F_{A''_k}$ for the cross-term part of N_e, where $\theta_k = \|f_k\|^2$ and A''_k is the compression to $\{e, f_k\}^\perp$. If $X_e(x) \leq 0$ for every weighted 2-plane family with $\mathrm{Adj} \preceq aI$, every unit e and every $x \geq 2\sqrt{a}$, then every such family satisfies the band $[a - 2\sqrt{a},\, a + 2\sqrt{a}]$.*

Proof. Induct on m. Put $R_e = F_A/F_{A^{(e)}}$ and $R'_k = F_{A^{(e)}}/F_{A''_k}$. Dividing Theorem 38 by $F_{A^{(e)}}$,

$$R_e = x - \sum_k \frac{\theta_k}{R'_k} - \frac{X_e}{F_{A^{(e)}}}.$$

The class is closed under compression, so the inductive hypothesis $R'_k \geq x/2$ applies, and $\sum_k \theta_k = \sum_k \|\iota_e \omega_k\|^2 = \langle e, \mathrm{Adj}(A)e\rangle \leq a$. Hence $R_e \geq x - 2a/x$, and $x - 2a/x \geq x/2$ exactly when $x \geq 2\sqrt{a}$. The base case $m \leq 1$ is $R_e = x$. So $R_e \geq x/2 > 0$ for $x \geq 2\sqrt{a}$, and $F_A = R_e F_{A^{(e)}}$ has no root there; F_A is even or odd, so all roots lie in $[-2\sqrt{a}, 2\sqrt{a}]$. ☐

The leading cross term can be evaluated exactly, and it is a square.

Theorem 40. *Let A be a weighted 2-plane family on $\mathbb{R}^m$ and e a unit vector, and put $f_k = \iota_e \omega_k$, $\omega'_k = \omega_k - e \wedge f_k$ and $u_k = \iota_{f_k} \omega'_k$. Then*

$$C_2 = \Big\| \sum_k f_k \otimes \omega'_k \Big\|^2 - \Big\| \sum_k u_k \Big\|^2,$$

and $\sum_k u_k$ is the off-diagonal $(e, e^\perp)$ block of $\mathrm{Adj}(A)$. So if $\mathrm{Adj}(A) = aI$ then

$$C_2 = \Big\| \sum_k f_k \otimes \omega'_k \Big\|^2 \;\geq\; 0,$$

with equality if and only if $\sum_k f_k \otimes \omega'_k = 0$.

Proof. For a vector u and forms α, γ one has $\iota_w(u \wedge \alpha) = \langle w, u\rangle \alpha - u \wedge \iota_w \alpha$, hence

$$\langle u \wedge \alpha,\; w \wedge \gamma \rangle = \langle u, w\rangle \langle \alpha, \gamma\rangle - \langle \iota_w \alpha, \iota_u \gamma\rangle. \tag{18}$$

Applying (18) to each cross term with $\alpha = \omega'_l$, $\gamma = \omega'_k$ gives $C_2 = \sum_{k \neq l} \big[\langle f_k, f_l\rangle \langle \omega'_k, \omega'_l\rangle - \langle u_k, u_l\rangle\big]$.

Now $f_k = \iota_e \omega_k$ lies in the plane of ω_k, and it is orthogonal to e, so it lies in the image of that plane under the projection to $e^\perp$, which is the plane of ω'_k. Hence $f_k \wedge \omega'_k = 0$, and since

$\|f_k\|^2\|\omega'_k\|^2 = \|\iota_{f_k}\omega'_k\|^2 + \|f_k \wedge \omega'_k\|^2$ for simple ω'_k, this says exactly $\|f_k\|^2\|\omega'_k\|^2 = \|u_k\|^2$. Therefore

$$\sum_{k\neq l}\langle f_k, f_l\rangle\langle\omega'_k,\omega'_l\rangle = \Big\|\sum_k f_k\otimes\omega'_k\Big\|^2 - \sum_k\|u_k\|^2, \qquad \sum_{k\neq l}\langle u_k, u_l\rangle = \Big\|\sum_k u_k\Big\|^2 - \sum_k\|u_k\|^2,$$

and subtracting gives the display. For the last claim, polarizing $\langle v, \Theta_k v\rangle = \|\iota_v\omega_k\|^2$ at $v = e$ against $v \in e^\perp$ gives $\langle e, \mathrm{Adj}(A)v\rangle = -\langle v, \sum_k u_k\rangle$, so $\mathrm{Adj}(A) = aI$ forces $\sum_k u_k = 0$. □

Theorem 40 is sharp in the way the data show: over coordinate families $\min_e C_2 = 0$ to machine precision, attained exactly at the coordinate directions, where each summand $f_k\otimes\omega'_k$ vanishes separately, every ω_k either contains e in its plane, so $\omega'_k = 0$, or is orthogonal to it, so $f_k = 0$. That is precisely why the classical recursion has no cross terms. Off that locus the minimum is strictly positive.

The proof reindexes, and doing so gives every level at once.

Theorem 41. *For every $r \geq 2$,*

$$C_r = \sum_{|S|=r-2}\Big[Q_S - \|W_S\|^2\Big], \qquad \begin{aligned} Q_S &= \sum_{k,l\notin S}\langle f_k, f_l\rangle\big\langle\omega'_k\wedge\omega'_S,\ \omega'_l\wedge\omega'_S\big\rangle, \\ W_S &= \sum_{k\notin S}\iota_{f_k}\big(\omega'_k\wedge\omega'_S\big), \end{aligned}$$

and $Q_S = \big\|\sum_{k\notin S} f_k\otimes(\omega'_k\wedge\omega'_S)\big\|^2 \geq 0$.

Proof. Writing $T = S\cup\{k,l\}$ reindexes C_r as $\sum_{|S|=r-2}\sum_{k\neq l\notin S}\langle f_k\wedge\omega'_l\wedge\omega'_S,\ f_l\wedge\omega'_k\wedge\omega'_S\rangle$. Apply (18) with $\alpha = \omega'_l\wedge\omega'_S$ and $\gamma = \omega'_k\wedge\omega'_S$. The wedge $\omega'_k\wedge\omega'_S$ is a product of vectors, hence simple, and $f_k\wedge(\omega'_k\wedge\omega'_S) = (f_k\wedge\omega'_k)\wedge\omega'_S = 0$ by the proof of Theorem 40; so $\|f_k\|^2\|\omega'_k\wedge\omega'_S\|^2 = \|\iota_{f_k}(\omega'_k\wedge\omega'_S)\|^2$, and the diagonal terms of the two double sums cancel exactly as before. The Gram matrix of the vectors $\omega'_k\wedge\omega'_S$ is positive semidefinite, so Q_S is the squared norm of $\sum_{k\notin S} f_k\otimes(\omega'_k\wedge\omega'_S)$. □

At $r = 2$ the set S is empty, $W_\emptyset = \sum_k \iota_{f_k}\omega'_k$ vanishes by tightness, and Theorem 41 is Theorem 40. At $r = 3$ it does not vanish, but it simplifies: since $\iota_{f_n}\omega'_n$ lies in the plane of ω'_n, the self term $(\iota_{f_n}\omega'_n)\wedge\omega'_n$ is zero, and using $\sum_k u_k = 0$,

$$W_{\{n\}} = \sum_{k\neq n}\omega'_k\wedge\iota_{f_k}\omega'_n.$$

So the obstruction at $r = 3$ is exactly the failure of the compressed planes to be mutually orthogonal, and it is the only obstruction: everything else is a square.

It is worth being exact about what this settles. Writing the condition $X_e \leq 0$ at $x = 2\sqrt{a}$ in the form

$$\sum_{r\geq 2}(-1)^r C_r\,(4a)^{2-r} \ \geq\ 0, \tag{19}$$

Theorem 40 is the leading coefficient of (19), that is the case $a\to\infty$. It gives $X_e \leq 0$ outright when $m \leq 5$, where C_2 is the only cross term. That range does not improve on the two-moment bounds above, so Theorem 40 does not by itself enlarge the unconditional range; what it supplies is the first structural reason for the sign.

The obvious next step, bounding C_3 by $4a\,C_2$ so as to reach $m \le 7$, is not available, and it fails for a reason worth recording rather than for lack of effort. The ratio $4a\,C_2/C_3$ decreases steadily in m, over cubic graphs it runs 3.49 at $m = 8$, 2.00 at $m = 10$ (Petersen), 1.38 at $m = 12$ (Franklin), 1.03 at $m = 14$ (Heawood), and crosses below 1 at $m = 16$, where the Möbius–Kantor graph gives 0.827, then 0.687 (Pappus, $m = 18$), 0.590 (Desargues, $m = 20$) and 0.457 (Nauru, $m = 24$). So $C_3 \le 4aC_2$ is false in general, and increasingly so.

The reason is visible one level down. For odd r the criterion wants C_r small, that is $\|W_S\|^2$ close to Q_S; but the ratio $\sum_n \|W_n\|^2/\sum_n Q_n$ decays, running 0.207, 0.190, 0.154, 0.119, 0.101 over the same graphs at $m = 8, \dots, 16$. So C_3 is essentially $\sum_n Q_n$, an undamped sum of Schur squares, and the contraction W supplies almost no cancellation. Whatever makes (19) hold is therefore the alternation *between* levels, not any cancellation *within* one.

What the same computation shows is that the truncation is what fails, not the conclusion. Writing Σ_R for the partial sum of (19) over $2 \le r \le R$:

G	m	Σ_2	Σ_3	Σ_4	Σ_5	Σ_6
cube	8	1.268	0.905	0.926		
Petersen	10	1.624	0.812	0.891	0.891	
Franklin	12	1.713	0.468	0.730	0.713	0.713
Heawood	14	2.007	0.060	0.679	0.603	0.607
Möbius–Kantor	16	2.139	−0.448	0.669	0.458	0.476
Pappus	18	2.142	−0.974	0.739	0.291	
Desargues	20	2.235	−1.552	0.969	0.127	
Nauru	24	2.231	−2.656	1.786		

The values are for one direction per graph, fixed by the seed in `code/hl_Cr.py`, which also regenerates Theorem 41 against the definition of C_r. The terms $|C_r|(4a)^{2-r}$ peak at $r = 3$ and then decay geometrically, at $m = 16$ they run 2.14, 2.59, 1.12, 0.211, 0.0176, so the series converges, but the number of terms needed grows with m: three nearly suffice at $m = 14$, four at $m = 16$, and at $m = 20$ and 24 the partial sums have not settled by Σ_5. So (19) is not a statement of bounded length, and in particular the two-term reading fails outright from $m = 16$. By Theorem 41 every term is a Schur square minus the square of a contraction, so what a proof must supply is control of the alternating sum of the Q_S across levels.

How much room the inequality actually has is worth measuring, and the answer is that there is a great deal, and that the recursion cannot use any of it. Evaluating the exact ratio $R_e = F_A/F_{A^{(e)}}$ at $x = 2\sqrt{a}$ on regular graphs, in units of the threshold $x/2 = \sqrt{a}$:

a	graphs	$R_e/(x/2)$	spread
3	cube, Petersen, Franklin, Heawood	1.3662–1.3692	$3.0 \cdot 10^{-3}$
4	$C_8(1,2)$, $C_{10}(1,2)$, $C_{12}(1,2)$	1.3488–1.3497	$8.6 \cdot 10^{-4}$
5	$C_8(1,2,4)$, $C_{10}(1,2,5)$	1.3246–1.3322	$7.5 \cdot 10^{-3}$

The ratio is a function of a alone to three digits, independent of the graph and of the dimension, and exceeds the threshold by a third. The slack $-X_e/(F_{A^{(e)}}\sqrt{a})$ likewise *increases* with m, running $0.033, 0.044, 0.049, 0.058$ over the four cubic graphs above.

That constant can be identified exactly. Averaging the vertex identity over directions gives $\mathbb{E}_e F_{A^{(e)}} = F'_A/m$, which is $F'_A = \sum_i F_{A^{(e_i)}}$ again, so the direction-averaged ratio is

$$\bar{R} = \frac{mF_A}{F'_A} = \Big(\frac{1}{m}\sum_i \frac{1}{x-\rho_i}\Big)^{-1} = \frac{1}{G(x)},$$

G the Stieltjes transform of the root measure of F_A. For a coordinate family F_A is the matching polynomial, and the Kesten–McKay measure of parameter a has Stieltjes transform $G(z) = 2(a-1)/\big((a-2)z + a\sqrt{z^2-4(a-1)}\big)$. At $z = 2\sqrt{a}$ the radical is *exactly* 2, and everything collapses:

Proposition 42. $\dfrac{1}{G(2\sqrt{a})}\Big/\sqrt{a} = \dfrac{\sqrt{a}+2}{\sqrt{a}+1} = 1 + \dfrac{1}{1+\sqrt{a}}.$

The passage from a finite graph to that value is usually made by citing convergence of the matching measure. It need not be: the evaluation point lies strictly outside the support, so the whole step can be made finite and effective, with an explicit rate. We are grateful to Péter Csikvári for pointing us to the mechanism, which is his with P. E. Frenkel [20], after Abért and Hubai for the chromatic polynomial.

Their theorem is that for a monic multiplicative graph polynomial of bounded exponential type, the power sum $p_j(G)$ of the roots is a fixed linear combination of counts of *connected* subgraphs of G; the matching polynomial is such a polynomial. In Godsil's formulation [22] the matching-measure moments count closed *tree-like* walks, and the trace of such a walk of length j carries at most $j/2$ edges. Since a subgraph with fewer than girth(G) edges is a forest, and since in an a-regular graph the number of copies of a fixed tree is $|V(G)|$ times a function of a alone, one gets an identity rather than a limit.

Proposition 43. *Let G be a-regular of girth g. Then $p_j(G)/|V(G)|$ equals the j-th moment of the Kesten–McKay measure of parameter a for every $j < 2g$. The threshold is sharp: for K_4, K_5, $K_{3,3}$, $K_{4,4}$, the cube, Petersen and Heawood, the first differing moment is exactly $2g$ whenever one exists at all (`code/moment_girth.py`, exact arithmetic).*

Because Heilmann–Lieb confines the roots to $[-\rho, \rho]$ with $\rho = 2\sqrt{a-1}$, while the evaluation point is $x = 2\sqrt{a}$, the ratio $\rho/x = \sqrt{1-1/a}$ is strictly below one. A finite geometric expansion with remainder, and no infinite series anywhere, then converts matching moments into matching transforms.

Proposition 44. *Let ν_1, ν_2 be probability measures on $[-\rho, \rho]$ whose moments agree up to order k, and let $x > \rho$. Then*

$$\big|G_{\nu_1}(x) - G_{\nu_2}(x)\big| \;\le\; \frac{2\rho^{k+1}}{x^{k+1}(x-\rho)},$$

and if both transforms are at least $c > 0$ the reciprocals differ by at most that quantity divided by c^2.

Proof. For $|t| \le \rho < x$ one has the identity $1/(x-t) = \sum_{j\le k} t^j/x^{j+1} + t^{k+1}/\big(x^{k+1}(x-t)\big)$, whose right-hand side does not depend on k. Integrating, the polynomial parts agree by hypothesis, and each remainder is at most $\rho^{k+1}/\big(x^{k+1}(x-\rho)\big)$ in absolute value. The statement for reciprocals is $|1/S_1 - 1/S_2| = |S_2 - S_1|/(S_1S_2)$. □

Remark 45. Csikvári asks (personal communication, August 2026) whether $\mu_G(2\sqrt{d-1})$ has a combinatorial meaning for d-regular G with $|V(G)|$ even, observing that it should at least be a positive integer. It does, and the answer is in his own paper with Bencs [26]. Their polynomial $R_G(z) = \sum_M (-z)^{|M|} \prod_{v \notin V(M)} (z + d_v - 1)$ is, for d-regular G, a transformation of the matching polynomial: comparing coefficients termwise,

$$R_G(z) = z^{n/2}\, \mu_G\Big(\frac{z+d-1}{\sqrt{z}}\Big).$$

The point $2\sqrt{d-1}$ is not an arbitrary argument of that substitution. Solving $(\sqrt{z} + (d-1)/\sqrt{z}) = 2\sqrt{d-1}$ gives $(\sqrt{z} - \sqrt{d-1})^2 = 0$, so it is exactly the *double root*, attained only at $z = d-1$. Their Corollary 2.7 expands $R_G(w) = \sum_{A \in \mathcal{PF}} 2^{c(A)} (w-1)^{n-|A|}$ over pseudo-forests A, edge sets each of whose components carries at most one cycle, with $c(A)$ the number of cycles. Putting $w = d-1$,

$$\mu_G\big(2\sqrt{d-1}\big) \;=\; (d-1)^{-n/2} \sum_{A \in \mathcal{PF}(G)} 2^{c(A)} (d-2)^{\,n-|A|}. \tag{20}$$

At $d = 3$ the weight is 1 and (20) reads: $2^{n/2}\mu_G(2\sqrt{2})$ is the number of pseudo-forests of G counted with multiplicity 2 per cycle. The values are 19, 74, 76, 297, 1162 for K_4, $K_{3,3}$, the prism, the cube and Petersen.

The transformation, the double root and the evaluation are machine-checked in `RamaLean/PseudoForest.lean`; Corollary 2.7 is [26] and enters as a hypothesis. Identity (20) is verified by brute force in `code/pseudoforest.py` on nine regular graphs of degrees 3 and 4 and orders 4 to 10, in exact arithmetic. The positivity and integrality Csikvári notes in passing are separately machine-checked in `RamaLean/EvenEval.lean`, together with the observation that the parity hypothesis is needed: for $|V(G)|$ odd the value is $2\sqrt{d-1}$ times an integer.

Propositions 43 and 44 together bound the distance from the finite-graph ratio to the value of Proposition 42 by a quantity decaying like $(1 - 1/a)^g$, with no convergence theorem invoked. The bound is not tight, since $|m_j| \le \rho^j$ discards the $j^{-3/2}$ decay of the Kesten–McKay moments at the edge: at $a = 3$ it gives $0.62, 0.41, 0.28$ at girths $4, 5, 6$ against observed errors of $3.2 \cdot 10^{-3}$, $9.2 \cdot 10^{-4}$, $2.2 \cdot 10^{-4}$. What it does give, and the citation did not, is an effective statement about a finite graph.

So the cavity ratio exceeds the threshold the induction can propagate by exactly $1/(1+\sqrt{a})$, and the excess tends to 0 as a grows, which is the analytic shadow of $2\sqrt{a}$ being asymptotically attained.

The limit appears to be attained from above, never crossed, which suggests that Proposition 42 states an infimum rather than merely a limit:

Conjecture 46. *For every weighted* 2*-plane family with* $\mathrm{Adj}(A) = aI$ *and every unit* e*,*

$$\frac{R_e}{x/2} \;\ge\; 1 + \frac{1}{1+\sqrt{a}} \qquad \text{at } x = 2\sqrt{a}.$$

This was not violated in any family tested, over graphs and over general weighted plane families, minimizing over directions; the smallest margin observed was $2.2 \cdot 10^{-4}$, at the highest-girth cubic graph, which is where Proposition 42 says it should be. Conjecture 46 is strictly stronger than $X_e \le 0$, which is its $c = 1$ case, and by Proposition 47 it cannot be proved by this induction either, the recursion propagates $c = 1$ and nothing else, whatever is true. Against the exact values the prediction is 1.366025 at $a = 3$, 1.333333 at $a = 4$ and 1.309017 at $a = 5$, and the error decays with the girth exactly as Kesten–McKay convergence requires: over cubic graphs $3.2 \cdot 10^{-3}$ at girth 4 (the cube), $9.2 \cdot 10^{-4}$ at girth 5 (Petersen), and $2.2 \cdot 10^{-4}$, $1.7 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$ at girth 6 (Heawood, Möbius–Kantor, Pappus, Desargues). None of that slack is available to the argument:

Proposition 47. *Fix $a > 0$ and suppose the inductive hypothesis of Proposition 39 is strengthened from $R \ge x/2$ to $R \ge c\,(x/2)$ for some $c > 0$. At $x = 2\sqrt{a}$ the recursion sustains this only for $c = 1$.*

Proof. Propagating $R \ge c\sqrt{a}$ requires $c\sqrt{a} \le 2\sqrt{a} - a/(c\sqrt{a})$, which after clearing the denominator is $a(c-1)^2 \le 0$, so $c = 1$. □

There is one hypothesis the argument has not used, and it changes what has to be proved. The class is *not* closed under compression in the tight sense:

Proposition 48. *For a weighted 2-plane family with $\operatorname{Adj}(A) = aI$ and any unit e,*

$$\operatorname{Adj}\big(A^{(e)}\big) = a\, I_{e^\perp} - F, \qquad F = \sum_k f_k f_k^{\mathsf{T}}, \qquad \operatorname{tr} F = a.$$

Proof. For $u, v \perp e$ one has $\iota_u \omega_k = -\langle u, f_k\rangle e + \iota_u \omega_k'$ with $\iota_u \omega_k' \perp e$, so $\langle \iota_u\omega_k, \iota_v\omega_k\rangle = \langle u, f_k\rangle\langle v, f_k\rangle + \langle \iota_u\omega_k', \iota_v\omega_k'\rangle$; summing and using $\operatorname{Adj}(A) = aI$ gives the display, and $\operatorname{tr} F = \sum_k \theta_k = \langle e, \operatorname{Adj}(A)e\rangle = a$. □

So compressing costs exactly a positive semidefinite matrix of trace a, and the children of a tight family are deficient by a definite amount. Spending that deficiency strengthens the induction. Write $d(e) = a - \langle e, \operatorname{Adj}(A)e\rangle \ge 0$ for the local deficiency, and replace the hypothesis $R \ge x/2$ of Proposition 39 by

$$\text{(H)} \qquad R_e \;\ge\; \frac{x}{2} + \frac{2\,d(e)}{x}.$$

This is consistent at the base ($m \le 1$: $R_e = x$, $d = a$, and $x/2 + 2a/x = x$ exactly), and at a tight family $d \equiv 0$, so it still delivers the band.

Proposition 49. *Let $g_k = \langle \hat f_k, F \hat f_k\rangle$ be the deficiency the children inherit. Then* (H) *propagates provided*

$$X_e \;\le\; \frac{2F_{A^{(e)}}}{x} \sum_k \frac{\theta_k\, g_k}{a + g_k},$$

and the estimate is lossless: the two sides of the resulting comparison are identically equal.

The right-hand side is *strictly positive*, since $g_k \ge \theta_k > 0$. So the strengthened induction does not require $X_e \le 0$; it requires X_e below an explicit positive quantity. On the graphs tested (H) held with margin 0.52 to 0.71, and the sufficient condition held with the right-hand side exceeding $|X_e|$ by a factor of five to twenty.

That is worth having because an upper bound on X_e does exist, and the deficiency supplies it. Applying the vertex identity a second time writes each child polynomial as $F_{A''_k} = \big(F_{A^{(e)}} - q_k\big)/x$ with $q_k = \langle \hat f_k, W^{(e)}(x)\hat f_k\rangle$, so $\sum_k \theta_k F_{A''_k} = \big(aF_{A^{(e)}} - \mathrm{tr}(FW^{(e)})\big)/x$, and substituting into $X_e = N_e - \sum_k \theta_k F_{A''_k}$ gives a relation between the vertex matrices at two consecutive levels:

Proposition 50. *$x\,X_e = \mathrm{tr}\big(F\,W^{(e)}(x)\big) - x\,\langle e, W(x)e\rangle - a\,F_{A^{(e)}}(x)$, and consequently*

$$x\,X_e \;\le\; a\,\lambda_{\max}\big(W^{(e)}(x)\big) - x\,\langle e, W(x)e\rangle - a\,F_{A^{(e)}}(x).$$

Proof. The identity is the substitution above. For the bound, $F = \sum_k \theta_k \hat f_k \hat f_k^{\mathsf T}$ is a nonnegative combination of unit rank-ones with $\sum_k \theta_k = \mathrm{tr}\,F = a$, so $\mathrm{tr}(FW^{(e)}) = \sum_k \theta_k q_k \le \big(\sum_k \theta_k\big)\lambda_{\max}(W^{(e)})$; no spectral theory is needed beyond the definition of $\lambda_{\max}$. $\square$

This is the first upper bound on X_e in the development, every earlier statement bounded it below or computed it exactly, and the trace step is not lossy where it matters: on K_5 and K_6, where $X_e = 0$, it holds with equality. Together with Proposition 49, whose target is strictly positive, the band is therefore reduced to a spectral bound on the child's vertex matrix.

That bound has a clean form in the cavity ratios, and the form explains what is left. Write $r = R_e/(x/2)$ and $r' = \big(\max_{e'} R'_{e'}\big)/(x/2)$. Since $\lambda_{\max}(W^{(e)}) = F_{A^{(e)}} - x\min_{e'} F_{A''}$ and $\min_{e'} F_{A''} = F_{A^{(e)}}/\max_{e'} R'_{e'}$, the bound $\lambda_{\max}(W^{(e)}) \le xF_A/a - 3F_{A^{(e)}}$ is exactly

$$r' \;\le\; \frac{1}{2-r}. \tag{21}$$

Proposition 51. *The Kesten–McKay value of Proposition 42 satisfies* (21) *automatically: with $u = 1/(1+\sqrt{a})$ and $r = r' = 1+u$, one has $(1+u)(1-u) = 1-u^2 \le 1$, and the slack is exactly*

$$\frac{1}{1-u} - (1+u) = \frac{u^2}{1-u} = \frac{1}{\sqrt{a}\,(1+\sqrt{a})}.$$

So the target is not merely consistent with the Kesten–McKay picture, it is *implied* by it, with a slack that vanishes like $1/a$, matching the asymptotic tightness of $2\sqrt{a}$. On the graphs tested (21) holds throughout, and it is tight exactly where $X_e = 0$: at K_5 and K_6 the two sides agree to four digits (1.5673 and 1.5194), while the larger-girth graphs have margin. What remains unproved is therefore not (21) in the abstract but the gap between the true ratios and their Kesten–McKay values, which is Conjecture 46 together with its analogue one level down.

The constant is therefore not an artefact of the estimate: $\sqrt{a}$ is the fixed point of $\rho \mapsto x - a/\rho$ at $x = 2\sqrt{a}$, the value at which the two roots of $\rho^2 - x\rho + a$ collide. For $x > 2\sqrt{a}$ the inductive hypothesis has an interval of admissible thresholds and the argument has slack; at $x = 2\sqrt{a}$ the interval is a point and it has none. That is exactly why $2\sqrt{a}$, and not the conjectured $2\sqrt{a-1}$, is the ceiling of every argument of this shape. The cross terms $C_r = N_r - \sum_k \theta_k M''_{k,r-1} = \sum_{|T|=r}\sum_{k\ne l\in T}\langle f_k \wedge \omega'_{T\setminus k}, f_l \wedge \omega'_{T\setminus l}\rangle$ were nonnegative in every family tested, and at $r = 2$ that is now Theorem 40. The nonnegativity is not termwise, and it is worth recording that it is not, since the termwise statement is the natural thing to try and it is false: single pairs (k,l) and even whole summands $\sum_{k\ne l\in T}$ take negative values, the smallest observed being $-1.9 \cdot 10^{-3}$ and $-3.8 \cdot 10^{-3}$ over weighted plane families in dimensions six and eight. If $C_r \geq 0$ holds it is therefore a statement about the sum over all T of a given size, not about its terms. Summed, it survives minimization over the sphere, and sharply: for coordinate families $\min_e C_2 = 0$ to machine precision, attained exactly at the coordinate directions, where the classical recursion lives, while for noncommuting projection and general plane families the minimum is strictly positive, growing from 0.048 at $m = 4$ to 0.777 at $m = 8$. Finally $X_e = \sum_{r \geq 2} (-1)^{r-1} C_r x^{m-2r}$ was nonpositive at $x = 2\sqrt{a}$ in every case under direct maximization over the sphere, zero to $4 \cdot 10^{-12}$ for coordinate families, strictly negative otherwise, over graph families in dimensions $4, 5, 6$, projection families in dimensions $4, 6$ and weighted plane families in dimension 4. The identity of Theorem 38 was checked to 10^{-13} on graph, projection and general weighted plane families at generic directions (`code/hl_Wspec.py`, `code/hl_Xsign.py`).

Both of the bounds that follow are rungs of a single ladder, and it is worth saying what the ladder reaches before giving them. Writing $y = x^2$, the M_r are the elementary symmetric functions of the y-roots, so with p_k the power sums one has $y_{\max}^k \leq p_k$, and any a priori bound $p_k \leq (4a)^k$ gives the band (`MomentLadder.le_of_power_sum_le`). Writing $p_k = (m/2)c_k^k$, the k-th rung covers dimension m exactly when

$$m \;\leq\; 2\Big(\frac{4a}{c_k}\Big)^k \tag{22}$$

(`reach_of_moment`). Measured on the class, c_k^k is the number of closed walks of length $2k$ from a root of the a-regular tree, so $c_1 = a$ and $c_2 = \sqrt{a(2a-1)}$, which are the two rungs below; that count grows like $\big(4(a-1)\big)^k k^{-3/2}$, the base being the square of the tree's spectral radius, against a target of $(4a)^k$. Since $4(a-1) < 4a$, the reach (22) is unbounded in k (`reach_unbounded`): at $a = 3$ it is 146, 509 and 1489 at $k = 5, 7, 9$, against the 8 and 18 of the first two rungs (`code/moments2.py`). What the reach costs is an a priori bound on p_k, and the natural candidate is $p_k \leq (m/2)W_{2k}$ with W_{2k} the tree walk count; for graphs that is Godsil's path-tree argument, the path tree being a subtree of the universal cover. Off the coordinate case it is false. Weighted families with $\mathrm{Adj}(A) = aI$ exceed it by 1.008, 1.134 and 1.315 at $m = 6, 8, 10$ and $a = 3$, the excess growing in both k and m, so no constant-factor version survives either; coordinate families obey it, with equality at $k = 1, 2$ (`code/momentbound.py`, `code/p16verify.py`). Weakening it to a rate, $p_k \leq (m/2)R^k W_{2k}$, leaves the reach unbounded exactly when $R < a/(a-1)$ (`rate_lt_target`, `reach_unbounded_of_rate`), and the measured rates sit far inside that: 0.97, 1.02, 1.05, 1.07, 1.09 at $m = 6, \ldots, 14$ and $a = 3$ (`code/excessrate.py`). That is still not a route, for a reason that is structural rather than numerical.

Proposition 52. *Let $y_1, \ldots, y_n \geq 0$, let $c > 0$, and suppose $\sum_i y_i^k \leq Cc^k$ for every $k \geq 1$, with C independent of k. Then $y_i \leq c$ for every i.*

Proof. If $y_i > c$ then $(y_i/c)^k \leq \big(\sum_j y_j^k\big)/c^k \leq C$ for every k, while the left side is unbounded (`MomentLadder.band_of_all_moments`). □

Applied with $c = 4(a-1)R$, a rate bound holding at every rung already forces $y_{\max} \leq 4(a-1)R$. So at $R < a/(a-1)$ the hypothesis is strictly stronger than the band it is meant to prove, and at $R \geq a/(a-1)$ every rung collapses to $m \leq 2$ (`reach_trivial_of_rate`). The ladder has content only at a single finite k, where it buys the finite range (22) and nothing more; the dimension restriction is intrinsic to it.

Two bounds follow unconditionally from real-rootedness alone. Writing $F_A = x^m - M_1x^{m-2} + M_2x^{m-4} - \cdots$, one has $M_1 = \frac{1}{2}\operatorname{tr}\operatorname{Adj}(A) \le am/2$ and, using $\|\omega_j \wedge \omega_l\|^2 \ge 1 - \operatorname{tr}(P_jP_l)$, $M_1^2 - 2M_2 \le \operatorname{tr}\operatorname{Adj}(A)^2 - \sum_k c_k^2$. Since the largest root λ satisfies $\lambda^2 \le M_1$ and $\lambda^4 \le M_1^2 - 2M_2$, the band $[a - 2\sqrt{a}, a + 2\sqrt{a}]$ holds whenever $m \le 8$, and whenever $m \le 16 + \big(\sum_k c_k^2\big)/a^2$, which for projection families is $m \le 32a/(2a-1)$. The second reproduces the known two-moment bound for projections and extends it to all weighted plane families. Adversarial search over rank-two projection families and over random biregular graphs reaches 0.83 of the threshold at $a = 3$ and 0.72 at $a = 4$, with no violation; graphs are marginally more extremal than general projections.

9.7 The reflection for projections

At $b = 2$ the whole question also collapses onto the known half, by a symmetry.

Proposition 53. *Let $P_1, \dots, P_q$ be rank-2 orthogonal projections on $\mathbb{R}^p$ with $\sum_k P_k = aI$. Then $\mu[P_1, \dots, P_q](2a - y) = (-1)^p \mu[P_1, \dots, P_q](y)$; in particular $\lambda_{\min} + \lambda_{\max} = 2a$.*

Proof. The mixed characteristic polynomial is affine-linear in each slot, so it commutes with taking expectations slotwise, and for rank-one slots $\mu[v_1v_1^\mathsf{T}, \dots] = \det\big(yI - \sum_k v_kv_k^\mathsf{T}\big)$, since a multi-affine polynomial is sent by $\prod_k(1 - \partial_k)$ to its value at $z = (-1, \dots, -1)$. Choose an orthonormal pair e_k, f_k spanning the range of P_k and set $v_k = e_k + \varepsilon_k f_k$ with ε_k independent uniform signs. Then $\mathbb{E}\, v_kv_k^\mathsf{T} = P_k$ and $v_kv_k^\mathsf{T} = P_k + \varepsilon_k S_k$ with $S_k = e_kf_k^\mathsf{T} + f_ke_k^\mathsf{T}$ and $S_k^2 = P_k$, so $\mu(y) = \mathbb{E}_\varepsilon \det\big((y-a)I - \sum_k \varepsilon_k S_k\big)$. That is an even or odd function of $t = y - a$ according to the parity of p, because $\varepsilon \mapsto -\varepsilon$ preserves the distribution. $\square$

The last step, which carries the whole proposition, is formalized in `RamaLean/SignReflection.lean` as `sign_avg_reflect`: for any family of matrices $S_1, \dots, S_q$ the sign average $\sum_{\varepsilon \in \{\pm1\}^q} \det\big(tI - \sum_k \varepsilon_k S_k\big)$ satisfies $\rho(-t) = (-1)^n \rho(t)$, with no hypothesis on the S_k; the shift and the exchange of band edges $2a - (\sqrt{a-1} + 1)^2 = (\sqrt{a-1} - 1)^2$ are `mixedChar_reflect` and `band_edges_swap`. The step identifying the sign average with μ is formalized in the same file: by `sum_prod_sgn` a nonempty sign monomial averages to zero over $\{\pm1\}^q$, proved by the same involution, so by `sum_multiaffine` the average of a multi-affine function returns 2^q times its constant term, which for $\det\big(yI + \sum_k z_kA_k\big)$ with rank-one A_k is what $\prod_k(1 - \partial_{z_k})$ extracts.

Since $2a - (\sqrt{a-1} + 1)^2 = (\sqrt{a-1} - 1)^2$, the two band edges are exchanged by the reflection, so at $b = 2$ the lower bound is *equivalent* to the upper one. For diagonal families the upper bound is Godsil's path-tree theorem and $b = 2$ is therefore settled, that is Theorem 8 again. For projections it is not: the path tree is a graph construction, and the only bound available in that generality is the Marchenko–Pastur edge $(\sqrt{a} + \sqrt{b})^2$, which is strictly larger than $(s + t)^2$. So at $b = 2$ the projection statement is reduced to its upper half, not closed. The mechanism in the diagonal case is transparent: there $\sum_k \varepsilon_k S_k$ is a signed adjacency matrix, the expectation is Godsil–Gutman, and one recovers $\nu_{S(H)}(y) = \mu_H(y - a)$.

This also locates the obstruction at $b \ge 3$ sharply. Writing $N_k = v_kv_k^\mathsf{T} - P_k$ one has $N_k^2 = (b-2)N_k + (b-1)P_k$, with spectrum $\{b-1, -1^{(b-1)}\}$ on the range of P_k; the reflection needs this invariant under $N_k \mapsto -N_k$, which happens exactly when $b = 2$. Equivalently:

the mean root is always a, since $\sum_i \lambda_i = \operatorname{tr} \sum_k P_k = pa$, while the band centre is $a+b-2$. The conjecture therefore asserts a skew of exactly $b-2$, which is the operator form of the displacement in Theorem 25 and is invisible to any reflection or first-moment argument.

The source of that skew can be named exactly. The jump N_k has spectrum $\{b-1, (-1)^{(b-1)}\}$ on the range of P_k, whose moments are

$$m_1 = 0, \qquad m_2 = b(b-1), \qquad m_3 = b(b-1)(b-2).$$

So the third moment vanishes precisely when $b=2$: Proposition 53 is not an accident of small b but the vanishing of the jump skewness, and for $b \geq 3$ any argument reaching the lower edge must see a third-order quantity.

10 The band is the support of a free convolution power

There is a single reason behind all of this, and it is worth stating because it accounts uniformly for the failure of every method tried. Let $\chi = (1-\frac{1}{b})\delta_0 + \frac{1}{b}\delta_b$, and let ν be the centred slot spectrum $\frac{1}{b}\delta_{b-1} + (1-\frac{1}{b})\delta_{-1}$, so that $\chi = \delta_1 \boxplus \nu$.

Proposition 54. $\operatorname{supp}\bigl(\chi^{\boxplus a}\bigr) = \bigl[(\sqrt{a-1} - \sqrt{b-1})^2,\ (\sqrt{a-1} + \sqrt{b-1})^2\bigr]$.

So the band is the support of a free convolution *power*. Every method that fails computes instead a compound free Poisson of rate a with jump ν, whose free cumulants are $a\, m_n(\nu)$ rather than $a\, \kappa_n(\nu)$, and

$$m_n(\nu) = \kappa_n(\nu) \ \ (n \leq 3), \qquad \kappa_4(\nu) - m_4(\nu) = -2(b-1)^2.$$

The two sequences agree through third order and part company at the fourth, by exactly $-2a(b-1)^2$. Order four is where every structure in this problem turns over, and it does so three independent times: the coefficients $E_0, \ldots, E_3$ are determined by the parameters while E_4 carries the 4-cycle count; the moment and free-cumulant sequences of the jump agree up to order three and split at order four; and the parallel-class construction for the finite free convolution reproduces the elementary symmetric functions e_1, e_2, e_3 of μ and first fails at e_4. That is why the barrier bound, the scalar free convolution and the finite free convolution all overshoot by the same margin of about 1.1, and it is the analytic form of the third-moment obstruction above.

The finite free cumulants of μ itself follow the free power, not the Poisson:

$$\kappa_1 = a, \qquad \kappa_2 = \frac{ap(b-1)}{p-1}, \qquad \kappa_3 = \frac{ap^2(b-1)(b-2)}{(p-1)(p-2)},$$

with κ_4 the first that depends on the family, matching the coefficient ladder, where $E_0, \ldots, E_3$ are universal and E_4 carries the 4-cycle count. Note $\kappa_3 = 0$ exactly when $b=2$: Proposition 53 is visible in the cumulants.

Two further facts delimit what can be done with this. Writing $\psi_0(x) = x^{p-p/b}(x-b)^{p/b}$ and $\Psi = \psi_0^{\boxplus_p a}$, the polynomial Ψ satisfies the band at every p, but $\mu \neq \Psi$ in general and there is no real-rooted ρ with $\mu = \Psi \boxplus_p \rho$ unless the two agree: finite free deconvolution is unique, so such a ρ would have $\kappa_1 = \kappa_2 = 0$, and κ_2 is a positive multiple of the variance of the roots, forcing $\rho = x^p$. The very agreement of the first three cumulants is what forbids the factorization. Divisibility by a single copy, $\mu = \psi_0 \boxplus_p \rho$ with ρ real-rooted, would by the Marcus–Spielman–Srivastava bound give a lower bound on $\lambda_{\min}$, and it holds in small cases; but it is not a property of the class. Taking $b = 2$ and the incidence families of 3-regular graphs, the deconvolution ρ is real-rooted for $S(K_4)$ at $p = 4$ and for $K_{3,3}$ at $p = 6$, and fails for the cube Q_3 at $p = 8$ and the Petersen graph at $p = 10$, where the largest imaginary part of a root of ρ is 0.133 and 0.290 respectively. The cube is the informative case: it is 3-regular and bipartite, hence 1-factorable, so the failure is not caused by the absence of a partition of the blocks into parallel classes. Whatever separates the projection class from the positive semidefinite relaxation, single-copy divisibility is not it: at $(p, a, b) = (12, 4, 2)$ the scalar family, which violates the band, admits the factorization while a projection family does not.

Two general facts about free convolution came out of the attempt, and they bound what any argument of this shape can achieve. For finitely supported probability measures,

$$\min \operatorname{supp} \omega + \min \operatorname{supp} \tau \;\leq\; \min \operatorname{supp}(\omega \boxplus \tau) \;\leq\; \min \operatorname{supp} \tau + m_1(\omega), \tag{23}$$

the lower bound from the operator model and the upper from $\min \operatorname{supp}(\omega \boxplus \tau) = \sup_{w<0}[K_\tau(w) + R_\omega(w)]$ together with the monotonicity of R_ω, which follows from Cauchy–Schwarz. And if $\min \operatorname{supp} \tau \geq c$ then $\tau(\{c\}) \leq \mu_2/\big(\mu_2 + (m_1 - c)^2\big)$, so by Bercovici–Voiculescu an atom of mass exceeding $\omega(\{\min \operatorname{supp} \omega\})$ pins the left edge of $\omega \boxplus \tau$ at c exactly.

Applied with $\omega = \chi$, for which $m_1 = 1$, (23) says that convolving with χ moves the left edge by at most 1. Any certificate that tries to reach the band edge by exhibiting μ as a free convolution with χ is therefore capped, and the first three cumulants do not constrain the other factor enough: there are measures with exactly the forced values of $\kappa_1, \kappa_2, \kappa_3$, real-rooted, supported in $[0, ab]$, whose convolution with χ has left edge 0. At $(a, b) = (5, 4)$, $p = 12$ one may take $x^4(x-4)^3(x-6)^4(x-12)$, whose mass at 0 exceeds $1/b$.

The same atom mechanism explains the hypothesis $a \geq b$: for $a < b$ the measure $\chi^{\boxplus a}$ carries an atom at 0 of mass $1 - a/b$, so its left edge is 0 and the band statement is empty.

11 Rank monotonicity, and what it explains

What does separate them can be said exactly, in terms of the representation (14). For a general positive semidefinite family with $\sum_k A_k = aI$ the marginal of s_k has probability generating function $\prod_i \big(\frac{a-\alpha_i}{a} + \frac{\alpha_i}{a} z\big)$ over the eigenvalues α_i of A_k, a Poisson binomial; it is exactly Binomial$(b, 1/a)$ if and only if every α_i lies in $\{0, 1\}$, that is if and only if A_k is a rank-b projection. Among Poisson binomials of fixed mean b/a with n equal parameters the variance is $(b/a)(1 - b/(na))$, increasing in n; so the projections, where $n = b$, are the *variance-minimizing* extreme point of the whole family. The scalar counterexample $A_k = (b/p)I$ sits at the opposite end, at the Poisson limit, with variance exceeding the projection value by exactly b/a^2.

This has a consequence that organizes the whole picture. Fix (a, b) and let the common rank n of the A_k vary, with $A_k = (b/n)Q_k$ for rank-n projections Q_k summing to $(an/b)I$, so that $\operatorname{tr} A_k = b$ and $\sum_k A_k = aI$ throughout. At $(a, b) = (4, 2)$, $p = 6$, $q = 12$ the extreme roots move monotonically:

n	least root	greatest root
2	1.129	6.871
3	1.004	7.706
4	0.951	8.090
5	0.921	8.317
6	0.903	8.467

against the band $[0.536, 7.464]$ at $n = b = 2$ and the Marchenko–Pastur interval $[0.343, 11.657]$ approached as $n \to p$. So the conjectured band is the endpoint at $n = b$ of a family of intervals increasing in n, whose other endpoint is Marchenko–Pastur, and the Marchenko–Pastur value is attained. That is the correct reading of the obstructions recorded above: each of them is sharp for the class it sees, namely the one with n unrestricted, and the conjecture is the assertion that restricting to $n = b$ contracts the interval all the way to the tree band.

The condition $\operatorname{rank} A_k \le b$ is therefore the condition that the occupancy of a block fluctuates as little as possible at its given mean, and the conjecture asserts that minimal marginal fluctuation propagates to the extreme confinement recorded above. That suggests the right statement is one about measures rather than matrices: for a random vector $(s_1, \ldots, s_q)$ of nonnegative integers with $\sum_k s_k = p$ almost surely, each s_k exactly Binomial$(b, 1/a)$, and joint generating function homogeneous of degree p and real stable, are the roots of $\mathbb{E} \prod_k (y - a s_k)$ confined to the band? Every projection family induces such a law, and the scalar family does not, failing the marginal condition.

That class is rigid, which is what makes the reformulation faithful rather than a weakening. The marginal condition is itself a rigidity statement: in the polarized picture it holds if and only if the b slots inside each block are exactly jointly independent Bernoulli$(1/a)$. Consequently, any *determinantal* law satisfying the two combinatorial conditions has a rank-p projection kernel whose diagonal $b \times b$ blocks are $\frac{1}{a} I_b$, that is, it is the Naimark kernel of a projection family; and any law given by independent balls in bins satisfying them comes from an (a, b)-biregular incidence matrix, that is, from a commuting family. So the two canonical constructions of stable laws produce nothing new. At $b = 2$ more is true: for $(p, q, a, b) = (4, 6, 3, 2)$ and $(3, 6, 4, 2)$ the affine hull of the stable laws in the class coincides with the affine hull of the projection families, of dimensions 54 and 14 against 78 and 38 for the two combinatorial conditions alone. The measure formulation is therefore not a relaxation at $b = 2$.

The two combinatorial conditions alone do not suffice, and the point at which they fail is sharp: over all laws satisfying them the largest root is confined to the band only for q below an explicit threshold, and it reaches the trivial value ab exactly at $q = a^b$, the smallest q at which the binomial profile $n_j = q\binom{b}{j}(a-1)^{b-j} a^{-b}$ is integral. Stability is what must supply the rest, and it cannot be seen by any argument that symmetrizes: the divisibility constraint that stability imposes, namely that $(w + a - 1)^{b-1}$ divides $\mathbb{E}\big[s_l\, w^{s_k}\big]$, becomes vacuous on exchangeable laws. Negative association alone is likewise insufficient, by an explicit permutation counterexample whose polynomial has a root at ab.

The size of the claim is also worth recording. Since E_1 and E_2 are universal, the root distribution of μ has mean a and variance $a(b-1)$ for every admissible family. Measured against that, the lower band edge sits between 1.35 and 1.92 standard deviations below the mean over the range $3 \le a \le 100$, tending to 2 as a grows. The conjecture is therefore a statement of extreme concentration: the least of p roots is confined to about two standard deviations, where an unstructured sample of that size would reach out to order $\sqrt{\log p}$.

11.1 The extremal problem in dimension four

Graphs are not always extremal in that class, however, and the smallest case can be settled exactly. This is no longer needed for the band, at $p = 4$, $b = 2$ the reflection gives $\mu(t + a) = t^4 - 2at^2 + \beta$, and real-rootedness alone forces $0 \le \beta \le a^2$, hence $t^2 \le 2a \le 4(a - 1)$, but it identifies the extremal families. Take $b = 2$, $p = 4$, so $q = 2a$ and $\sum_k P_k = aI_4$. Writing $g_{ij} = \operatorname{tr}(P_iP_j)$ and $h_{ij} = \operatorname{tr}\big((P_iP_j)^2\big)$,

Proposition 55. *For rank-2 orthogonal projections $P_1, \dots, P_q$ on $\mathbb{R}^4$ with $\sum_k P_k = \frac{q}{2}I$,*

$$E_4 = \frac{q^4}{16} - \frac{q^3}{4} + \frac{q}{2} + \frac{1}{4}\sum_{i \ne j}\big(g_{ij}^2 - h_{ij}\big),$$

and every summand is nonnegative. Equality holds exactly when $\operatorname{rank}(P_iP_j) \le 1$ *for all $i \ne j$.*

Nonnegativity is immediate once the right object is named: for $i \ne j$ put $X = P_jP_iP_j \succeq 0$; then $\operatorname{tr}\big((P_iP_j)^m\big) = \operatorname{tr}(X^m)$ for every $m \ge 1$, so $g_{ij}^2 - h_{ij} = (\operatorname{tr} X)^2 - \operatorname{tr} X^2 = 2\sum_{r<s}\lambda_r\lambda_s \ge 0$. The last step is formalized in `RamaLean/TracePSD.lean`: `sq_sum_sub_sum_sq` gives the identity $\big(\sum f\big)^2 - \sum f^2 = \sum_i \sum_{j \ne i} f_if_j$ over any finite index set and `sum_sq_le_sq_sum` the inequality under nonnegativity, with the spectral input carried as explicit hypotheses on the matrix form.

Identifying a 2-plane $U \subset \mathbb{R}^4$ with its unit simple bivector $\omega_U \in \Lambda^2\mathbb{R}^4 \cong \mathbb{R}^6$ gives $g_{ij}^2 - h_{ij} = 2\langle\omega_i, \omega_j\rangle^2$, so the Welch bound for q unit vectors in a 6-dimensional space yields

$$E_4 \;\ge\; \frac{q^4}{16} - \frac{q^3}{4} + \frac{q^2}{12},$$

with equality exactly when $\{\omega_k\}$ is a tight frame of *simple* bivectors. This gives 30, 400/3, 1150/3 and 876 at $q = 6, 8, 10, 12$, matching the values found numerically.

At $q = 6$ the bound reads $E_4 \ge 30$, attained by the incidence family of K_4; and there the tight-frame condition degenerates to "orthonormal basis", which is the one value of q at which it can be met by simple bivectors alone. That is why graphs are extremal at $q = 6$ and not at $q = 8$, where the minimum 400/3 lies strictly below the graph minimum 134: the coincidence is the degeneration, not a principle. Even at $q = 6$ the graphs are only a slice of the minimizers, the equality locus is 4-dimensional modulo $O(4)$, and taking the ω_k from the six icosahedral axes, with the two $\mathbb{R}^3$ components Galois conjugate over $\mathbb{Q}(\sqrt{5})$, produces a noncommuting family with all $g_{ij} = 4/5$ and $E_4 = 30$. A lower bound of the same shape is known in the random setting: Brito, Dumitriu and Harris [9] show that a random bipartite biregular graph satisfies $\eta^+_{\min} \ge \sqrt{d_1 - 1} - \sqrt{d_2 - 1} - \varepsilon_n$ with high probability. That is an asymptotic statement about adjacency eigenvalues of random graphs; Conjecture 1 is a deterministic statement about matching polynomial roots of every graph.

12 What a proof must contain

Each of the methods that does not reach the band fails for a reason that can be named exactly, and each such reason is a property that a proof must have. We collect them here as a specification rather than as a list of failures: a proof must not induct on vertex deletion, must not be a side-constant specialization of the multivariate matching polynomial, must control deficient vertices of the path tree, and must not pass through Floquet theory. Every one of these is sharp, in the sense that the method in question is exactly optimal for the class of objects it can see; that is the content of Section 11.

The lower bound, unlike the upper bound, is not monotone under vertex deletion. The Heilmann–Lieb argument for $|x| \leq 2\sqrt{\Delta - 1}$ inducts on vertex deletion, which is available because Δ only decreases. The quantity $|\sqrt{a-1} - \sqrt{b-1}|$ has no such monotonicity: deleting one vertex of the larger side of $K_{3,4}$ produces $K_{3,3}$, which is regular, and the gap collapses to zero, for $a = b$ one has $c = 2\sqrt{m}$, so the interval $[c - 2\sqrt{m},\, c + 2\sqrt{m}]$ has its lower end at 0 and the statement is empty. So the naive induction by vertex deletion, which is what yields the upper bound, cannot prove the lower one.

That obstruction can be strengthened, and in a way that rules out the whole family of such arguments rather than one instance of it. Every b-regular bipartite graph is an induced subgraph of some (a, b)-biregular graph, by an explicit padding construction; and over b-regular bipartite graphs the least root of ν_G tends to 0, in closed form $2\sin(\pi/2N)$ for $b = 2$. So on the class obtained by closing the biregular graphs under vertex deletion, the infimum of the least root is 0, and the inequality to be proved is simply false there. Any argument that deletes vertices and applies an inductive hypothesis, cavity recursions, stability transfer, the Heilmann–Lieb induction, is an induction over that closure, and therefore cannot work, not because biregularity is lost but because the statement itself fails. An induction that carries something other than the target statement is not excluded, and §2.5 gives one: its hypothesis $q - r \geq 1$ is precisely a hypothesis the padding family above violates. For $\mathrm{AG}(2, 3)$, where $(s - t)^2 = 0.10102$, the minimum over all choices of d deleted vertices on the larger side runs 0.3593, 0.2424, 0.1383, 0.0549 for $d = 0, 1, 2, 3$: it drops below the target at depth three.

A second route, substituting side-constant values into the multivariate matching polynomial of Heilmann and Lieb, fails for a reason that is visible in one line. Writing $M_G(z; x) = \sum_M z^{|M|} \prod_{v \notin M} x_v$ and setting $x_v = \alpha$ on the p-side and $x_v = \beta$ on the q-side gives $\sum_k m_k z^k \alpha^{p-k} \beta^{q-k}$, in which the exponent pair contributes c^{p-q} under $(\alpha, \beta) \mapsto (c\alpha, \beta/c)$ independently of k; so the specialization depends on α and β only through the product $\alpha\beta$. Since the image of a product of two upper half-planes is $\mathbb{C} \setminus [0, \infty)$, stability under this substitution says exactly that the roots are real and nonnegative, and nothing more. In particular it says the same thing when $a = b$, where the conclusion to be proved is empty. No choice of α and β can see $a - b$.

Two obstructions remain in the direct approach, both caused by cycles in G. First, a leaf of the path tree that is an A-vertex carries $R_A = y > 0$, of the wrong sign. This one is disposed of: a maximal self-avoiding walk ending at $w \in A$ contains all a neighbours of w, hence at least a vertices of B and therefore at least a vertices of A, so

$$p < a \implies \text{the path tree has no } A\text{-leaf},$$

which covers every $K_{p,q}$ with $p < q$. Second, and this is what blocks the argument, an internal A-vertex of the path tree may have $k < P$ children, the walk having already consumed part of its neighbourhood; the bound $R_A \leq y - P\alpha/(\alpha + Q)$ then fails. With $k = 1$ closure would require $\alpha(1 - \alpha - Q)/(\alpha + Q) \geq y$, impossible for $b \geq 3$ since $Q \geq 2$. Deficiency occurs already in $K_{3,4}$, where the walk $a_1 b_1 a_2 b_2 a_3$ leaves a_3 with two children rather than three. What the argument as it stands establishes is that a *full* truncated biregular tree carries no eigenvalue in the gap, consistent with the deficient subtree P_8 carrying one; that is a statement about the tree and not about G. Controlling deficient A-vertices is the open problem.

The case $b \geq 3$ of the biregular family is the remaining open one, since $b = 2$ is Theorem 8; note that every graph named below has $a \neq b$, so the statement being tested is not the empty one. It has been tested at $d = 1$ on $K_{3,4}, K_{3,5}, K_{3,6}, K_{3,7}, K_{3,8}, K_{4,5}, K_{4,6}, K_{4,7}, K_{5,6}$ and a $(3,2)$ design incidence graph, with smallest-root ratios between 1.51 and 3.33 and no root in a gap.

The natural attack is Godsil's recursion $R_v = x - \sum_u 1/R_u$, whose fixed points on the (a,b)-biregular tree are real precisely off the spectrum. Asking that the region $R_b \in [x, P/x)$, $P = a-1$, $Q = b-1$, be closed under the recursion reduces to $u^2 - (2P+Q)u + P^2 > 0$ at $u = x^2$, i.e. to $u < \bigl(2P + Q - \sqrt{Q^2 + 4PQ}\bigr)/2$. This is satisfied throughout the gap with room to spare: the ratio of that threshold to $(\sqrt{P} - \sqrt{Q})^2$ ranges from 2.7 to 37.7 over the cases above. So the analytic part of an induction is not the difficulty.

What blocks it is the boundary. A leaf of the path tree carries $R = x > 0$, while the argument wants $R < 0$ at a-vertices, and mixed signs among the children of a b-vertex leave the sum uncontrolled. One might hope to exploit that the counterexample to the naive subtree argument, P_8 inside the $(3,2)$-biregular tree, is deficient at *internal* vertices, whereas a path tree is full at internal vertices; but that is false for graphs of small girth, where a self-avoiding walk can be blocked early, in K_4 the walk $0, 1, 2$ admits a single extension. We therefore leave $b \geq 3$ open.

A proof of Conjecture 1 cannot go through Floquet theory. For $b_1 = b$ the deck group of the universal cover is the free group F_b, and Floquet theory is exactly the statement that for an *abelian* deck group the fibres are indexed by the characters of that group; the two coincide only at $b = 1$, where $F_1 = \mathbb{Z}$. For $b \geq 2$ the correct index set is the unitary dual of F_b, and the relevant object is the adjacency operator in the reduced C^*-algebra of F_b. This is not merely a technical obstruction: the computations of [27] show that the per-component weight in the exponential formula, a single graph for $b = 1$, becomes a sum over the many non-isomorphic connected ℓ-covers for $b \geq 2$.

That framework is not empty. As $d \to \infty$, independent uniform random permutations are strongly asymptotically free [15], and consequently the spectrum of a random d-lift of G converges to $\mathrm{spec}(T)$. Conjecture 1 is a finite-d statement about the expected characteristic polynomial rather than an asymptotic statement about typical lifts, and its content is precisely that the containment holds uniformly in d, with no exceptional set. What the $b_1 = 1$ case shows is that the interval in [4] is not the natural receptacle for the roots, and that the correct one is spectral rather than metric.

13 What this leaves

The biregular case is not a leftover from the refutation; it is where the phenomenon actually lives, and this paper has tried to say how much room it has.

On the positive side the case is genuinely constrained. Complete bipartite graphs never refute it, by a margin $\sqrt{d-1} - h_d/2$ that Gershgorin on the Hermite Jacobi matrix makes positive for every d, and the ratio route closes the inner edge on a nontrivial class. Those are the two proved statements here, and they are not vacuous: the first covers a family whose gap can be made arbitrarily wide, and the second reaches the edge exactly, its certificate degenerating at the gap edge and nowhere earlier.

On the negative side the margin decays like $n^{-2/3}$. The statement is true but tight, and that single fact disqualifies most of what one would try next, including the Gershgorin constant that proves the complete bipartite case: any argument yielding a bound independent of the size is known in advance to be insufficient. What is needed is a Friedman-type edge theorem, in which the roots fill out $\mathrm{spec}(T)$ without escaping it, at the soft-edge scale.

The obstructions are the part we would most want a reader to take away, because each one closes a route that looks open from outside.

- A barrier argument that sees only the rank, the trace and $\sum_k P_k = aI$ is reading the Marchenko–Pastur law, and $A_k = (b/p)I_p$ satisfies all three while exceeding the tree band already at $p = 4$. No such argument can reach either edge.

- Compressions cannot reach the inner end at all: a compression takes values the original excludes, so no argument phrased in Rayleigh quotients can exclude them.

- The coefficient ladder's reach is unbounded in the moment order, but a bound holding at every rung with a constant free of the order already gives the conclusion (`MomentLadder.band_of_all_moments`). The dimension restriction on the two unconditional bounds is therefore intrinsic to the method, not an artefact of stopping early, and the natural a priori input — the tree walk count — is false off the coordinate case.

One question the paper does close, rather than obstruct, is the value of the constant. On the coordinate locus the mixed characteristic polynomial of a tight rank-b family is the matching polynomial of an (a, b)-biregular incidence graph in the squared variable (Proposition 35), so Xu's inequality is a theorem there at every b and, by Proposition 37, the constant $(\sqrt{a-1}+\sqrt{b-1})^2$ is attained and cannot be lowered. That removes a possibility the numerics had left open — that the constant is honest at $b = 2$ and slack at $b \geq 3$, the adversarial search having stalled at 0.739 and 0.726 there — and it fixes what any proof of the general statement has to deliver. It also locates the difficulty: the bridge fails off the coordinate locus, and that is where the whole content of the conjecture now sits.

What survives all three obstructions is the ratio recursion, which bounds ratios of matching polynomials rather than Rayleigh quotients, and that is where we would put further effort. The remaining gap is stated plainly: the certificate closes for six of the seven families tested, and the defect in the seventh is a right vertex whose child ratio sits well below the interval a uniform certificate can express. Closing that is the concrete open problem this paper ends on.

14 Related work

Two ingredients are classical and are used, not claimed. The factorization of the characteristic polynomial of a cyclic cover into twisted determinants at roots of unity is the block-circulant diagonalization [17, 18]; in the infinite-cover limit it is the Floquet–Bloch decomposition of a periodic discrete operator, and $y = -P/(2Q)$ is the corresponding Floquet discriminant. The identity $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ is [4]. Closest in form is McSorley and Feinsilver [16].

For a path whose edges are cyclically labelled with t labels they establish a (τ, Δ)-recurrence $\Phi_N = \tau\Phi_{N-1} - \Delta\Phi_{N-2}$, which is the algebraic skeleton of (2), and their Theorem 4.4 solves it as $\Delta^{N/2}U_N(\tau/2\sqrt{\Delta})$, exactly the form used here. In the univariate specialization their τ is the matching polynomial of the cycle C_t and $\Delta = (-1)^t x_1 \cdots x_t$ is a monomial; passing between their counting normalization and the characteristic one used here is the homogenization $V_d \mapsto c^d V_d$, under which a monomial Δ becomes $\mu_{G-V(C)} = 1$. Their Theorems 4.4 and 4.5 together give $U_{dn+n-1} = U_d(T_n)U_{n-1}$, the identity invoked in Section 1 to pass between Theorem 3 and Hall's conjecture, so that bridge was already available in 2009. Their parameter is the number of label periods along a path and their graphs are paths, cycles and rooted trees; there is no covering and no cycle with trees attached. What is added here is the pair $(\mu_G, \mu_{G-V(C)})$, whose second entry is a genuine polynomial recording the attached trees rather than a monomial, together with the covering interpretation that makes the recurrence a statement about $\mu_{d,G}$.

A The formalization

The core algebraic derivation, the exponential formula, the closed form of Theorem 3 for all n, r (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 7, is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula`, the exponential formula (3) in integer form $r\,A_r = \sum_k r^{(k)} f(k) A_{r-k}$ over any commutative ring, proved via Mathlib's Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType_mul_eq`);
- `Paper2.thm1_Sr`, Theorem 3 in the form: the expected cycle weight (1) equals the closed form, for all n, r, over any $\mathbb{Q}$-algebra domain;
- `Paper2.cG_factored`, `Paper2.quotient_parity`, the factorization and Corollary 7.

The three remaining corollaries are now formalized in full. Corollary 4 is proved as a literal identity of formal power series, `Paper2GenFun.genU_mul` giving $(1 - 2YX + X^2)\sum_d U_d(Y)X^d = 1$ and `genPhi_mul` the companion form with numerator $(1 - X)^2$, by coefficient comparison against the Chebyshev recurrence. Corollary 5 is proved as the statement about roots, `Paper2Roots.Phi_ne_zero_of_two_lt_abs`: if $|x| > 2$ then $\Phi_{n,r}(x) \neq 0$, via $|T_n(x/2)| > 1$ (`one_lt_abs_cT`) and the vanishing of neither factor of the factorization. Corollary 6 is proved for all n and r, `Paper2FibLucas.cor2_phi3_general`, replacing the bounded `native_decide`; the route is the identity $F_{j+2m} + F_j = L_m F_{j+m}$ for even m (`fib_shift`), whose base cases are $F_{2m} = L_m F_m$ and, using Cassini, $F_{2m+1} + 1 = L_m F_{m+1}$, and which is stated without truncated subtraction by writing $L_m = 2F_{m+1} - F_m$.

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No `sorry`. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles, elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

For the biregular half, `RamaLean/BipartiteMatchingPoly.lean` supplies a matching polynomial for bipartite graphs, Mathlib having none, and formalizes: the deletion recursion

$Q_G = Q_{G-v} - z\sum_{u\sim v} Q_{G-v-u}$ (`bipQ_delete_left`); the general Taylor shift $[X^j]f(X+c) = \sum_i \binom{i+j}{j} f_{i+j}c^i$ (`taylor_coeff_of_natDegree_le`), from which the coefficient formulas at $p-1$, $p-2$ and $p-4$ follow; the count $2m_2 + |\mathrm{confL}| + |\mathrm{confR}| = |E|(|E|-1)$ with the two conflict classes as degree sums; and the combinatorial inputs to Theorems 26 and 27, that three pairwise-meeting edges of a bipartite graph share a vertex (`three_meeting_share`), that the conflict degree of an edge is $(d_u - 1) + (d_v - 1)$ (`conflict_degree`), the trichotomy for four-cycles of the conflict graph (`four_cycle_trichotomy`), and the identity the whole expansion runs on, that the matching indicator is the alternating sum over subsets of the conflicting pairs (`alt_sum_conflictPairs`). Several of these, including the trichotomy, depend on no axioms at all. `RamaLean/Paper2Unicyclic.lean` further formalizes that the lower band edge is $(\sqrt{P}-\sqrt{Q})^2$ and so vanishes exactly when $P = Q$ (`gap_eq_zero_iff`), which is the triviality recorded above.

The inclusion–exclusion itself is formalized as a chain. The matching indicator is the alternating sum over subsets of the conflicting pairs of a set of edges (`alt_sum_conflictPairs`); substituting it turns m_k into a double sum (`mCount_eq_sum_alt`); because conflict is intrinsic to a pair of edges, the inner index set depends on the outer one only through a containment (`powerset_conflictPairs_eq`), so the two summations exchange (`mCount_inclusion_exclusion`); the resulting weight is a binomial coefficient, by a bijection between the k-subsets containing a fixed V and the $(k - |V|)$-subsets of the complement (`card_powersetCard_filter_superset`), giving

$$m_k = \sum_S (-1)^{|S|}\binom{|E| - |V(S)|}{k - |V(S)|}$$

over sets S of conflicting pairs (`mCount_closed_form`); and since the weight sees S only through $|S|$ and $|V(S)|$, the sum fibers over that pair (`mCount_grouped`, `mCount_grouped_range`), which is the grouping by subgraph type. The chain is exercised end to end at $k = 1$ (`mCount_one_via_inclusion_exclusion`): every nonempty set of conflicting pairs has at least two endpoints, so only the empty set survives the binomial and contributes $\binom{|E|}{1}$, reproving $m_1 = |E|$ through the inclusion–exclusion rather than by the direct bijection, so that two independent routes cross-check each link.

What is *not* formalized is the evaluation of the ten resulting counts as explicit functions of (p, a, b) and $C_4(G)$, the step at which `three_meeting_share` and `four_cycle_trichotomy` are applied, together with the numerical verifications. One obstacle there is worth recording: the formalized conflict set consists of *ordered* pairs, so each edge of the conflict graph occurs twice. This leaves the inclusion–exclusion untouched, since the alternating sum vanishes over any nonempty index set whether or not it double counts, but it means the grouping enumerates more configurations than the ten unordered types above, and aligning the two would require rebuilding the conflict set on unordered pairs.

The operator side of Sections [27]–11 is formalized in four further files, all on the standard three axioms and with no `sorry`. `RamaLean/SignReflection.lean` carries Proposition 53 to the point where only the rank-one expansion of μ is left by hand: `sign_avg_reflect` proves that $\sum_{\varepsilon\in\{\pm1\}^q} \det(tI - \sum_k \varepsilon_k S_k)$ is even or odd in t according to the parity of the dimension, with *no* hypothesis on the S_k, by the involution $\varepsilon \mapsto -\varepsilon$; `sum_prod_sgn` is character orthogonality on the Boolean cube, proved by the same involution applied to one coordinate; `sum_multiaffine` deduces that averaging a multi-affine function over signs returns 2^q times its constant term, which is exactly what $\prod_k(1-\partial_{z_k})$ extracts; and `mixedChar_reflect` and `band_edges_swap` supply the shift and the identity $2a-(\sqrt{a-1}+1)^2=(\sqrt{a-1}-1)^2$ that exchanges the band edges. `RamaLean/ProductBound.lean` proves Proposition 32 (`no_root_above`: a convex combination of products of positive numbers is positive) together with the exact overshoot $ab-(s+t)^2=(\sqrt{m}-1)^2$ (`edge_gap`) and its vanishing exactly at $m=1$ (`edge_gap_eq_zero_iff`), which is the case $a=b=2$ where the band is proved outright; it also proves Proposition 33, that at $b=2$ the two band edges are the single condition $\pi_{\max}\le 4(a-1)/a^2$ (`band_iff_pi`), and that the strictly weaker $\pi_{\max}\le 4/a$ already yields $a+2\sqrt{a}$, which beats the Marchenko–Pastur edge for every a and the product bound ab for $a>4$ (`weak_bound`, `weak_beats_mp`, `weak_beats_ab`). `RamaLean/GramDet.lean` discharges the exterior-algebra inputs by writing them in Gram coordinates, where each becomes a determinant statement. `det_border` is the bordered determinant formula $\det\left[\begin{smallmatrix} a & r^{\mathsf{T}} \\ c & G\end{smallmatrix}\right]=a\det G-r^{\mathsf{T}}\operatorname{adj}(G)c$, which is exactly (18); `det_gram_eq_zero_of_dep` is $f_k\wedge\omega'_k=0$, the vanishing of a Gram determinant on a dependent family; and `hsimple_of_border_zero` combines them into $\|f_k\|^2\|\omega'_k\|^2=\|\iota_{f_k}\omega'_k\|^2$, which is precisely the hypothesis `hsimple` used below, so that hypothesis is now proved rather than assumed. `RamaLean/Tightness.lean` supplies the other input, `htight`: that $\operatorname{Adj}(A)=aI$ forces $\sum_k\iota_{f_k}\omega'_k=0$. Working in coordinates, with a rank-two block presented by its two spanning vectors, the polarization step becomes three elementary facts, `iota_antisymm`, that $\langle a,\iota_b\omega\rangle=-\langle b,\iota_a\omega\rangle$, which is a two-term `ring` identity; `iota_proj_of_orthogonal`, that compressing the block to $e^\perp$ does not change the pairing against f_k; and that each $\iota_{f_k}\omega'_k$ already lies in $e^\perp$. `tight_sum_contraction_eq_zero` assembles them, and `crossTerm_nonneg_of_tight` composes the result with `CrossTerm.crossTerm_nonneg`, so that Theorem 40 is machine-checked from the tightness hypothesis alone. `RamaLean/Plucker.lean` closes the last link, the identification of `CrossTerm`'s coordinate vector `w` with the Plücker coordinates of ω'_k. Representing $b_1\wedge b_2$ by its antisymmetric array on *ordered* pairs scaled by $1/\sqrt{2}$ makes `pl_dot`, that the resulting inner product is the Gram determinant, hold with no $i<j$ bookkeeping: the double count and the two factors of $1/\sqrt{2}$ cancel. `iota_eq_comb` writes $f_k=\iota_e\omega_k$ in the compressed plane with explicit coefficients $-\langle e,b_2\rangle$ and $\langle e,b_1\rangle$, and `hsimple_of_comb` then makes `hsimple` a `ring` identity rather than an assumption. `crossTerm_nonneg_plucker` assembles everything: **for a tight family of rank-two blocks and any unit vector, $C_2\ge 0$ is machine-checked with no hypothesis remaining**.

The same holds at every level. `GramDet.border_gram` identifies the bordered Gram matrix of $[v\mid M]$ with the Gram matrix of the bordered family; `border_det_eq_zero_of_mem_span` then gives $f\wedge\omega=0$ whenever f lies in the column span, which it always does; and `hsimple_of_mem_span` combines them into $\|f\|^2\|\omega\|^2=\|\iota_f\omega\|^2$ for a wedge of *any* number of vectors. So the hypothesis `hsimple` of Theorem 41 is discharged at every r, not only at $r=2$, and what Theorem 41 asserts, that each S-summand of C_r is a Schur square minus the square of a contraction, is machine-checked unconditionally. Finally `gram_det_basis_of_injective` and `gram_det_basis_of_not_injective` prove that for a coordinate family the Gram determinant of a set of blocks is 1 on matchings and 0 otherwise, that is $F_A=\mu_G$; that is the first of the two inputs to Proposition 42, and the second, the Kesten–McKay limit, is the only analytic ingredient anywhere in this section and is cited rather than proved. `RamaLean/CrossTerm.lean` carries Theorems 40 and 41. `crossTerm_eq_sq_sub` is the general level: with no hypothesis beyond `hsimple` it proves

$\sum_{k\neq l}[\langle f_k, f_l\rangle\langle w_k, w_l\rangle - \langle u_k, u_l\rangle] = \|\sum_k f_k \otimes w_k\|^2 - \|\sum_k u_k\|^2$, which is the S-summand of Theorem 41 once $w_k = \omega'_k \wedge \omega'_S$ and $u_k = \iota_{f_k}(\omega'_k \wedge \omega'_S)$. Its two geometric inputs are hypotheses, `hsimple`, that $f_k \wedge \omega'_k = 0$ in its metric form $\|f_k\|^2\|\omega'_k\|^2 = \|u_k\|^2$, and `htight`, that $\sum_k u_k = 0$, and `crossTerm_eq_sq` specializes it to $r = 2$, where tightness kills the second square, and proves that they force C_2 to equal $\sum_{i,j}\big(\sum_k (f_k)_i(\omega'_k)_j\big)^2$, with `crossTerm_nonneg` and `crossTerm_eq_zero_iff` the consequences. The supporting `sum_gram_prod_eq_sum_sq` is the Schur product theorem in the case needed, that $\sum_{k,l}\langle f_k, f_l\rangle\langle w_k, w_l\rangle$ is a sum of squares. No exterior algebra is involved: the bivectors enter only through their coordinates. `RamaLean/VertexSplit.lean` carries the Gram form of Theorem 38. Its core, `det_add_vecMulVec`, is the adjugate form (17) of the matrix determinant lemma; Mathlib has the lemma only as $\det(A + uv^\mathsf{T}) = \det A \cdot \det(1 + v^\mathsf{T} A^{-1} u)$ and records the adjugate form as an explicit `TODO`, so this supplies it, under the hypothesis that $\det A$ is a unit, which is what the application needs, the Gram matrix of an independent family being positive definite. `det_le_det_add_vecMulVec` is the inequality the recursion consumes, $M_r(A) \geq M_r(A^{(e)})$, deduced from `AdjugatePSD.adjugate_posDef`. The singular case is scoped out as in `AdjugatePSD`: both sides of (17) are polynomial in the entries, so it extends by continuity from $A + \varepsilon I$, and that limit is not formalized. `RamaLean/CavityThreshold.lean` carries Proposition 39: `cavity_step` is the induction step, that child ratios at least $x/2$, nonnegative weights of total at most a, a nonpositive remainder and $4a \leq x^2$ force the parent ratio to be at least $x/2$ again; `threshold_iff` is its sharpness, that $x - 2a/x \geq x/2$ holds *if and only if* $x \geq 2\sqrt{a}$; `double_root_at_threshold` is the structural form of the same statement, the factorization $\rho^2 - 2\sqrt{a}\,\rho + a = (\rho - \sqrt{a})^2$ that collapses the interval of admissible thresholds to a point; and `no_slack_propagates` is Proposition 47; `cavity_step_deficient` is Proposition 49, the strengthened step; `compression_sum`, `trace_le_of_rayleigh` and `remainder_upper` are Proposition 50; and `ratio_form`, `one_add_le_inv_one_sub`, `inv_one_sub_sub_one_add`, `km_slack_calc` and `km_satisfies_target` are (21) and Proposition 51. Proposition 48 is `Tightness.adj_compressed`, with the per-block splitting `Tightness.iota_split`. `RamaLean/KestenMcKay.lean` carries Proposition 42: `radical_at_edge` that the radical is exactly 2 at the threshold, `G_at_edge` the closed form there, `inv_G_div_edge` the constant $1 + 1/(1 + \sqrt{a})$, and `excess_eq`, `excess_pos`, `excess_lt` that the excess is exactly $1/(1 + \sqrt{a})$, positive for every $a > 1$ and below any ε once $\sqrt{a} > 1/\varepsilon$. The two inputs it does not formalize are the ones that are not arithmetic: that F_A is the matching polynomial, which is Theorem 34. The Kesten–McKay step is no longer cited as a limit: `RamaLean/MomentTransfer.lean` carries Proposition 44 in full, namely the exact geometric expansion with remainder (`geom_remainder`), the remainder bound (`remainder_le`), the moment transfer (`stieltjes_close`), its form for reciprocals (`inv_close`) and the positivity of the gap between the support and the evaluation point (`edge_gap_pos`). What remains cited is the locality of root moments, which is [20] after [21], and Godsil's tree-like walk formulation [22]. The vertex recursion of Theorem 38 and the sign of X_e enter as hypotheses; what is machine-checked is that they close the induction, and that they close it exactly at $2\sqrt{a}$. `RamaLean/TracePSD.lean` and `RamaLean/AdjugatePSD.lean` carry the two positivity facts the extremal computation and the kernel recursion run on. In both, the standard input is a visible hypothesis rather than a hidden one: `trace_sq_le_sq_trace` takes the spectral decomposition of a positive semidefinite matrix as an explicit assumption, and `adjugate_posDef` is proved in the definite case, the semidefinite limit $A + \varepsilon I$ being left by hand.

`RamaLean/LaguerreBand.lean` carries Theorem 9 in full on the inequality side: `sqrt_amgm` and `sqrt_step` are the arithmetic–geometric mean step (4), `rowRadius_le` and `lastRow_le` are the two Gershgorin row bounds, `band_endpoints` identifies the disc with the biregular tree band, and `laguerre_band` assembles them through Mathlib's `eigenvalue_mem_ball`. What is not formalized is the classical passage from $\mu_{K_{p,q}}$ to the Laguerre polynomial and thence to its Jacobi matrix; that identification is a coefficient comparison and is done by hand above.

The three developments of [27] are formalized to the extent stated there. `RamaLean/AbelianCover.lean` proves that the integrand is real (`det_sub_smul_one_im`), that a continuous real function with vanishing average on a connected space has a zero (`exists_zero_of_integral_zero`), and the resulting localization and reduction; the Godsil–Gutman expansion is assumed as the hypothesis `haverage`. `RamaLean/TreeSubstitution.lean` proves the substitution identity as a homogenization (`homogenize`), fixes its orientation on the star (`star_ratio`), and transports Heilmann–Lieb through a subdivision (`subdivision_root`, `subdivision_abs`); the covering combinatorics is by hand. `RamaLean/FeedbackVertex.lean` proves the combinatorial core (`no_bypass`, via Mathlib's characterization of acyclicity by path uniqueness), the resolvent diagonal, the algebraic collapse of the trace formula, and the trace step; the Pimsner–Voiculescu input appears as an explicit hypothesis of `feedback_one_of`, since it is not in Mathlib. Every theorem in all four files depends only on the three standard axioms, and none uses `native_decide`.

Remark 56. Everything here is stated for the uniform permutation model; by [4] the quotient coincides with the $(r-1)$-matching polynomial, so Theorem 3 can alternatively be deduced from [5], with the caveat about that preprint's status recorded in [27]'s preamble. We believe the generating function of Corollary 4, the evaluation of Corollary 6, and the formalization are new.

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